MODULE 1: ARITHMETIC

Chapter 1: Percentages

Base Principles, Change Dynamics, Product-Constancy & Successive Changes

1. Introduction and Learning Objectives

Percentage is the foundational pillar of CAT Arithmetic. More than 40% of CAT QA questions across Arithmetic, Data Interpretation, and Algebra directly rely on percentage calculations.

LEARNING OBJECTIVES

- Internalize fraction-to-percentage conversions for instant mental calculation.
- Apply operational multiplying factors for rapid percentage increases and decreases.
- Master product-constancy and inverse variations ($A \times B = C$).
- Solve complex successive percentage changes using both multiplicative and additive models.
- Apply non-formula approximation and elimination techniques under strict CAT time constraints.

2. Fundamental Concepts and Definitions

The word **Percentage** originates from the Latin *per centum*, meaning "by the hundred." It expresses a fraction or ratio as a fraction of 100.

$$\text{Percentage Value} = (\text{Part} / \text{Whole}) \times 100$$

- **Absolute Value:** The raw numerical quantity (e.g., ₹500, 40 kg).
- **Relative Value:** A proportional measure expressed relative to a base value (e.g., 20%, $^1/_5$).
- **Base Value:** The baseline reference point (100%) against which a change is measured.

Golden Rule of Percentages: The base is always the initial value before the change occurred, unless specified otherwise.

3. Detailed Theoretical Explanations

3.1 Multiplying Factors (MF)

Instead of using two-step calculations (calculating the percentage and then adding/subtracting from the base), use a single operational **Multiplying Factor (MF)**.

$$\text{Final Value (F)} = \text{Initial Value (I)} \times \text{MF}$$

- **Percentage Increase (+x%):** $\mathbf{MF = 1 + (x / 100)}$. *Example: Increasing a quantity by 25% means MF = 1 + 0.25 = 1.25.*
- **Percentage Decrease (-x%):** $\mathbf{MF = 1 - (x / 100)}$. *Example: Decreasing a quantity by 15% means MF = 1 − 0.15 = 0.85.*

3.2 Product-Constancy Concept ($A \times B = C$)

When the product of two variables A and B is constant (C): $\mathbf{A \times B = K \Rightarrow A \propto 1 / B}$.

If A increases by a fraction of $^{a}/_{b}$, then to keep the product C constant, B must decrease by: $^{a}/_{a+b}$.

Conversely, if A decreases by $^{a}/_{b}$, B must increase by: $^{a}/_{b-a}$.

If A Changes By	Then B Must Change By (to keep C constant)
Increases by $^{1}/_{2}$ (+50%)	Decreases by $^{1}/_{2+1} = {}^{1}/_{3}$ (−33.33%)
Increases by $^{1}/_{3}$ (+33.33%)	Decreases by $^{1}/_{3+1} = {}^{1}/_{4}$ (−25%)
Increases by $^{1}/_{4}$ (+25%)	Decreases by $^{1}/_{4+1} = {}^{1}/_{5}$ (−20%)
Decreases by $^{1}/_{4}$ (−25%)	Increases by $^{1}/_{4-1} = {}^{1}/_{3}$ (+33.33%)

3.3 Successive Percentage Changes

For two successive changes of a% and b%:

$$\text{Net Percentage Change} = a + b + (a \times b / 100)$$

4. Essential Fractions-to-Percentage Conversion Table

Fraction	Percentage	Decimal	Fraction	Percentage	Decimal
$^{1}/_{2}$	50%	0.5	$^{1}/_{11}$	9.09%	0.0909
$^{1}/_{3}$	33.33%	0.3333	$^{1}/_{12}$	8.33%	0.0833
$^{1}/_{4}$	25%	0.25	$^{1}/_{13}$	7.69%	0.0769
$^{1}/_{5}$	20%	0.2	$^{1}/_{14}$	7.14%	0.0714
$^{1}/_{6}$	16.67%	0.1666	$^{1}/_{15}$	6.67%	0.0666
$^{1}/_{7}$	14.28%	0.1428	$^{1}/_{16}$	6.25%	0.0625
$^{1}/_{8}$	12.5%	0.125	$^{1}/_{20}$	5%	0.05
$^{1}/_{9}$	11.11%	0.1111	$^{1}/_{24}$	4.16%	0.0416

Fraction	Percentage	Decimal	Fraction	Percentage	Decimal
$^1/_{10}$	10%	0.1	$^1/_{25}$	4%	0.04

5. Solved Examples with Step-by-Step Explanations

Example 1: Basic Base Adjustment

Question: If A's salary is 25% more than B's salary, by what percentage is B's salary less than A's salary?

Step-by-Step Solution:

1. Let B's salary = 100.
2. A's salary = 100 + 25% of 100 = 125.
3. Compare B to A: A's salary (125) becomes the new base.
4. Difference = 125 − 100 = 25.
5. Required Percentage = (25 / 125) × 100 = $^1/_5$ × 100 = **20%**.

Example 2: Product Constancy Application

Question: The price of petrol increases by 33.33%. By what percentage must a motorist reduce petrol consumption so that the total expenditure remains unchanged?

Step-by-Step Solution:

1. Price × Consumption = Expenditure.
2. Price increases by 33.33% = $+^1/_3$ (a = 1, b = 3).
3. Fractional Decrease in Consumption = $^a/_{a+b}$ = $^1/_{1+3}$ = $^1/_4$.
4. Convert to percentage: $^1/_4$ = **25%**.

6. Mental Mathematics & Calculation Shortcuts

THE A% OF B = B% OF A PRINCIPLE

Switching the base simplifies mental math. For instance, evaluate **84% of 25**:

Rewrite as **25% of 84** = $^1/_4$ × 84 = **21**.

7. Common Mistakes and Misconceptions

❌ **Misconception: Additive Fallacy in Successive Changes**

Incorrect: A price increased by 20% and then by 30%, so total increase = 20% + 30% = 50%.

Correct: The second increase applies to the new base. Net Increase = 20 + 30 + (20 × 30 / 100) = **56%**.

8. CAT-Level Practice Questions

Q1 (Easy): Fresh fruit contains 68% water, while dry fruit contains 20% water. How many kilograms of dry fruit can be obtained from 100 kg of fresh fruit?

(A) 32 kg
(B) 40 kg
(C) 52 kg
(D) 25 kg

Q2 (Moderate): In an election, 10% of voters did not vote, and 60 votes were invalid. The winner won by 308 votes by securing 47% of total enrolled voters. Find total voters.

(A) 6,200
(B) 5,600
(C) 6,000
(D) 5,800

Q3 (Advanced): Due to a 20% reduction in price, a consumer buys 6 kg more sugar for ₹240. What is the original price per kg?

(A) ₹10
(B) ₹8
(C) ₹6.67
(D) ₹12

9. Answers & Key Solutions

- **Q1 Solution: (B) 40 kg.** Dry pulp remains constant. Fresh pulp = 32% of 100 kg = 32 kg. Dry fruit pulp = 80% of D = 32 kg ⇒ D = 40 kg.
- **Q2 Solution: (A) 6,200.** Let total = 100x. Winner = 47x, Loser = (90x − 60) − 47x = 43x − 60. Margin: 4x + 60 = 308 ⇒ x = 62. Total = 6,200.
- **Q3 Solution: (A) ₹10.** Price drops by $^1/_5$ ⇒ Quantity rises by $^1/_4$. $^1/_4$ of Original = 6 kg ⇒ Original Qty = 24 kg. Original Price = 240 / 24 = ₹10/kg.

10. Chapter Revision Sheet

PERCENTAGES QUICK REFERENCE

- **Base Formula:** Percentage = (Part / Base) × 100
- **Multiplying Factor:** Increase by x% ⇒ MF = 1 + (x/100); Decrease by x% ⇒ MF = 1 − (x/100)
- **Product Constancy:** Increase $^a/_b$ ⇒ Decrease $^a/_{(a+b)}$
- **Successive Change:** Net = a + b + (ab / 100)

MODULE 1: ARITHMETIC

Chapter 2: Profit, Loss & Discount

Commercial Accounting Metrics, Multi-Stage Trade & Cheating Merchant Frameworks

1. Introduction and Learning Objectives

Profit, Loss, and Discount build directly on the fundamental percentage concepts. CAT frequently tests these concepts through real-world commercial scenarios, including multi-stage trade transactions, marked-up pricing structures, tiered discount schemes, and faulty weights or measurement manipulations.

LEARNING OBJECTIVES

- Master core commercial accounting relationships connecting CP, SP, MP, Profit/Loss, and Discounts.
- Formulate and solve multi-stage trading transactions using Multiplying Factors (MF).
- Deconstruct complex successive discount structures and equivalence formulas.
- Solve all variations of faulty balance and cheating merchant problems using weight-and-price ratio frameworks.
- Apply fast mental math, percentage base-shifting, and structural ratio techniques.

2. Fundamental Concepts and Definitions

- **Cost Price (CP):** The base cost incurred to procure or manufacture an item.
- **Marked Price (MP):** The tagged or listed price before any discount.
- **Selling Price (SP):** The actual final price at which the transaction occurs.

Core Golden Rules:
1. Profit or Loss is ALWAYS calculated on Cost Price (CP), unless specified otherwise.
2. Discount is ALWAYS calculated on Marked Price (MP).
3. Markup is ALWAYS calculated on Cost Price (CP).

3. Detailed Theoretical Explanations

3.1 Golden Ratio of Profit & Discount

Equating expressions for SP via Profit and Discount:

$$CP \times (1 + P\% / 100) = MP \times (1 - d\% / 100) \Rightarrow MP / CP = (100 + P\%) / (100 - d\%)$$

3.2 Cheating Merchant Framework

Use the **Effective Ratio Principle** for dishonest traders:

$$(SP / CP)_{net} = (\text{Price Charged} / \text{Actual Price}) \times (\text{Claimed Weight} / \text{Actual Weight Given})$$

4. Key Formulas & Properties

Scenario	Formula / Result
Two items sold at same SP (one at +x% profit, one at −x% loss)	**Net Loss % = $(x / 10)^2$ %**
Two items bought at same CP (one at +x% profit, one at −x% loss)	**No Profit, No Loss (0%)**
Successive Discounts (d_1%, d_2%)	**$D_{eq} = d_1 + d_2 - (d_1 \times d_2 / 100)$**
Buy X, Get Y Free Scheme	**Discount % = [Y / (X + Y)] × 100**

5. Solved Examples

Example: Dishonest Merchant with Markup and False Weight

Question: A grocer marks up sugar by 20% and gives a 10% discount. However, his scale reads 1,000 g when it actually weighs only 800 g. What is his actual profit %?

Step-by-Step Solution:

1. Markup Ratio = $^{6}/_{5}$, Discount Ratio = $^{9}/_{10}$.
2. Weight Ratio = $^{1000}/_{800} = {}^{5}/_{4}$.
3. Net Ratio (SP / CP) = $(^{6}/_{5}) \times (^{9}/_{10}) \times (^{5}/_{4}) = {}^{54}/_{40} = 1.35$.
4. Net Profit % = (1.35 − 1) × 100 = **35%**.

6. Common Mistakes and Misconceptions

❌ **Misconception: "Buy 2 Get 1 Free" equals 50% Discount**

Correct: The customer receives 3 total items while paying for 2. Discount % = $^{1}/_{3}$ = **33.33%**.

7. CAT-Level Practice Questions

Q1 (Moderate): A dealer marks up goods by 50% and offers two successive discounts of 20% and 10%. What is his net profit or loss %?

(A) 8% profit
(B) 10% profit
(C) 5% loss
(D) 4% profit

Q2 (Advanced): A trader buys goods using a scale reading 900 g per kg. He sells them using a scale reading 1 kg per 800 g delivered, marking up by 10%. What is his net profit %?

(A) 45%
(B) 52.78%
(C) 40.5%
(D) 48.25%

8. Answers & Key Solutions

- **Q1 Solution: (A) 8% profit.** MF = (1.50) × (0.80) × (0.90) = 1.08 ⇒ 8% profit.
- **Q2 Solution: (B) 52.78%.** Net Ratio = $(^{10}/_{9}) \times (^{5}/_{4}) \times (^{11}/_{10}) = {}^{55}/_{36}$ ⇒ Profit = $^{19}/_{36} \times 100 \approx 52.78\%$.

9. Chapter Revision Sheet

PROFIT & LOSS QUICK REFERENCE

- **Link Formula:** MP / CP = (100 + P%) / (100 − d%)
- **Dishonest Merchant Ratio:** (SP / CP) = (Price Ratio) × (Weight Ratio)
- **Equal SP Loss:** Net Loss % = $(x / 10)^2$ %

MODULE 1: ARITHMETIC

Chapter 3: Simple Interest & Compound Interest

Compounding Cycles, Installments & Growth Rates

1. Introduction and Learning Objectives

Interest calculations represent the mathematical foundation of financial dynamics, time value of money, and exponential growth models. In the CAT exam, interest-related problems frequently test your ability to navigate varying compounding frequencies, changing interest rates across periods, loan amortization through equal periodic installments, and population growth or depreciation mechanics.

LEARNING OBJECTIVES

- Deconstruct the fundamental distinctions between linear growth (Simple Interest) and exponential growth (Compound Interest).
- Calculate Effective Annual Rates (EAR) across arbitrary compounding cycles (semi-annual, quarterly, monthly, continuous).
- Apply structural multiplying factors to solve multi-year varying rate problems rapidly without tedious algebraic expansion.
- Derive and solve loan installment problems for both SI and CI using geometric series principles.
- Apply population growth and asset depreciation formulas to CAT-style real-world scenarios.

2. Fundamental Concepts and Definitions

Every interest framework operates on four core variables:

- **Principal (P):** The initial sum of money invested, borrowed, or lent.
- **Rate of Interest (R):** The percentage rate per annum (p.a.) charged or earned.
- **Time Period (T or n):** The duration for which money is lent/invested (in years).
- **Amount (A):** The total sum accumulated at the end of time T ($\mathbf{A = P + I}$).

Core Distinctions:
• **Simple Interest (SI):** Interest is calculated *only on the original Principal (P)* throughout the entire loan tenure. The interest earned per unit time remains constant.
• **Compound Interest (CI):** Interest earned during each period is added to the principal to form the *new base principal* for the subsequent period ("interest on interest").

3. Detailed Theoretical Explanations

3.1 Simple Interest (SI) Dynamics

Since Simple Interest accumulates uniformly over time:

$$\mathbf{SI = (P \times R \times T) / 100} \quad | \quad \mathbf{A = P + SI = P \times [1 + (R \times T) / 100]}$$

Key Properties of Simple Interest:

1. **Constant Annual Yield:** The absolute interest accrued per year is constant: $\Delta I_{annual} = (P \times R) / 100$
2. **Linear Growth:** The Total Amount A(T) is a linear function of time T.
3. **Difference between Amounts across years:** $A_{T2} - A_{T1} = [P \times R \times (T2 - T1)] / 100$

3.2 Compound Interest (CI) Dynamics

Compound Interest follows an exponential multiplication model:

$$A = P \times (1 + R / 100)^n \quad | \quad CI = A - P = P \times [(1 + R / 100)^n - 1]$$

Compounding Frequency & Effective Annual Rate (EAR)

If interest is compounded *k* times per year at an annual nominal rate of R%:

- **Periodic Rate (R'):** $R' = (R / k)\%$
- **Total Periods (N):** $N = n \times k$
- **Accumulated Amount:** $A = P \times (1 + R / 100k)^{n \times k}$

Compounding Frequency (k)	Rate per Period (R'%)	Number of Periods (N)	Accumulated Amount (A)
Annual (k=1)	R%	n	$P(1 + R/100)^n$
Semi-Annual (k=2)	R / 2%	2n	$P(1 + R/200)^{2n}$
Quarterly (k=4)	R / 4%	4n	$P(1 + R/400)^{4n}$
Monthly (k=12)	R / 12%	12n	$P(1 + R/1200)^{12n}$

Effective Annual Rate (EAR): The actual equivalent annual yield generated by multi-period compounding:

$$EAR\% = [(1 + R / 100k)^k - 1] \times 100\%$$

3.3 SI vs. CI Comparison & Structural Differences

For a given principal P and interest rate R% per annum:

- **For Year 1 (n = 1):** $SI_1 = CI_1 = (P \times R) / 100$ $\Rightarrow$ Difference = 0
- **For Year 2 (n = 2):** $\text{Difference } (\Delta_2) = CI_2 - SI_2 = P \times (R / 100)^2$
- **For Year 3 (n = 3):** $\text{Difference } (\Delta_3) = CI_3 - SI_3 = P \times (R / 100)^2 \times [(300 + R) / 100]$

3.4 Installment Frameworks (SI & CI)

Simple Interest Installments

When a debt of amount D due after T years is discharged in T equal annual installments of ₹x each at R% p.a.:

The 1st installment pays interest for (T−1) years, the 2nd for (T−2) years, up to the last installment paying 0 interest.

$$\text{Total Debt } (D) = T \times x + (R \times x / 100) \times [T(T-1) / 2]$$

$$x = 100D / [100T + R \times T(T-1) / 2]$$

Compound Interest Installments (Loan Amortization)

When a borrowed amount P is repaid in n equal annual installments of ₹x each at R% p.a. compounded annually, the present value (PV) of all future installments equals the borrowed Principal P:

$$P = x / (1 + R/100)^1 + x / (1 + R/100)^2 + \ldots + x / (1 + R/100)^n$$

4. Important Formulas and Mathematical Properties

Scenario / Description	Formula
Sum doubles itself in T years under SI:	$R \times T = 100 \Rightarrow R = (100 / T)\%$
Sum becomes m times itself in T years under SI:	$R = [(m - 1) \times 100] / T\%$
Rule of 72 (Doubling time under CI):	$T_{double} \approx 72 / R$ years
Rule of 114 (Tripling time under CI):	$T_{triple} \approx 114 / R$ years
Population Growth ($P_0 \rightarrow P_n$ at rate +r%):	$P_n = P_0 \times (1 + r / 100)^n$
Asset Depreciation ($V_0 \rightarrow V_n$ at rate −d%):	$V_n = V_0 \times (1 - d / 100)^n$

5. Solved Examples with Step-by-Step Explanations

Example 1: SI Period Differences

Question: A sum of money put at Simple Interest amounts to ₹756 in 2 years and to ₹873 in 3.5 years. Find the principal and the rate of interest per annum.

Step-by-Step Solution:

1. Time difference = 3.5 − 2 = 1.5 years. Interest in 1.5 years = 873 − 756 = ₹117.
2. Annual interest (SI_{annual}) = 117 / 1.5 = ₹78/year.
3. SI for 2 years = 2 × 78 = ₹156.
4. Principal (P) = $A_2 - SI_{for\ 2\ years}$ = 756 − 156 = **₹600**.
5. Rate of Interest (R) = (78 / 600) × 100 = **13% p.a.**

Example 2: CI Multiplicative Scaling

Question: A sum of money lent at Compound Interest amounts to ₹4,800 after 4 years and to ₹7,200 after 8 years. Find the amount to which it will accumulate after 12 years.

Step-by-Step Solution:

1. $A_4 = P \times k^4 = 4800$, $A_8 = P \times k^8 = 7200$.
2. Growth Factor per 4-year block (g) = $A_8 / A_4 = 7200 / 4800 = {}^3/_2 = 1.5$.
3. Since time intervals are equal (0 → 4 → 8 → 12 years), amounts form a Geometric Progression (GP).
4. $A_{12} = A_8 \times g = 7200 \times 1.5 =$ **₹10,800**.

Example 3: Compound Interest Installment

Question: A man borrows ₹25,200 from a bank at 10% per annum compound interest, compounded annually. He agrees to clear the loan in two equal annual installments paid at the end of each year. Calculate the value of each installment.

Step-by-Step Solution:

1. P = 25,200, R = 10% ⇒ Discount Factor k = 1 + 10/100 = 1.1 = ${}^{11}/_{10}$.
2. $P = x / k^1 + x / k^2 \Rightarrow 25{,}200 = x / (1.1) + x / (1.1)^2 = (10x / 11) + (100x / 121)$.
3. 25,200 = x × (210 / 121).
4. x = (25,200 × 121) / 210 = 120 × 121 = **₹14,520**.

6. Mental Mathematics and Calculation Techniques

PASCAL'S TRIANGLE METHOD FOR CI CALCULATION

Instead of $(1 + R/100)^n$, use Pascal's binomial coefficients for integer years:

- **2 Years Coefficients:** (2, 1) ⇒ $\mathbf{CI_2 = 2A + 1B}$ (where A = R% of P, B = R% of A)
- **3 Years Coefficients:** (3, 3, 1) ⇒ $\mathbf{CI_3 = 3A + 3B + 1C}$ (where C = R% of B)

Example: Find CI on ₹10,000 at 5% p.a. for 3 years.

A = 5% of 10,000 = 500; B = 5% of 500 = 25; C = 5% of 25 = 1.25.

$CI_3 = 3(500) + 3(25) + 1(1.25) = 1500 + 75 + 1.25 =$ **₹1,576.25**.

7. Shortcuts, Tricks, and Time-Saving Approaches

SHORTCUT 1: EFFECTIVE RATE FORMULA FOR 2-YEAR CI

$$\text{Effective Rate } R_{\text{eff}}\% = 2R + (R^2 / 100)\%$$

Example: If R = 8%, effective 2-year CI rate = 2(8) + (64/100) = 16.64%.

8. Multiple Approaches to Solving the Same Question

Question: The difference between CI and SI on a certain sum of money for 2 years at 10% per annum is ₹65. Find the principal.

Approach 1 (Direct Formula): $\Delta_2 = P \times (R/100)^2 \Rightarrow 65 = P \times (10/100)^2 = P / 100 \Rightarrow P =$ **₹6,500**.

Approach 2 (Effective Percentage): SI = 20%, CI = 10 + 10 + (10×10/100) = 21%. Difference = 1%. 1% of P = 65 ⇒ P = **₹6,500**.

9. Common Mistakes and Misconceptions

❌ Misconception: Confusing Semi-Annual Rate with Annual Rate

Incorrect: Using R = 10% and n = 2 directly for semi-annual compounding over 1 year.

Correct: Halve the rate (R' = 5%) and double the compounding periods (N = 2).

10. CAT-Level Practice Questions

Q1 (Easy): A sum of money invested at Simple Interest becomes triple itself in 8 years. In how many years will it become 7 times itself at the same rate of interest?

(A) 16 years
(B) 24 years
(C) 20 years
(D) 28 years

Q2 (Easy): What is the difference between the compound interest on ₹5,000 for 1.5 years at 4% per annum calculated yearly and half-yearly?

(A) ₹2.04
(B) ₹3.06
(C) ₹1.50
(D) ₹4.80

Q3 (Moderate): A sum of ₹12,000 deposited at compound interest becomes double after 5 years. How much will it become after 20 years?

(A) ₹96,000
(B) ₹192,000
(C) ₹144,000
(D) ₹240,000

Q4 (Moderate): An annual debt of ₹1,720 due after 4 years is to be discharged in 4 equal annual installments at 5% per annum simple interest. Find the amount of each installment.

(A) ₹400 (B) ₹420
(C) ₹450 (D) ₹380

Q5 (Advanced): A person borrowed ₹100,000 at 10% per annum compound interest. He repaid ₹30,000 at the end of the first year and ₹40,000 at the end of the second year. How much amount must he pay at the end of the third year to clear his entire remaining debt?

(A) ₹48,400 (B) ₹52,800
(C) ₹52,000 (D) ₹46,200

Q6 (Advanced): The simple interest accrued on a sum of money for 3 years at 12% per annum is ₹4,320. What will be the compound interest on the same principal for 2 years at 15% per annum, compounded semi-annually?

(A) ₹3,985 (B) ₹4,146
(C) ₹4,026 (D) ₹4,250

11. Detailed Answer Keys and Complete Solutions

- **Q1 Solution: (B) 24 years.** Amount = 3P ⇒ SI = 2P in 8 years. Annual SI = 2P / 8 = P / 4. For amount to become 7P, required SI = 6P. Time required = 6P / (P / 4) = 6 × 4 = **24 years**.
- **Q2 Solution: (A) ₹2.04.** Compounded Yearly: Year 1 = ₹200; Half Year (2% on 5200) = ₹104. Total = ₹304. Compounded Half-Yearly: $A = 5000 \times (1.02)^3$ = ₹5,306.04 ⇒ CI = ₹306.04. Difference = 306.04 − 304 = **₹2.04**.
- **Q3 Solution: (B) ₹192,000.** Multiplicative growth factor = 2 every 5 years. Total period = 20 years = 4 blocks of 5 years. Accumulated Amount = $12,000 \times 2^4 = 12,000 \times 16$ = **₹192,000**.
- **Q4 Solution: (A) ₹400.** $D = T \cdot x + (R \cdot x / 100) \times [T(T-1) / 2] \Rightarrow 1720 = 4x + (5x / 100) \times 6 = 4.30x \Rightarrow x = 1720 / 4.30$ = **₹400**.
- **Q5 Solution: (B) ₹52,800.** Year 1 End = 100,000 × 1.10 = 110,000. Repay 30,000 ⇒ Balance = 80,000. Year 2 End = 80,000 × 1.10 = 88,000. Repay 40,000 ⇒ Balance = 48,000. Year 3 End = 48,000 × 1.10 = **₹52,800**.
- **Q6 Solution: (C) ₹4,026.** Principal P = 4,320 / (0.12 × 3) = ₹12,000. Semi-annual rate R' = 7.5% = $^3/_{40}$, N = 4. Amount $A = 12,000 \times (43 / 40)^4$ = ₹16,025.63. CI = 16,025.63 − 12,000 = ₹4,025.63 ≈ **₹4,026**.

12. Chapter Summary and Quick-Revision Sheet

SIMPLE & COMPOUND INTEREST QUICK REFERENCE

- **2-Year Difference:** $\Delta_2 = P(R/100)^2$
- **3-Year Difference:** $\Delta_3 = P(R/100)^2 \times [(300 + R) / 100]$
- **Doubling Time (Rule of 72):** $T \approx 72 / R$ years
- **Tripling Time (Rule of 114):** $T \approx 114 / R$ years
- **SI Installment:** Debt $= T \cdot x + (R \cdot x / 100) \times [T(T-1) / 2]$
- **CI Installment:** Loan $P = x / (1 + R/100)^1 + x / (1 + R/100)^2 + \ldots$

MODULE 1: ARITHMETIC

Chapter 4: Ratio, Proportion & Variation

Direct, Inverse & Joint Variations, Componendo-Dividendo & Partnerships

1. Introduction and Learning Objectives

Ratio, Proportion, and Variation form the analytical backbone of quantitative aptitude. They provide the fundamental language for comparing magnitudes, scaling quantities, establishing relative relationships, and modeling physical and economic dependencies. In CAT, ratio techniques serve as high-speed substitutes for tedious algebraic equations across Mixture-Alligations, Time-Speed-Distance, Time & Work, and Data Interpretation.

LEARNING OBJECTIVES

- Deconstruct and manipulate complex multi-variable ratio expressions and compound ratios.
- Apply fundamental properties of proportions (Componendo, Dividendo, Alternendo, Invertendo).
- Solve joint, direct, and inverse variation problems using proportionality constants and scale factors.
- Apply the constant-total and constant-difference principles to age and distribution problems.
- Solve income-expenditure-savings systems and partnership profit distributions effortlessly.

2. Fundamental Concepts and Definitions

2.1 Ratio

A **Ratio** is a mathematical comparison of two homogeneous quantities of the same kind, expressed in the same units: $\textbf{Ratio } A : B = A / B$.

- **Antecedent & Consequent:** In A : B, A is the antecedent and B is the consequent.
- **Scale Invariance:** Multiplying or dividing both terms by a non-zero constant k leaves the ratio unchanged: $A / B = (k \cdot A) / (k \cdot B)$.
- **Absolute Values:** A ratio A : B = 3 : 5 implies A = 3x and B = 5x for some real multiplier x ≠ 0.

2.2 Proportion

A **Proportion** is an equation stating that two ratios are equal: $A / B = C / D$ (or $A : B :: C : D$).

- **Extremes & Means:** A, D are *extremes*; B, C are *means*. Product of Extremes = Product of Means ⇒ $A \cdot D = B \cdot C$.
- **Mean Proportional:** If A : B :: B : C, then B is the Mean Proportional between A and C: $B^2 = A \cdot C \Rightarrow B = \sqrt{(A \cdot C)}$.

3. Detailed Theoretical Explanations

3.1 Algebraic Properties of Proportions

If $a / b = c / d$, the following theorems hold strictly:

- **Invertendo:** $b / a = d / c$
- **Alternendo:** $a / c = b / d$
- **Componendo:** $(a + b) / b = (c + d) / d$
- **Dividendo:** $(a - b) / b = (c - d) / d$
- **Componendo-Dividendo (C&D):** $(a + b) / (a - b) = (c + d) / (c - d)$
- **Theorem on Equal Ratios:** If $a / b = c / d = e / f = k$, then $k = (a + c + e) / (b + d + f) = (p \cdot a + q \cdot c + r \cdot e) / (p \cdot b + q \cdot d + r \cdot f)$.

3.2 Combine Ratios (Bridge Method)

When given individual ratios A : B = x : y and B : C = m : n, combine them into A : B : C by equalizing the common term B using $LCM(y, m)$.

Example: If A : B = 2 : 3 and B : C = 4 : 5, LCM(3, 4) = 12 ⇒ A : B = 8 : 12, B : C = 12 : 15 ⇒ A : B : C = 8 : 12 : 15.

3.3 Variation Theory

- **Direct Variation ($A \propto B$):** $A = k \cdot B \Rightarrow A_1 / A_2 = B_1 / B_2$
- **Inverse Variation ($A \propto 1 / B$):** $A = k / B \Rightarrow A_1 \cdot B_1 = A_2 \cdot B_2$
- **Joint Variation:** $A \propto B / C \Rightarrow A = k \cdot (B / C) \Rightarrow (A_1 \cdot C_1) / B_1 = (A_2 \cdot C_2) / B_2$

3.4 Partnership Accounting Framework

Profit distribution is based on the product of Investment Amount (I) and Investment Time Duration (T):

$$\text{Profit Ratio } (P_1 : P_2 : P_3) = (I_1 \cdot T_1) : (I_2 \cdot T_2) : (I_3 \cdot T_3)$$

4. Important Formulas and Mathematical Properties

Concept / Relationship	Formula / Expression
Duplicate / Sub-duplicate Ratio of A : B	Duplicate: $A^2 : B^2$ | Sub-duplicate: $\sqrt{A} : \sqrt{B}$
Triplicate / Sub-triplicate Ratio of A : B	Triplicate: $A^3 : B^3$ | Sub-triplicate: $\sqrt[3]{A} : \sqrt[3]{B}$
Fourth Proportional to A, B, C	$x = (B \cdot C) / A$ $(A : B :: C : x)$
Mean Proportional to A, B	$x = \sqrt{(A \cdot B)}$ $(A : x :: x : B)$
Third Proportional to A, B	$x = B^2 / A$ $(A : B :: B : x)$

5. Solved Examples with Step-by-Step Explanations

Example 1: Componendo & Dividendo Application

Question: If $[\sqrt{(x+5)} + \sqrt{(x-2)}] / [\sqrt{(x+5)} - \sqrt{(x-2)}] = 3$, find the value of x.

Step-by-Step Solution:

1. Apply Componendo & Dividendo to (a/b = 3/1): $[(a+b) / (a-b)] = (3+1) / (3-1)$.
2. $2\sqrt{(x+5)} / 2\sqrt{(x-2)} = 4 / 2 = 2 \Rightarrow \sqrt{(x+5)} / \sqrt{(x-2)} = 2$.
3. Square both sides: $(x+5) / (x-2) = 4 \Rightarrow x + 5 = 4x - 8$.
4. $3x = 13 \Rightarrow x = 13 / 3$.

Example 2: Income-Expenditure Ratio Dynamics

Question: The ratio of incomes of A and B is 5 : 4, and their expenditures ratio is 3 : 2. If each saves ₹1,600 monthly, find A's income.

Step-by-Step Solution:

1. Income Ratio = 5 : 4, Expenditure Ratio = 3 : 2.
2. Numerical gap in both ratios: (5 − 3) = 2 units and (4 − 2) = 2 units.
3. 2 ratio units = ₹1,600 ⇒ 1 ratio unit = ₹800.
4. A's Income = 5 units = 5 × 800 = **₹4,000**.

Example 3: Joint Variation Problem

Question: Project cost C is partly fixed (F) and partly varies as the square of workers employed (N). For N=10, C=₹2,500. For N=20, C=₹7,000. Find C for N=30.

Step-by-Step Solution:

1. Cost equation: $C = F + k \cdot N^2$.
2. Form equations: $F + 100k = 2500$ (Eq 1) and $F + 400k = 7000$ (Eq 2).
3. Subtract Eq 1 from Eq 2: $300k = 4500 \Rightarrow k = 15$. Substitute: $F = 1000$.
4. For N=30: $C = 1000 + 15(30^2) = 1000 + 13500 = ₹14,500$.

6. Mental Mathematics & Calculation Shortcuts

EQUALIZING GAPS METHOD FOR AGE RATIOS

When comparing age ratios across time, the absolute difference between ages stays constant.

Example: 5 yrs ago ratio was 2:3 (gap=1). 5 yrs hence ratio is 3:4 (gap=1). Time elapsed = 10 yrs. Change in units = 3−2 = 1 unit = 10 yrs. Present ages: A = 20+5 = 25, B = 30+5 = 35.

7. Common Mistakes and Misconceptions

❌ Misconception: Misinterpreting Inverse Proportionality Ratios

Incorrect: If A : B : C = 2 : 3 : 4, their inverse ratio is 4 : 3 : 2.

Correct: Inverse ratio is $^1/_2 : ^1/_3 : ^1/_4$. Multiply by LCM(2,3,4)=12 ⇒ **6 : 4 : 3**.

8. CAT-Level Practice Questions

Q1 (Easy): If A : B = 3 : 4, B : C = 8 : 9, and C : D = 15 : 16, find the ratio A : D.

(A) 5 : 8
(B) 5 : 6
(C) 3 : 5
(D) 1 : 2

Q2 (Easy): Find the third proportional to 0.36 and 0.9.

(A) 2.25
(B) 1.8
(C) 2.4
(D) 1.35

Q3 (Moderate): Containers have volumes in ratio 2 : 3 : 4 with milk to water ratios 1:5, 3:5, and 5:7. If mixed together, find final milk to water ratio.

(A) 25 : 47
(B) 157 : 275
(C) 53 : 91
(D) 4 : 7

Q4 (Moderate): A, B, C enter partnership. A invests 3× B, B invests $^2/_3$ of C. Total profit = ₹6,600. Find B's share.

(A) ₹1,200
(B) ₹1,800
(C) ₹3,600
(D) ₹1,500

Q5 (Advanced): Diamond price P varies as square of weight W ($P = k\,W^2$). A diamond breaks into 3 pieces with weights 1 : 2 : 3. Total value drops by ₹44,000. Find original price.

(A) ₹72,000
(B) ₹96,000
(C) ₹108,000
(D) ₹144,000

Q6 (Advanced): Revenue $R = k \cdot P \cdot \sqrt{S}$. For P=200 and S=400, R=₹80,000. If ticket price increases by 25% and occupied seats drop by 36%, find new revenue.

(A) ₹80,000
(B) ₹75,000
(C) ₹84,000
(D) ₹72,000

9. Detailed Answer Keys and Complete Solutions

- **Q1 Solution: (A) 5 : 8.** A / D = (A/B) × (B/C) × (C/D) = (3/4) × (8/9) × (15/16) = (3×8×15) / (4×9×16) = 360 / 576 = **5 : 8**.
- **Q2 Solution: (A) 2.25.** 0.36 / 0.9 = 0.9 / x ⇒ x = (0.9 × 0.9) / 0.36 = 0.81 / 0.36 = **2.25**.
- **Q3 Solution: (A) 25 : 47.** Container vols 2, 3, 4. $Milk_1 = 2\times(1/6) = {}^{1}/_{3}$; $Milk_2 = 3\times(3/8) = {}^{9}/_{8}$; $Milk_3 = 4\times(5/12) = {}^{5}/_{3}$. Total Milk = ${}^{25}/_{8}$. Total Water = $9 - {}^{25}/_{8} = {}^{47}/_{8}$. Ratio = **25 : 47**.
- **Q4 Solution: (A) ₹1,200.** C = x, B = ${}^{2}/_{3}$x, A = 2x. Investment Ratio A : B : C = 6 : 2 : 3. Sum = 11 units = ₹6,600 ⇒ 1 unit = ₹600. B's share = 2 × 600 = **₹1,200**.
- **Q5 Solution: (A) ₹72,000.** Total W = 6x ⇒ Original P = $k(6x)^2 = 36kx^2$. Broken pieces P = $k(1x)^2 + k(2x)^2 + k(3x)^2 = 14kx^2$. Drop = $22kx^2 = 44{,}000 \Rightarrow kx^2 = 2{,}000$. Original P = 36 × 2,000 = **₹72,000**.
- **Q6 Solution: (A) ₹80,000.** Initial state: 80,000 = k × 200 × 20 ⇒ k = 20. New P = 250, New S = 256 (√256 = 16). New Revenue R = 20 × 250 × 16 = **₹80,000**.

10. Chapter Summary and Quick-Revision Sheet

RATIO, PROPORTION & VARIATION QUICK REFERENCE

- **Componendo-Dividendo:** (a + b) / (a − b) = (c + d) / (c − d)
- **Mean Proportional:** b = √(a · c)
- **Direct / Inverse Variation:** A = k·B | A = k / B
- **Partnership Profit:** Profit Ratio = $(I_1 \cdot T_1) : (I_2 \cdot T_2)$
- **Bridge Method:** Equalize common term using LCM to merge individual ratios.

MODULE 1: ARITHMETIC

Chapter 5: Averages, Weighted Averages & Mixture-Alligation

Deviation Methods, Alligation Cross-Rules & Successive Replacement Dynamics

1. Introduction and Learning Objectives

Averages, Weighted Averages, and Mixture-Alligation form a tightly coupled analytical ecosystem in CAT Arithmetic. Averages represent the central tendency of a dataset, while Weighted Averages extend this concept when subsets carry different numerical weights. The Alligation rule is an elegant cross-multiplication framework derived directly from weighted averages that allows CAT aspirants to solve complex multi-component mixture, replacement, and blend problems in seconds.

LEARNING OBJECTIVES

- Understand the core principles of Arithmetic Mean, Deviations from Assumed Mean, and Sequential Averages.
- Master Weighted Averages and derive the Alligation Cross-Rule systematically.
- Solve complex replacement and dilution problems using the successive replacement formula.
- Apply Alligation to diverse quantitative domains, including Profit & Loss blends, Interest rate mixtures, and Speed-Time-Distance scenarios.
- Master Maxima-Minima concepts within constrained group averages.

2. Fundamental Concepts and Definitions

2.1 Simple Average (Arithmetic Mean)

The Simple Average of n observations is defined as the sum of all observations divided by total count: $\bar{X} = (\sum x) / n$.

- **Deviation Principle:** Sum of algebraic deviations from arithmetic mean is always zero: $\sum (x_i - \bar{X}) = 0$.
- **Linear Shift:** If every observation is changed by constant k, the average changes by k.
- **Inclusion/Exclusion:** Adding an element equal to average leaves it unchanged; larger element increases it; smaller element decreases it.

2.2 Weighted Average

When subsets carry weights w_1, w_2, ... w_k, the Weighted Average is: $\bar{X}_w = (w_1x_1 + w_2x_2 + ...) / (w_1 + w_2 + ...)$.

2.3 The Mixture-Alligation Framework

Alligation cross-rule for two elements of values A_1 and A_2 producing mean A_m:

$$n_1 / n_2 = (A_2 - A_m) / (A_m - A_1)$$

> **Golden Rule of Alligation:** All three values (A_1, A_2, A_m) must be expressed in the exact same units and calculated with respect to the exact same base reference (e.g., all Cost Prices per kg, all concentration percentages, or all speeds).

2.4 Successive Replacement & Dilution Formula

When x units are drawn from V units of pure liquid A and replaced with liquid B, repeated n times:

$$\textbf{Remaining Qty of A} = \mathbf{V \times (1 - x / V)^n}$$

3. Detailed Theoretical Explanations

3.1 Deviations & Assumed Mean Technique

Choose an Assumed Mean (A) close to expected average: $\bar{\mathbf{X}} = \mathbf{A} + (\sum \mathbf{d_i}) / \mathbf{n}$.

Example: Find average of 84, 88, 92, 95, 81. Let A=90. Deviations: −6, −2, +2, +5, −9 (Sum = −10). Average = 90 + (−10/5) = 88.

3.2 Consecutive Numbers & APs

For numbers in Arithmetic Progression (AP): $\textbf{Average} = \textbf{(First Term + Last Term)} / \mathbf{2}$.

- First n consecutive integers starting at 1: Average = $\mathbf{(n + 1) / 2}$.
- First n consecutive even integers: Average = $\mathbf{n + 1}$.
- First n consecutive odd integers: Average = $\mathbf{n}$.

4. Important Formulas and Mathematical Properties

Scenario / Description	Formula
Simple / Weighted Average	$\bar{X} = (\sum x) / n$ | $\bar{X}_w = (n_1A_1 + n_2A_2) / (n_1 + n_2)$
Alligation Ratio Formula	$n_1 / n_2 = (A_2 - A_m) / (A_m - A_1)$
Successive Dilution (Equal Steps)	$Q_{final} = Q_{initial} \times (1 - x / V)^n$
Average Speed (Equal Distances)	$S_{avg} = (2\ S_1\ S_2) / (S_1 + S_2)$
Average Speed (Equal Time Intervals)	$S_{avg} = (S_1 + S_2) / 2$

5. Solved Examples with Step-by-Step Explanations

Example 1: Inclusion / Exclusion in Averages

Question: Average weight of 24 students is 45 kg. Including the teacher increases average weight by 1 kg. Find teacher's weight.

Step-by-Step Solution:

1. Initial sum = 24 × 45 = 1080 kg.
2. Final sum with teacher (25 people, avg 46 kg) = 25 × 46 = 1150 kg.
3. Teacher's weight = 1150 − 1080 = **70 kg**.
4. *Deviation shortcut:* Old Avg + (New Total Count × Increase) = 45 + (25 × 1) = **70 kg**.

Example 2: Mixture Alligation with Profit & Loss

Question: A merchant sells part of 100 kg sugar at 8% profit and rest at 18% profit, gaining 14% overall. Find sugar sold at 18% profit.

Step-by-Step Solution:

1. Apply Alligation: Ratio (Qty at 8% : Qty at 18%) = (18 − 14) / (14 − 8) = 4 / 6 = 2 / 3.
2. Total parts = 2 + 3 = 5 parts = 100 kg ⇒ 1 part = 20 kg.
3. Sugar at 18% profit = 3 parts = 3 × 20 = **60 kg**.

Example 3: Repeated Dilution / Replacement

Question: A container has 80 L of pure milk. 8 L are drawn and replaced with water, repeated twice more. Find remaining milk.

Step-by-Step Solution:

1. V = 80, x = 8, n = 3.
2. Remaining Milk = $80 \times (1 - 8/80)^3 = 80 \times (9/10)^3 = 80 \times (729 / 1000)$.
3. Remaining Milk = 5832 / 100 = **58.32 liters**.

6. Mental Mathematics & Calculation Shortcuts

WRONG ENTRY CORRECTION RULE

$$\text{Correct Average} = \text{Wrong Average} + (\text{Correct Value} - \text{Wrong Value}) / n$$

7. Common Mistakes and Misconceptions

❌ Misconception: Misidentifying the Base Ratio in Alligation

Incorrect: Assuming Alligation of speeds gives the ratio of speeds or distances.

Correct: Speed = Distance / Time. Because Time is in the denominator, Alligation of speeds yields the ratio of **Time taken**.

8. CAT-Level Practice Questions

Q1 (Easy): Average of 5 consecutive numbers is 27. Find the product of the smallest and largest numbers.

(A) 725 (B) 720
(C) 700 (D) 729

Q2 (Easy): In what ratio must water be mixed with milk costing ₹40/L so that resulting mixture is worth ₹32/L?

(A) 1 : 4 (B) 1 : 5
(C) 2 : 3 (D) 1 : 3

Q3 (Moderate): A vessel has refined oil mixture with 80% oil. Half is drawn and filled with kerosene, repeated once more. Find final oil %.

(A) 20% (B) 30%
(C) 40% (D) 25%

Q4 (Moderate): Section A has 40 students with avg 65 marks; Section B has 60 students with avg 75 marks. Find combined average mark.

(A) 70 (B) 71
(C) 69 (D) 72

Q5 (Advanced): 100 L container has pure alcohol. 20 L drawn and replaced with water, then 25 L withdrawn and replaced, then 50 L withdrawn and replaced. Find final alcohol %.

(A) 30% (B) 24%
(C) 36% (D) 40%

Q6 (Advanced): 120 candidates scored overall avg 35. Passed avg = 39, Failed avg = 15. How many candidates passed?

(A) 100 (B) 90
(C) 105 (D) 96

9. Detailed Answer Keys and Complete Solutions

- **Q1 Solution: (A) 725.** Middle number = 27. Numbers are 25, 26, 27, 28, 29. Product = 25 × 29 = **725**.
- **Q2 Solution: (A) 1 : 4.** Water cost = 0, Milk cost = 40, Mean = 32. Ratio Water : Milk = (40 − 32) / (32 − 0) = 8 / 32 = **1 : 4**.
- **Q3 Solution: (A) 20%.** Initial oil = 80%. Fraction remaining each step = 1/2. Final oil = 80% × $(1/2)^2$ = 80% × (1/4) = **20%**.
- **Q4 Solution: (B) 71.** Student ratio A : B = 40 : 60 = 2 : 3. Combined Avg = (2×65 + 3×75) / 5 = (130 + 225) / 5 = 355 / 5 = **71**.
- **Q5 Solution: (A) 30%.** Final Conc = 1.0 × (1 − 20/100) × (1 − 25/100) × (1 − 50/100) = (4/5) × (3/4) × (1/2) = $^3/_{10}$ = **30%**.

- **Q6 Solution: (A) 100.** Passed : Failed = (35 − 15) / (39 − 35) = 20 / 4 = 5 : 1. Total parts = 6 = 120 ⇒ 1 part = 20. Passed candidates = 5 × 20 = **100**.

10. Chapter Summary and Quick-Revision Sheet

AVERAGES & ALLIGATION QUICK REFERENCE

- **Alligation Ratio:** $n_1 / n_2 = (A_2 - A_m) / (A_m - A_1)$
- **Successive Replacement:** $Q_{final} = Q_{initial} \times (1 - x / V)^n$
- **AP Average:** (First Term + Last Term) / 2
- **Corrected Avg:** Wrong Avg + (Correct Value − Wrong Value) / n

MODULE 1: ARITHMETIC

Chapter 6: Time & Work, Pipes & Cisterns

Efficiency Dynamics, Alternating Work Schedules & Group Work Chains

1. Introduction and Learning Objectives

Time & Work and Pipes & Cisterns form one of the most frequently tested sub-domains of CAT Arithmetic. The core principle relies on inverse proportionality: assuming a fixed quantity of total work, the time required to complete the task is inversely proportional to the worker's efficiency. Mastering these concepts allows you to handle complex group dynamics, alternating work schedules, negative work (drain pipes), and efficiency scaling seamlessly.

LEARNING OBJECTIVES

- Master the LCM Method (Unit Work Framework) to eliminate fraction addition during time calculations.
- Establish exact mathematical relations between Efficiency, Work, and Time ($\mathbf{W = E \times T}$).
- Handle alternating work schedules and determine exact cycle completion times.
- Model group work dynamics ($\mathbf{M_1D_1H_1E_1 = M_2D_2H_2E_2}$).
- Solve Pipes & Cisterns scenarios involving filling pipes, emptying pipes, and variable leakages.

2. Fundamental Concepts and Definitions

2.1 The Core Equation

$$\textbf{Total Work (W)} = \textbf{Efficiency (E)} \times \textbf{Time (T)}$$

- **Work (W):** Total output required (measured in units, liters, or job fractions).
- **Efficiency (E):** Rate of doing work per unit time (units/day, liters/hour).
- **Time (T):** Duration taken to complete assigned work.

2.2 The Unit Work (LCM) Framework

Set **Total Work = LCM of individual time periods** to work with whole numbers.

Example: A takes 12 days, B takes 15 days. Total Work = LCM(12, 15) = 60 units. E_A = 5 units/day, E_B = 4 units/day. Combined Time = 60 / 9 = 6 $^2/_3$ days.

3. Detailed Theoretical Explanations

3.1 Alternating Work Schedules

1. Calculate Work done per cycle (e.g., $E_A + E_B$ for a 2-day cycle).
2. Compute full cycles: $\textbf{Full Cycles} = \lfloor \textbf{Total Work / Work per Cycle} \rfloor$.
3. Allocate remaining work turn-by-turn according to sequence.

3.2 Pipes & Cisterns (Inlets & Outlets)

Inlets do positive work (+E), while outlets do negative work (−E).

$$\text{Net Efficiency } (E_{net}) = E_{inlet1} + E_{inlet2} - E_{outlet}$$

3.3 The Man-Day-Hour Chain Rule

$$(M_1 \cdot D_1 \cdot H_1 \cdot E_1) / W_1 = (M_2 \cdot D_2 \cdot H_2 \cdot E_2) / W_2$$

4. Important Formulas and Mathematical Properties

Scenario / Description	Formula
Two Workers (A in a days, B in b days)	$T_{A+B} = (a \cdot b) / (a + b)$ days
Three Workers (A, B, C in a, b, c days)	$T_{A+B+C} = (a \cdot b \cdot c) / (ab + bc + ca)$ days
Efficiency Ratio to Time Ratio	$E_A : E_B = x : y \Rightarrow T_A : T_B = y : x$
Wages Share Principle	Wages Ratio = $(E_1 \cdot T_1) : (E_2 \cdot T_2)$

5. Solved Examples with Step-by-Step Explanations

Example 1: Alternating Work Schedule

Question: A finishes a task in 10 days, B in 15 days. They work on alternate days starting with A. Find total days.

Step-by-Step Solution:

1. Total Work = LCM(10, 15) = 30 units. $E_A = 3$, $E_B = 2$.
2. 1 cycle (2 days) = 3 + 2 = 5 units.
3. Cycles needed = 30 / 5 = 6 full cycles. Total Time = 6 × 2 = **12 days**.

Example 2: Pipes with Leakage

Question: Pipe A fills a tank in 8 hours. With a leak, it takes 10 hours. How long will the leak take to empty a full tank?

Step-by-Step Solution:

1. Capacity = LCM(8, 10) = 40 L. E_A = +5 L/hr, Net E = +4 L/hr.
2. Leak Efficiency (E_L) = 5 − 4 = 1 L/hr (draining).
3. Time for leak to empty = 40 / 1 = **40 hours**.

Example 3: Group Work (Men & Women)

Question: 3 men or 6 women complete a job in 16 days. In how many days will 12 men and 8 women complete it?

Step-by-Step Solution:

1. $3M \times 16 = 6W \times 16 \Rightarrow 1M = 2W \Rightarrow E_M : E_W = 2 : 1$.
2. Total Work = $3(2) \times 16 = 96$ units.
3. E_{target} (12M + 8W) = 12(2) + 8(1) = 32 units/day. Time = 96 / 32 = **3 days**.

6. Mental Mathematics & Calculation Shortcuts

EFFICIENCY SCALING RULE

If A is 50% more efficient than B $\Rightarrow E_A : E_B = 3 : 2 \Rightarrow T_A : T_B = 2 : 3$.

7. Common Mistakes and Misconceptions

❌ Misconception: Averaging Time Periods Directly

Incorrect: If A takes 10 days and B takes 20 days, together they take (10+20)/2 = 15 days.

Correct: Work rates add! Combined rate = $^1/_{10} + {}^1/_{20} = {}^3/_{20} \Rightarrow$ Time = $^{20}/_3$ = **6.67 days**.

8. CAT-Level Practice Questions

Q1 (Easy): A completes work in 12 days, B in 24 days. Working together, in how many days will they finish?

(A) 8 days
(B) 6 days
(C) 9 days
(D) 7.5 days

Q2 (Easy): Pipes A and B fill a cistern in 15 mins and 20 mins. How long to fill together?

(A) $8\ {}^4/_7$ mins
(B) 9 mins
(C) $8\ {}^1/_2$ mins
(D) $7\ {}^1/_2$ mins

Q3 (Moderate): A and B finish work in 12 and 18 days. A works alone for 3 days, then B joins. Total days taken?

(A) 8.4 days
(B) 6.6 days
(C) 9.6 days
(D) 7.5 days

Q4 (Moderate): A is twice as efficient as B. Together they finish in 14 days. How many days for A alone?

(A) 21 days
(B) 28 days
(C) 42 days
(D) 14 days

Q5 (Advanced): Pipes A, B fill in 20, 30 mins; C empties in 15 mins. Opened at t=0, 1, 2 sequentially. At what minute is tank full?

(A) 167 mins
(B) 173 mins
(C) 165 mins
(D) 180 mins

Q6 (Advanced): 10M + 15W complete work in 6 days; 12M + 8W in 7 days. How many days for 4M + 6W?

(A) 15 days
(B) 18 days
(C) 12 days
(D) 20 days

9. Detailed Answer Keys and Complete Solutions

- **Q1 Solution: (A) 8 days.** Work = 24. $E_A = 2$, $E_B = 1$. Combined E = 3. Time = 24 / 3 = **8 days**.
- **Q2 Solution: (A) 8 $^4/_7$ mins.** Capacity = 60. $E_A = 4$, $E_B = 3$. Time = 60 / 7 = **8 $^4/_7$ mins**.
- **Q3 Solution: (A) 8.4 days.** Work = 36. $E_A = 3$, $E_B = 2$. A does 9 units in 3 days. Remaining = 27 units. Time = 3 + (27/5) = **8.4 days**.
- **Q4 Solution: (A) 21 days.** $E_A : E_B = 2 : 1$. Combined E = 3. Total Work = 3 × 14 = 42. A's time = 42 / 2 = **21 days**.
- **Q5 Solution: (B) 173 mins.** Capacity = 60 L. $E_A = +3$, $E_B = +2$, $E_C = -4$. Cycle = 3 mins (+1 L). Sequential timing pushes fill point to **173 minutes**.
- **Q6 Solution: (A) 15 days.** (10M + 15W) × 6 = 60M + 90W = 84M + 56W ⇒ 12M = 17W. 4M + 6W group is $^1/_{2.5}$ of first group ⇒ Time = 6 × 2.5 = **15 days**.

10. Chapter Summary and Quick-Revision Sheet

TIME & WORK QUICK REFERENCE

- **Core Equation:** Work = Efficiency × Time
- **Two-Worker Shortcut:** $T = (a \cdot b) / (a + b)$
- **Pipes Net Efficiency:** $E_{net} = \sum(\text{Inlets}) - \sum(\text{Outlets})$
- **Chain Rule:** $(M_1D_1H_1E_1) / W_1 = (M_2D_2H_2E_2) / W_2$

MODULE 1: ARITHMETIC

Chapter 7: Time, Speed & Distance

1. Fundamental Principles & Proportionality

The core relationship of kinematics is governed by $\text{Distance} = \text{Speed} \times \text{Time}$ ($D = S \times T$). Master application of this topic relies entirely on recognizing proportional relationships when one parameter is held constant.

CORE PROPORTIONALITY LAWS

- **Constant Distance (D is fixed):** $S \propto 1 / T$. Speed and Time are inversely proportional. If speed changes in ratio $a : b$, time taken changes in ratio $b : a$.
- **Constant Time (T is fixed):** $D \propto S$. Distance and Speed are directly proportional. Ratio of distances covered equals ratio of speeds ($D_1 / D_2 = S_1 / S_2$).
- **Constant Speed (S is fixed):** $D \propto T$. Distance covered is directly proportional to time taken ($D_1 / D_2 = T_1 / T_2$).

UNIT CONVERSION & AVERAGE SPEED FORMULAS

- **Conversion:** 1 km/h $= 5 / 18$ m/s and 1 m/s $= 18 / 5$ km/h.
- **Average Speed (General):** $S_{avg} = \text{Total Distance} / \text{Total Time}$.
- **Equal Distances (Two Speeds a and b):** Harmonic mean of speeds: $S_{avg} = 2ab / (a + b)$.
- **Equal Distances (Three Speeds a, b, c):** $S_{avg} = 3abc / (ab + bc + ca)$.
- **Equal Time Intervals:** Arithmetic mean of speeds: $S_{avg} = (S_1 + S_2 + \ldots + S_n) / n$.

2. Relative Speed & Meeting Points

Relative speed is the rate at which distance between two moving bodies decreases or increases.

- **Opposite Directions (Towards or Away):** $S_{rel} = S_1 + S_2$.
- **Same Direction (Overtaking):** $S_{rel} = |S_1 - S_2|$.
- **Time to Meet:** $T_{meet} = \text{Initial Distance} / S_{rel}$.

ADVANCED MEETING POINT THEOREMS

- **Time Taken After Crossing:** If two bodies start from A and B simultaneously toward each other, and after meeting take time t_1 and t_2 to reach B and A respectively:
  $S_1 / S_2 = \sqrt{(t_2 / t_1)}$
- **Distance from A at Meeting Point:** $D_{meet} = D \times [\, S_1 / (S_1 + S_2) \,]$.

3. Boats & Streams

Motion in a fluid medium depends on water current velocity.

- Let speed of boat in still water = u km/h, speed of stream/current = v km/h.
- **Downstream Speed (S_d):** Moving along the current → $S_d = u + v$.
- **Upstream Speed (S_u):** Moving against the current → $S_u = u - v$ (where $u > v$).
- **Boat Speed in Still Water:** $u = (S_d + S_u) / 2$.
- **Stream Velocity:** $v = (S_d - S_u) / 2$.

4. Escalator Kinematics

Escalator problems follow the exact structure of Boats & Streams, where steps represent distance.

- Let N = total visible steps on a stationary escalator.
- Let person speed = e_p steps/sec, escalator speed = e_e steps/sec.
- **Walking In Same Direction as Escalator (Moving Up):** Total steps seen $N = t \times (e_p + e_e)$. Steps taken by person = $t \times e_p$.
- **Walking Opposite to Escalator Direction:** Total steps seen $N = t \times (e_p - e_e)$. Steps taken by person = $t \times e_p$.

5. Linear & Circular Races

LINEAR RACES CONCEPT

- **"A beats B by x meters":** When A reaches finish line (distance L), B has covered $L - x$ meters.
- **"A beats B by t seconds":** B takes t seconds more than A to complete length L.
- **Start Head-start:** "A gives B a head-start of x meters" means A starts at start line, B starts x meters ahead.

CIRCULAR RACES FUNDAMENTALS

Two runners with speeds a and b ($a > b$) on a track of length L starting simultaneously from the same point:

Scenario	Same Direction	Opposite Direction
First Meeting Anywhere	$T = L / (a - b)$	$T = L / (a + b)$
First Meeting at Starting Point	$\text{LCM}(L/a, L/b)$	$\text{LCM}(L/a, L/b)$
Number of Distinct Meeting Points	$\lvert a - b \rvert$ (in simplified ratio a:b)	$a + b$ (in simplified ratio a:b)

6. High-Yield Shortcuts & Mental Math Strategies

SHORTCUT 1: SHIFT IN SPEED FORMULA

If a person travels at speed S_1, he is late by t_1. If he travels at speed S_2, he is early by t_2.
Exact Distance: $D = (S_1 \times S_2 / |S_1 - S_2|) \times \text{Total Time Difference } (t_1 + t_2)$.

SHORTCUT 2: STOPPAGE TIME CALCULATION

If speed excluding stoppages is S_{fast} and speed including stoppages is S_{slow}:
Stoppage time per hour: $T_{stop} = [\ (S_{fast} - S_{slow}) / S_{fast}\ ] \times 60$ minutes.

7. Common Pitfalls & Mistakes

- **Wrong Average Speed calculation:** Adding speeds and dividing by 2 $((a+b)/2)$ when distances are equal instead of taking the harmonic mean $(2ab/(a+b))$.
- **Unit Mismatch:** Adding speed in km/h to time in minutes without multiplying by 60 or converting units first. Always unify units to (m, m/s, seconds) or (km, km/h, hours).
- **Escalator Step Confusion:** Forgetting that total steps N on an escalator equals the sum or difference of steps taken by the person and steps moved by the escalator during the exact same time window.
- **Ratio Simplification in Circular Races:** For finding the number of distinct meeting points, the speeds MUST be reduced to their simplest coprime ratio $a : b$ before taking $|a - b|$ or $a + b$.

8. CAT-Level Solved Practice Questions

Question 1 (Relative Speed & Late/Early Shift)

A man covers a certain distance between his house and office. If he drives at 30 km/h, he reaches 10 minutes late. If he drives at 40 km/h, he reaches 5 minutes early. Find the distance between his house and office, and the correct speed to reach on time.

Solution:

Method 1: Proportionality Approach
Speed Ratio = $30 : 40 = 3 : 4$.
Since Distance is constant, Time Ratio = $4 : 3$.
Difference in time units = $4 - 3 = 1$ unit.
Actual time difference = 10 mins late - (-5 mins early) = 15 minutes = 1/4 hour.
So 1 unit = 15 mins. Time taken at 30 km/h = $4 \times 15 = 60$ mins = 1 hour.
Distance = 30 km/h $\times$ 1 hr = 30 km.
Scheduled On-Time Arrival = 60 mins - 10 mins late = 50 mins = 5/6 hours.
Required Speed = 30 km / $(5/6$ hrs$) = 36$ km/h.

Question 2 (Circular Races - Advanced)

Two runners A and B run on a circular track of length 1200 m with speeds 18 km/h and 27 km/h respectively. If they start from the same point at the same time in opposite directions, after how much time will they meet for the first time, and how many distinct meeting points are there on the track?

Solution:

Convert speeds to m/s:
$S_A = 18 \times (5/18) = 5$ m/s
$S_B = 27 \times (5/18) = 7.5$ m/s
1. First Meeting Time: Since they move in opposite directions, relative speed $S_{rel} = 5 + 7.5 = 12.5$ m/s.
T_{first} = Track Length / S_{rel} = 1200 / 12.5 = 96 seconds.
2. Distinct Meeting Points:
Ratio of speeds $S_A : S_B = 5 : 7.5 = 2 : 3$ (coprime integers $a = 2, b = 3$).
Opposite direction → Number of distinct meeting points = $a + b = 2 + 3 = 5$ points.

Question 3 (Escalator Problem)

An escalator is moving downwards at a constant speed. A boy walks down the escalator and takes 40 steps to reach the bottom. A girl walks up the downward-moving escalator and takes 120 steps to reach the top. If the boy's speed is double the girl's speed, find the total number of visible steps on the stationary escalator.

Solution:

Let speed of escalator = e steps/sec.
Let speed of girl = g steps/sec → speed of boy = 2g steps/sec.
Boy moves down (same direction as escalator):
Time taken by boy $t_b = 40 / 2g = 20/g$ seconds.
Total escalator steps $N = (\text{Boy steps}) + (\text{Escalator steps}) = 40 + e \times (20/g) = 40 + 20(e/g)$ --- (Eq 1)
Girl moves up (opposite direction to downward escalator):
Time taken by girl $t_g = 120 / g$ seconds.
Total escalator steps $N = (\text{Girl steps}) - (\text{Escalator steps}) = 120 - e \times (120/g) = 120 - 120(e/g)$ --- (Eq 2)
Equating Eq 1 and Eq 2:
$40 + 20(e/g) = 120 - 120(e/g) \Rightarrow 140(e/g) = 80 \Rightarrow e/g = 4/7$.
Substitute e/g in Eq 1:
$N = 40 + 20 \times (4/7) = 40 + 80/7 = 360/7 \approx 51.4$ steps. (If integral step assumption: N = 51 or exact fraction 360/7 steps).

9. Quick-Revision Cheat Sheet

Topic	Key Formula / Rule	Quick Tip
Proportionality	If D is constant, $S_1/S_2 = T_2/T_1$	Ratio inversion instantly solves late/early problems.
Average Speed	$S_{avg} = 2ab / (a + b)$ for equal distance	Never use simple arithmetic mean for equal distances!
After Meeting Time	$S_1 / S_2 = \sqrt{(t_2 / t_1)}$	t_1, t_2 are times taken after crossing each other.
Boats & Streams	$u = (S_d + S_u)/2$, $v = (S_d - S_u)/2$	$S_d = u + v$, $S_u = u - v$
Stoppages	$T_{stop} = [\ (S_{fast} - S_{slow}) / S_{fast}\ ] \times 60$ min/hr	Direct application saves 2 minutes per question.
Circular Races	Distinct points: Same dir = $\|a - b\|$, Opp = $a + b$	Always reduce speed ratio $a:b$ to lowest coprime form first.

MODULE 2: ALGEBRA

Chapter 8: Linear & Higher-Degree Equations

1. Systems of Linear Equations & Consistency

A system of two linear equations in two variables, $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, represents two lines in a plane. The nature of their intersection determines the number of solutions.

CONDITIONS FOR SOLUTIONS

- **Unique Solution (Consistent & Independent):** Intersecting lines. Ratio condition: $a_1 / a_2 \neq b_1 / b_2$.
- **Infinitely Many Solutions (Consistent & Dependent):** Coincident lines. Ratio condition: $a_1 / a_2 = b_1 / b_2 = c_1 / c_2$.
- **No Solution (Inconsistent):** Parallel lines. Ratio condition: $a_1 / a_2 = b_1 / b_2 \neq c_1 / c_2$.

2. Linear Diophantine Equations & Integral Solutions

An equation of the form $ax + by = c$ where a, b, c are integers and we seek integer solutions (x, y) is a Linear Diophantine Equation.

SOLVABILITY & GENERAL SOLUTION STRUCTURE

- **Existence Theorem:** An integral solution to $ax + by = c$ exists if and only if $\gcd(a, b)$ divides c.
- **General Solution Family:** If (x_0, y_0) is a particular initial integer solution to $ax + by = c$, then all integer solutions are given by:
  $x = x_0 + (b / g) \times k$ and $y = y_0 - (a / g) \times k$, where $g = \gcd(a, b)$ and $k \in \mathbb{Z}$.
- **Number of Non-Negative Integer Solutions:** For positive integers a, b, c in $ax + by = c$: test the minimum non-negative value for x, find corresponding y, then step x by $+b$ while stepping y by $-a$ until $y < 0$.

3. Higher-Degree Equations & Polynomial Algebra

A polynomial equation of degree n is given by $P(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_0 = 0$.

FUNDAMENTAL POLYNOMIAL THEOREMS

- **Remainder Theorem:** When a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$.
- **Factor Theorem:** $(x - a)$ is a factor of $P(x)$ if and only if $P(a) = 0$.
- **Vieta's Formulas for Cubic Equations:** For $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ:
  • Sum of roots: $\alpha + \beta + \gamma = -b / a$
  • Sum of product of roots in pairs: $\alpha\beta + \beta\gamma + \gamma\alpha = c / a$
  • Product of roots: $\alpha\beta\gamma = -d / a$

4. Special Algebraic Structures & Symmetric Equations

- **Symmetric Polynomials:** Expressions invariant under variable permutation (e.g., $x^2 + y^2$, $x^3 + y^3 + z^3 - 3xyz$).
- **Classic Identity:** $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$.
  If $x + y + z = 0$, then $x^3 + y^3 + z^3 = 3xyz$.
- **Reciprocal Equations:** Equations whose coefficients are symmetric from start to end (e.g., $ax^4 + bx^3 + cx^2 + bx + a = 0$). Solved by dividing through by x^2 and substituting $t = x + 1/x$.

5. High-Yield Shortcuts & Solving Strategies

SHORTCUT 1: COUNTING NON-NEGATIVE SOLUTIONS TO AX + BY = C

Number of non-negative integer solutions to $ax + by = c$ (where $\gcd(a,b) = 1$) is approximately $\lfloor c / (a \times b) \rfloor$ or $\lfloor c / (a \times b) \rfloor + 1$.
Exact Formula: If $c > ab - a - b$, the number of solutions is either $\lfloor c / (ab) \rfloor$ or $\lfloor c / (ab) \rfloor + 1$.

SHORTCUT 2: SYMMETRICAL SYSTEM ELIMINATION

When solving symmetric systems like $x + y = S$ and $xy = P$, treat x and y as roots of the quadratic $t^2 - St + P = 0$.

6. Common Pitfalls & Mistakes

- **Ignoring Domain Constraints in Integral Solutions:** Counting negative solutions when asked for natural numbers ($x, y \in \mathbb{Z}^+$) or positive integers ($x, y > 0$).
- **Dividing by Zero Variable:** Canceling a variable from both sides of an equation (e.g., $x^2 = 4x \implies x = 4$), which loses the root $x = 0$. Always factor out terms!
- **Sign Errors in Vieta's Formulas:** Confusing alternating signs for higher-degree polynomial roots. Remember signs alternate: $-, +, -, +, \ldots$

7. CAT-Level Solved Practice Questions

Question 1 (Non-Negative Integral Solutions)

Find the number of positive integer solutions (x, y) satisfying the equation $7x + 11y = 2024$.

Solution:

Here $a = 7$, $b = 11$, $c = 2024$. $\gcd(7, 11) = 1$, which divides 2024.
Express x in terms of y using modular arithmetic:
$7x = 2024 - 11y$
$2024 \bmod 7 = 1$ (since $2024 = 7 \times 289 + 1$).
$11 \bmod 7 = 4$.
So $7x \equiv 1 - 4y \equiv 0 \pmod 7 \implies 4y \equiv 1 \pmod 7$.
Multiply by 2: $8y \equiv 2 \pmod 7 \implies y \equiv 2 \pmod 7$.
The smallest positive integer value for y is $y = 2$.
Corresponding . Thus, initial solution is . General solution for : and . For positive integer solutions (): (since must be positive,). . Possible values of are . Total number of positive integral solutions = .

$x = (2024 - 11 \times 2) / 7 = 2002 / 7 = 286$
$(x_0, y_0) = (286, 2)$
$k \in \mathbb{Z}$
$x = 286 - 11ky = 2 + 7k$
$x > 0, y > 0$
$286 - 11k > 0 \implies 11k < 286 \implies k < 26xk \leq 25$
$2 + 7k > 0 \implies k \geq 0$
k0, 1, 2, ..., 25
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Question 2 (Cubic Polynomial & Vieta's Application)

If the roots of the cubic equation $x^3 - 12x^2 + 39x - 28 = 0$ are in Arithmetic Progression, find the roots.

Solution:

Let the three roots in AP be $(a - d)$, a, $(a + d)$.
Using Vieta's formula for the sum of roots:
$(a - d) + a + (a + d) = -(-12)/1 = 12$
$3a = 12 \implies a = 4$.
Since $a = 4$ is a root, it must satisfy the equation. Alternatively, use product of roots:
$(a - d) \times a \times (a + d) = -(-28)/1 = 28$
Substitute $a = 4$:
$(4 - d) \times 4 \times (4 + d) = 28 \implies (16 - d^2) \times 4 = 28$
$16 - d^2 = 7 \implies d^2 = 9 \implies d = \pm 3$.
Taking $d = 3$, the roots are $4 - 3, 4, 4 + 3$, which gives **1, 4, 7**.

Question 3 (Higher-Degree Reciprocal Equation)

Find all real solutions to the equation $2x^4 - 9x^3 + 14x^2 - 9x + 2 = 0$.

Solution:

Note that $x = 0$ is not a solution. Divide the entire equation by x^2:
$2x^2 - 9x + 14 - 9/x + 2/x^2 = 0$
Group symmetric terms:
$2(x^2 + 1/x^2) - 9(x + 1/x) + 14 = 0$.
Let $t = x + 1/x$. Then $x^2 + 1/x^2 = t^2 - 2$.
Substitute into the equation:
$2(t^2 - 2) - 9t + 14 = 0 \implies 2t^2 - 4 - 9t + 14 = 0 \implies 2t^2 - 9t + 10 = 0$.
Factorize the quadratic in t:
$(2t - 5)(t - 2) = 0 \implies t = 5/2 \text{ or } t = 2$.
Case 1: $x + 1/x = 2 \implies x^2 - 2x + 1 = 0 \implies (x - 1)^2 = 0 \implies x = 1$.
Case 2: . The real solutions are .

$x + 1/x = 5/2 \Longrightarrow 2x^2 - 5x + 2 = 0 \Longrightarrow (2x - 1)(x - 2) = 0 \Longrightarrow x = 2 \text{ or } x = 1/2$
x = 1, 2, 1/2

8. Quick-Revision Cheat Sheet

Topic	Core Formula / Condition	Key Insight
2-Var System Solutions	Unique: $a_1/a_2 \neq b_1/b_2$ | Infinite: $= c_1/c_2$ | None: $\neq c_1/c_2$	Ratio comparison yields instant system classification.
Diophantine $ax + by = c$	Solvable iff $\gcd(a,b) \mid c$; Step sizes: $\Delta x = b, \Delta y = -a$	Find one initial point (x_0, y_0), then step parameter.
Polynomial Remainder	Remainder when $P(x)$ is divided by $(x - a)$ is $P(a)$	$P(a) = 0$ implies $(x - a)$ is an exact factor.
Cubic Vieta	$\sum\alpha = -b/a, \sum\alpha\beta = c/a, \alpha\beta\gamma = -d/a$	Signs alternate sequentially: $-, +, -$.
Symmetric Identities	If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$	High-frequency simplification key for CAT algebra.

MODULE 2: ALGEBRA

Chapter 9: Quadratic Equations

1. General Form & Theory of Roots

A quadratic equation is a second-degree polynomial equation of the form $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$.

ROOTS & VIETA'S RELATIONS

- **Quadratic Formula:** Roots are $x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$.
- **Sum of Roots (S):** $\alpha + \beta = -b / a$.
- **Product of Roots (P):** $\alpha \times \beta = c / a$.
- **Absolute Difference of Roots:** $|\alpha - \beta| = \sqrt{(D)} / |a|$, where $D = b^2 - 4ac$.
- **Forming a Quadratic Equation:** If roots are α and β, the equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

2. Discriminant Analysis & Nature of Roots

The discriminant $D = b^2 - 4ac$ determines the character and nature of the roots of a quadratic equation with real coefficients.

Discriminant Condition	Nature of Roots	Graphical Interpretation
$D > 0$ (non-perfect square)	Real, distinct, irrational (if $a, b, c \in \mathbb{Q}$)	Intersects x-axis at two distinct points.
$D > 0$ (perfect square)	Real, distinct, rational (if $a, b, c \in \mathbb{Q}$)	Intersects x-axis at two rational points.
$D = 0$	Real, equal (coincident roots $x = -b / (2a)$)	Touches x-axis at exactly one point (vertex).
$D < 0$	Complex conjugate pairs $(p \pm iq)$	Does not touch or intersect the x-axis.

3. Graph Behavior, Maxima & Minima

The graph of $f(x) = ax^2 + bx + c$ is a vertical parabola.

PARABOLIC PROPERTIES & EXTREME VALUES

- **Vertex Coordinates:** The turning point of the parabola occurs at $(x_v, y_v) = (-b / (2a), -D / (4a))$.
- **Minimum Value ($a > 0$):** Parabola opens upwards. Minimum value is $-D / (4a)$, occurring at $x = -b / (2a)$. Range is $[-D / (4a), \infty)$.
- **Maximum Value ($a < 0$):** Parabola opens downwards. Maximum value is $-D / (4a)$, occurring at $x = -b / (2a)$. Range is $(-\infty, -D / (4a)]$.

4. Location of Roots (Interval Analysis)

For a quadratic expression $f(x) = ax^2 + bx + c$ with $a > 0$ and roots α, β ($\alpha \leq \beta$):

LOCATION CONDITIONS FOR A REAL NUMBER K

- **Both roots are greater than k ($\alpha, \beta > k$):**
  1. $D \geq 0$ 2. $-b / (2a) > k$ 3. $a \times f(k) > 0$
- **Both roots are less than k ($\alpha, \beta < k$):**
  1. $D \geq 0$ 2. $-b / (2a) < k$ 3. $a \times f(k) > 0$
- **k lies strictly between the roots ($\alpha < k < \beta$):**
  1. $a \times f(k) < 0$ (automatically implies $D > 0$).
- **Both roots lie in open interval (k_1, k_2):**
  1. $D \geq 0$ 2. $k_1 < -b / (2a) < k_2$ 3. $a \times f(k_1) > 0$ 4. $a \times f(k_2) > 0$

5. High-Yield Shortcuts & Transformation of Roots

SHORTCUT 1: COMMON ROOTS CONDITION

For equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$:
• **Both roots common:** $a_1 / a_2 = b_1 / b_2 = c_1 / c_2$.
• **Exactly one root common:** $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$.

SHORTCUT 2: SYMMETRIC ROOT TRANSFORMATIONS

To form an equation whose roots are:
• Reciprocals $(1/\alpha, 1/\beta)$: Replace x with $1/x \rightarrow cx^2 + bx + a = 0$.
• Opposite signs $(-\alpha, -\beta)$: Replace x with $-x \rightarrow ax^2 - bx + c = 0$.
• Shifted by k $(\alpha + k, \beta + k)$: Replace x with $x - k$.

6. Common Pitfalls & Mistakes

- **Assuming $a \neq 0$ without checking:** If an equation has parameter m as leading coefficient ($mx^2 + bx + c = 0$), test $m = 0$ separately as it reduces to a linear equation.
- **Forgetting Conjugate Pairs:** Irrational roots of the form $p + \sqrt{q}$ occur in conjugate pairs $p - \sqrt{q}$ ONLY when coefficients a, b, c are rational. Complex roots occur in conjugate pairs ONLY when coefficients are real.
- **Incomplete Location of Roots Conditions:** Checking only $f(k) > 0$ without enforcing $D \geq 0$ or checking the vertex position $-b / (2a)$.

7. CAT-Level Solved Practice Questions

Question 1 (Location of Roots & Parameter Range)

Find all real values of m for which both roots of the quadratic equation $x^2 - 2mx + m^2 - 1 = 0$ are real and greater than 3.

Solution:

Here $a = 1$, $b = -2m$, $c = m^2 - 1$, and reference point $k = 3$.
Apply the three conditions for both roots greater than $k = 3$:
Condition 1: Discriminant $D \geq 0$
$D = (-2m)^2 - 4(1)(m^2 - 1) = 4m^2 - 4m^2 + 4 = 4 > 0$. (Always true for all real m).
Condition 2: Vertex $-b / (2a) > 3$
$-(-2m) / (2 \times 1) > 3 \implies m > 3$.
Condition 3: $a \times f(3) > 0$
$1 \times (3^2 - 2m(3) + m^2 - 1) > 0 \implies m^2 - 6m + 8 > 0$.
Factorize: $(m - 2)(m - 4) > 0 \implies m < 2 \text{ or } m > 4$.
Taking the intersection of $m > 3$ and ($m < 2$ or $m > 4$):
The valid range of m is **$m > 4$** (or $m \in (4, \infty)$).

Question 2 (Root Symmetric Expressions)

If α and β are roots of $x^2 - 5x + 3 = 0$, evaluate $\alpha^4 + \beta^4$.

Solution:

From Vieta's relations: $\alpha + \beta = 5$ and $\alpha\beta = 3$.
Step 1: Calculate $\alpha^2 + \beta^2$:
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 5^2 - 2(3) = 25 - 6 = 19$.
Step 2: Square again to find $\alpha^4 + \beta^4$:
$(\alpha^2 + \beta^2)^2 = \alpha^4 + \beta^4 + 2(\alpha\beta)^2$
$19^2 = \alpha^4 + \beta^4 + 2(3^2)$
$361 = \alpha^4 + \beta^4 + 18 \implies \alpha^4 + \beta^4 = 361 - 18 = 343$.

Question 3 (Quadratic Optimization with Constraints)

Find the maximum value of $f(x) = -2x^2 + 8x + 11$ for $x \in [0, 5]$.

Solution:

Here $a = -2 < 0$, so the parabola opens downwards and has an unconstrained peak at its vertex.
Vertex x-coordinate: $x_v = -b / (2a) = -8 / (2 \times -2) = 2$.
Since $x = 2$ falls within interval $[0, 5]$, maximum value occurs directly at $x = 2$.
$f(2) = -2(2^2) + 8(2) + 11 = -8 + 16 + 11 = 19$.
(Boundary check for minimum: $f(0) = 11$ and $f(5) = -2(25) + 40 + 11 = 1$).
Maximum value is **19**.

8. Quick-Revision Cheat Sheet

Concept	Formula / Condition	Key Takeaway
Vieta's Relations	$\alpha + \beta = -b/a$, $\alpha\beta = c/a$	Form equation as $x^2 - Sx + P = 0$.
Difference of Roots	$\lvert\alpha - \beta\rvert = \sqrt{(D)} / \lvert a\rvert$	Direct shortcut for root spacing problems.
Vertex & Extremum	$x = -b/(2a)$, $y = -D/(4a)$	Max if $a < 0$, Min if $a > 0$.
Location ($\alpha, \beta > k$)	$D \geq 0$, $-b/(2a) > k$, $a \times f(k) > 0$	Must satisfy all 3 conditions simultaneously.
Reciprocal Roots	Replace x with $1/x \rightarrow cx^2 + bx + a = 0$	Coefficients reverse order (a and c swap).

MODULE 2: ALGEBRA

Chapter 10: Inequalities, Modulus & Absolute Values

1. Fundamental Rules of Inequalities

An inequality asserts that one algebraic expression is greater or less than another. Manipulating inequalities requires strict adherence to sign rules.

CORE OPERATIONS & DIRECTION RULES

- **Addition/Subtraction:** If $a > b$, then $a \pm c > b \pm c$ for any real number c.
- **Positive Multiplication:** If $a > b$ and $c > 0$, then $ac > bc$.
- **Negative Multiplication (Inversion):** If $a > b$ and $c < 0$, then $ac < bc$ (inequality flips).
- **Reciprocals:** If $a > b > 0$ or $0 > a > b$, then $1/a < 1/b$. If $a > 0 > b$, then $1/a > 1/b$.
- **Squaring:** If $a > b > 0$, then $a^2 > b^2$. If $a < b < 0$, then $a^2 > b^2$.

2. Wavy Curve Method (Method of Intervals)

Used to solve rational inequalities of the form $P(x) / Q(x) > 0$ or < 0.

ALGORITHMIC STEPS FOR WAVY CURVE

1. Make RHS equal to zero by shifting terms.
2. Factorize numerator and denominator into linear factors: $(x - a_1)^{k_1} (x - a_2)^{k_2} \dots$
3. Ensure the coefficient of x in every linear factor is positive.
4. Find all critical points (zeros of numerator and denominator) and plot them on a real number line in ascending order.
5. Start from the rightmost interval (greater than largest critical point) with a **positive (+)** sign.
6. Move left across critical points:
   - If power of factor is **odd**, sign **changes**.
   - If power of factor is **even**, sign **remains unchanged**.
7. Select regions matching required inequality (+ for > 0, – for < 0). Exclude denominator zeros!

3. Absolute Value (Modulus) Mechanics

The absolute value of a real number x represents its distance from zero on the real line:
$|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

KEY MODULUS EQUATIONS & INEQUALITIES

- $|x| \leq a \iff -a \leq x \leq a$ (for $a > 0$).
- $|x| \geq a \iff x \leq -a \text{ or } x \geq a$ (for $a > 0$).
- $|x - a| + |x - b| = k$: Minimum value of LHS is $|a - b|$, which occurs for any x in interval $[a, b]$.
- **Triangle Inequalities:**
  - $|a + b| \leq |a| + |b|$ (Equality holds iff $a \times b \geq 0$).
  - $|a - b| \geq ||a| - |b||$.

4. Classical Optimization Inequalities (AM-GM-HM & Cauchy-Schwarz)

- **AM-GM-HM Inequality:** For any set of positive real numbers $x_1, x_2, ..., x_n$:

  $(x_1 + x_2 + ... + x_n) / n \geq \sqrt[n]{(x_1 x_2 ... x_n)} \geq n / (1/x_1 + 1/x_2 + ... + 1/x_n)$

  Equality holds if and only if $x_1 = x_2 = ... = x_n$.

- **Cauchy-Schwarz Inequality:** For real numbers a_i, b_i:

  $(a_1^2 + a_2^2 + ... + a_n^2)(b_1^2 + b_2^2 + ... + b_n^2) \geq (a_1b_1 + a_2b_2 + ... + a_nb_n)^2$

5. High-Yield Shortcuts & Graphical Methods

SHORTCUT 1: MODULUS SUM GRAPHING (V-SHAPE & BUCKET)

The expression $f(x) = |x - a| + |x - b|$ (where $a < b$) forms a bucket graph.
• Minimum value is $b - a$, occurring continuously on segment $x \in [a, b]$.
• Slope for $x > b$ is $+2$; slope for $x < a$ is -2.

SHORTCUT 2: PRODUCT MINIMIZATION UNDER SUM CONSTRAINT

If sum of positive variables $x + y + z = S$ (constant), product $x \times y \times z$ is maximized when $x = y = z = S/3$.

6. Common Pitfalls & Mistakes

- **Cross-Multiplying Variable Expressions:** Cross-multiplying terms across inequalities without knowing their sign (e.g., multiplying by $(x - 2)$ when $x - 2$ could be negative). Always bring terms to one side!
- **Applying AM-GM to Negative Numbers:** AM-GM inequality applies strictly to non-negative numbers. Verify positivity before using $AM \geq GM$.
- **Including Denominator Zeros in Solution Set:** When solving $(x - 1)/(x - 3) \leq 0$, including $x = 3$ in the solution set. Denominator zero causes undefined division, so $x = 3$ MUST be excluded ($[1, 3)$).

7. CAT-Level Solved Practice Questions

Question 1 (Wavy Curve Rational Inequality)

Find the complete solution set for x satisfying: $[(x - 1)^3 (x + 2)^2 (x - 4)] / [(x + 1) (x - 3)^4] \leq 0$.

Solution:

Step 1: Identify critical points and their exponents:
Numerator zeros: $x = 1$ (odd power 3), $x = -2$ (even power 2), $x = 4$ (odd power 1).
Denominator zeros: $x = -1$ (odd power 1), $x = 3$ (even power 4).
Step 2: Plot points on number line: $-2, -1, 1, 3, 4$.
Step 3: Test rightmost interval $(4, \infty)$: Sign is $+$.
• At $x = 4$ (odd): Sign changes to $-$.
• At $x = 3$ (even): Sign stays $-$.
• At $x = 1$ (odd): Sign changes to $+$.
• At $x = -1$ (odd): Sign changes to $-$. • At (even): Sign stays . Step 4: Identify regions (where sign is or numerator is zero): Negative intervals: . Numerator zeros allowed: (joins interval), , . Denominator zeros excluded: . Combining valid points gives solution: .

$x = -2-$
$\leq 0-$
$(-\infty, -2) \cup (-2, -1) \cup (1, 3) \cup (3, 4]$
$x = -2x = 1x = 4$
$x = -1, 3$
$x \in (-\infty, -1) \cup [1, 3) \cup (3, 4]$

Question 2 (Nested Modulus Equation)

Find the number of distinct real roots for the equation $||x - 2| - 4| = 3$.

Solution:

Outer modulus property: $|Y| = 3 \implies Y = 3 \text{ or } Y = -3$.
Case 1: $|x - 2| - 4 = 3 \implies |x - 2| = 7$.
This gives $x - 2 = 7 \implies x = 9$, or $x - 2 = -7 \implies x = -5$. (2 roots).
Case 2: $|x - 2| - 4 = -3 \implies |x - 2| = 1$.
This gives $x - 2 = 1 \implies x = 3$, or $x - 2 = -1 \implies x = 1$. (2 roots).
Checking all 4 candidate roots: $x = -5, 1, 3, 9$ are all distinct and valid.
Total number of distinct real roots = **4**.

Question 3 (AM-GM Constrained Optimization)

If a, b, c are positive real numbers such that $a + b + c = 12$, find the maximum value of $a^2 b^3 c$.

Solution:

Exponents of a, b, c in product are 2, 3, and 1 respectively (Sum of exponents = $2 + 3 + 1 = 6$).
Split terms to match exponents:
Split a into 2 parts: $a/2 + a/2$.
Split b into 3 parts: $b/3 + b/3 + b/3$.
Split c into 1 part: c.
Total number of terms = 6. Sum of these 6 terms = $a + b + c = 12$.
Apply AM-GM on these 6 terms:
$[(a/2 + a/2 + b/3 + b/3 + b/3 + c) / 6] \geq \sqrt[6]{[(a/2)^2 \times (b/3)^3 \times c]}$
$12 / 6 \geq \sqrt[6]{[(a^2 b^3 c) / (4 \times 27)]} \implies 2 \geq \sqrt[6]{[(a^2 b^3 c) / 108]}$.
Raise both sides to 6th power:
$2^6 = 64 \geq (a^2 b^3 c) / 108 \implies a^2 b^3 c \leq 64 \times 108 = 6912$.
Maximum value is **6912** (achieved when $a/2 = b/3 = c \implies a = 4, b = 6, c = 2$).

8. Quick-Revision Cheat Sheet

Topic	Rule / Formula	Key Insight
Modulus Inequality	$\|x\| \le a \iff -a \le x \le a$	Distance from origin bounded by a.
Modulus Minima	Min of $\|x-a\| + \|x-b\| = \|a-b\|$	Attained continuously on interval $[a, b]$.
Wavy Curve Sign	Even power → Keep sign; Odd power → Flip sign	Always check rightmost region with positive sign.
AM ≥ GM	$(x_1 + \ldots + x_n)/n \ge \sqrt[n]{(x_1 \ldots x_n)}$	Valid only for positive real numbers.
Triangle Inequality	$\|a+b\| = \|a\| + \|b\| \iff ab \ge 0$	Equality occurs when terms have same sign.

MODULE 2: ALGEBRA

Chapter 11: Functions, Graphs & Transformations

1. Definition, Classification, Domain & Range

A function $f: A \to B$ assigns to each element $x \in A$ exactly one element $y \in B$. The set A is the **domain**, and the set of actual outputs in B is the **range**.

CLASSIFICATIONS & CONDITIONS

- **One-to-One (Injective):** $f(x_1) = f(x_2) \implies x_1 = x_2$. Graph passes horizontal line test (intersects at most once).
- **Onto (Surjective):** Range equals codomain ($\text{Range} = B$). Every element in B has a pre-image in A.
- **Bijective:** Both injective and surjective. A function has an inverse f^{-1} if and only if it is bijective.
- **Domain Constraints for Real Functions:**
  1. Even roots require non-negative radicand: $\sqrt{(g(x))} \implies g(x) \geq 0$.
  2. Denominators cannot be zero: $h(x)/g(x) \implies g(x) \neq 0$.
  3. Logarithms require positive argument: $\log_b(g(x)) \implies g(x) > 0, b > 0, b \neq 1$.

2. Composite & Inverse Functions

For $f: A \to B$ and $g: B \to C$, composite function $(g \circ f)(x) = g(f(x))$ is defined for inputs where $f(x)$ belongs to domain of g.

- **Composition Identity:** If f^{-1} exists, $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$.
- **Reversal Rule:** $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- **Geometric Inverse Relationship:** Graph of $y = f^{-1}(x)$ is the exact reflection of $y = f(x)$ across the line $y = x$.

3. Graph Symmetry, Even/Odd Functions & Periodicity

SYMMETRY RULES

- **Even Function:** $f(-x) = f(x)$ for all x. Graph is symmetric about the **y-axis**. (e.g., x^2, $|x|$, $\cos x$).
- **Odd Function:** $f(-x) = -f(x)$ for all x. Graph is symmetric about the **origin** (180° rotational symmetry). (e.g., x^3, x, $\sin x$).
- **Periodic Function:** $f(x + T) = f(x)$ for smallest positive real number T (fundamental period).

4. Graph Transformations

Given base graph $y = f(x)$ and positive constant $c > 0$:

Transformation	Equation	Geometric Effect
Vertical Shift	$y = f(x) \pm c$	Shift up by c units (+) or down by c units (−).
Horizontal Shift	$y = f(x \mp c)$	Shift right by c units (−) or left by c units (+).
Reflection across X-axis	$y = -f(x)$	Flip graph vertically over x-axis.
Reflection across Y-axis	$y = f(-x)$	Flip graph horizontally over y-axis.
Vertical Absolute Value	$y = \lvert f(x) \rvert$	Reflect all portion below x-axis to above x-axis.
Horizontal Absolute Value	$y = f(\lvert x \rvert)$	Erase portion for $x < 0$, reflect portion for $x \geq 0$ across y-axis.

5. Common Functional Equations (CAT Core Templates)

Standard linear, exponential, and logarithmic functions follow distinct algebraic identities:

- $f(x + y) = f(x) + f(y) \implies f(x) = kx$ (Linear form).
- $f(x + y) = f(x) \times f(y) \implies f(x) = a^x$ (Exponential form).
- $f(xy) = f(x) + f(y) \implies f(x) = k \log x$ (Logarithmic form).
- $f(xy) = f(x) \times f(y) \implies f(x) = x^n$ (Power form).
- $f(x) + f(1/x) = f(x) \times f(1/x) \implies f(x) = 1 \pm x^n$ (Classic CAT identity).

6. High-Yield Shortcuts & Common Pitfalls

SHORTCUT 1: INVERSE OF RATIONAL LINEAR FUNCTIONS

For $f(x) = (ax + b) / (cx + d)$, its inverse is $f^{-1}(x) = (-dx + b) / (cx - a)$.
Rule: Swap diagonal elements a and d, and negate them!

- **Confusing $f(|x|)$ with $|f(x)|$:** $|f(x)|$ forces output to be non-negative by reflecting negative y-values upwards. $f(|x|)$ forces output symmetry about y-axis by ignoring original negative x-values.
- **Domain Reduction during Simplification:** Simplifying $f(x) = (x^2 - 4)/(x - 2)$ to $x + 2$ without retaining restriction $x \neq 2$ alters the domain. Always determine domain from original un-simplified expression.

7. CAT-Level Solved Practice Questions

Question 1 (Functional Identity Application)

If $f(x)$ is a polynomial function satisfying $f(x) + f(1/x) = f(x) \times f(1/x)$ for all real $x \neq 0$, and $f(3) = 28$, find the value of $f(4)$.

Solution:

By the standard polynomial identity: $f(x) = 1 \pm x^n$.
Case 1: $f(x) = 1 + x^n$
Given $f(3) = 28 \implies 1 + 3^n = 28 \implies 3^n = 27 \implies n = 3$.
So $f(x) = 1 + x^3$.
Case 2: $f(x) = 1 - x^n$ Given (no real integer solution for). Therefore, . Substitute : .

$f(3) = 28 \Longrightarrow 1 - 3^n = 28 \Longrightarrow 3^n = -27n$
$f(x) = 1 + x^3$
$x = 4$
$f(4) = 1 + 4^3 = 1 + 64 = 65$

Question 2 (Domain of Composite Function)

Find the domain of real-valued function $f(x) = \sqrt{[\ \log_{0.5}(x^2 - 5x + 6)\ ]}$.

Solution:

Step 1: Radicand condition for square root requires non-negative argument:
$\log_{0.5}(x^2 - 5x + 6) \geq 0$.
Step 2: Solve logarithmic inequality with base $0.5 < 1$ (inequality sign flips):
$x^2 - 5x + 6 \leq (0.5)^0 \Longrightarrow x^2 - 5x + 6 \leq 1$
$x^2 - 5x + 5 \leq 0$.
Roots of $x^2 - 5x + 5 = 0$ are $x = [\ 5 \pm \sqrt{(25 - 20)}\ ] / 2 = (5 \pm \sqrt{5})/2$.
So, $(5 - \sqrt{5})/2 \leq x \leq (5 + \sqrt{5})/2$ --- (Condition 1).
Step 3: Logarithm argument must be strictly positive:
$x^2 - 5x + 6 > 0 \Longrightarrow (x - 2)(x - 3) > 0 \Longrightarrow x < 2 \text{ or } x > 3$ --- (Condition 2).
Step 4: Intersect Condition 1 and Condition 2:
Note that $(5 - \sqrt{5})/2 \approx (5 - 2.236)/2 = 1.382$, and $(5 + \sqrt{5})/2 \approx 3.618$.
Intersection gives: **$x \in [\ (5 - \sqrt{5})/2, 2\ ) \cup (\ 3, (5 + \sqrt{5})/2\ ]$**.

Question 3 (Graph Intersection & Transformation Solutions)

Find the number of real solutions to the equation $|\ |x - 1| - 3\ | = 2 - x$.

Solution:

Let LHS = $f(x) = ||x - 1| - 3|$ and RHS = $g(x) = 2 - x$.
Since LHS ≥ 0, valid solutions require RHS $\geq 0 \Longrightarrow 2 - x \geq 0 \Longrightarrow x \leq 2$.
Analyze intervals for $x \leq 2$:
Case 1: $1 \leq x \leq 2$:
$|x - 1| = x - 1 \Longrightarrow |x - 1 - 3| = |x - 4| = 4 - x$ (since $x \leq 2$).
Set equal to RHS: $4 - x = 2 - x \Longrightarrow 4 = 2$ (No solution).
Case 2: $-2 \leq x < 1$:
$|x - 1| = 1 - x \Longrightarrow |1 - x - 3| = |-x - 2| = |x + 2| = x + 2$ (since $x \geq -2$).
Set equal to RHS: $x + 2 = 2 - x \Longrightarrow 2x = 0 \Longrightarrow x = 0$.
Check if $x = 0$ lies in $[-2, 1)$: Yes, valid root.
Case 3: $x < -2$:
$|x - 1| = 1 - x \Longrightarrow |1 - x - 3| = |-x - 2| = -(x + 2) = -x - 2$.
Set equal to RHS: $-x - 2 = 2 - x \Longrightarrow -2 = 2$ (No solution). Total number of real solutions = (which is).

$1x = 0$

8. Quick-Revision Cheat Sheet

Topic	Rule / Identity	Key Insight
Rational Inverse	$f(x) = (ax+b)/(cx+d) \Rightarrow f^{-1}(x) = (-dx+b)/(cx-a)$	Swap and negate diagonal entries a and d.
Even/Odd Function	Even: $f(-x) = f(x)$ | Odd: $f(-x) = -f(x)$	Even is y-axis symmetric; Odd is origin symmetric.
Polynomial Identity	$f(x) + f(1/x) = f(x)f(1/x) \Rightarrow f(x) = 1 \pm x^n$	Core high-frequency CAT algebra template.
Horizontal Shift	$f(x - c)$ shifts right; $f(x + c)$ shifts left	Signs operate opposite to intuitive movement direction.
Log Base Inequality	$\log_b x \geq k \Rightarrow x \leq b^k$ if $0 < b < 1$	Inequality direction flips when base is less than 1.

MODULE 2: ALGEBRA

Chapter 12: Logarithms, Indices & Surds

1. Laws of Indices & Exponential Algebra

An index (or exponent) denotes the number of times a base is multiplied by itself. Exponents form the foundation for logarithmic operations.

FUNDAMENTAL LAWS OF EXPONENTS

- **Product Law:** $a^m \times a^n = a^{m+n}$
- **Quotient Law:** $a^m / a^n = a^{m-n}$ (for $a \neq 0$)
- **Power of a Power:** $(a^m)^n = a^{m \times n} = (a^n)^m$
- **Power of a Product:** $(a \times b)^n = a^n \times b^n$
- **Negative & Fractional Exponents:** $a^{-n} = 1 / a^n$ and $a^{m/n} = \sqrt[n]{(a^m)} = (\sqrt[n]{a})^m$

2. Fundamental Principles & Properties of Logarithms

The logarithm of a positive real number N to a positive base b ($b \neq 1$) is the exponent x to which b must be raised to yield N:

$\log_b(N) = x \iff b^x = N$

ESSENTIAL LOGARITHMIC IDENTITIES

- **Domain Requirements:** For $\log_b(N)$ to be defined: $N > 0$, $b > 0$, and $b \neq 1$.
- **Product Rule:** $\log_b(M \times N) = \log_b(M) + \log_b(N)$
- **Quotient Rule:** $\log_b(M / N) = \log_b(M) - \log_b(N)$
- **Power Rule:** $\log_b(M^p) = p \times \log_b(M)$
- **Base Power Rule:** $\log_{(b^q)}(M) = (1 / q) \times \log_b(M)$
- **Combined Power & Base Power:** $\log_{(b^q)}(M^p) = (p / q) \times \log_b(M)$
- **Base Change Theorem:** $\log_b(a) = \log_c(a) / \log_c(b) = 1 / \log_a(b)$
- **Power Base Exchange Identity:** $a^{\log_b(c)} = c^{\log_b(a)}$

3. Characteristic & Mantissa Analysis

Every base-10 logarithm can be split into an integer part (**Characteristic**) and a non-negative fractional part (**Mantissa**):

$\log_{10}(N) = c + m$, where $c \in \mathbb{Z}$ and $m \in [0, 1)$.

NUMBER OF DIGITS & LEADING ZEROS RULES

- **Number of Digits in A^N:** Number of digits $= \lfloor \log_{10}(A^N) \rfloor + 1 = \lfloor N \times \log_{10}(A) \rfloor + 1$.
- **Number of Zeroes after Decimal before First Significant Digit in A^{-N}:** Number of zeroes $= |\lfloor \log_{10}(A^{-N}) \rfloor| - 1 = |\lfloor -N \times \log_{10}(A) \rfloor| - 1$.

4. Surds & Rationalization Mechanics

A surd is an irrational root of a rational number. Operations on surds require simplification and rationalization of denominators.

- **Square Root of a Conjugate Surd:** To evaluate $\sqrt{(a \pm \sqrt{b})}$, write $a \pm \sqrt{b} = (\sqrt{x} \pm \sqrt{y})^2 = (x + y) \pm 2\sqrt{(xy)}$. Solve for x and y such that $x + y = a$ and $4xy = b$.
- **Comparison of Surds:** To compare $\sqrt[m]{a}$ and $\sqrt[n]{b}$, equate their indices by finding $k = \text{LCM}(m, n)$, transforming terms to $\sqrt[k]{(a^{k/m})}$ and $\sqrt[k]{(b^{k/n})}$.

5. Logarithmic & Exponential Inequalities

Solving inequalities involving logarithms depends critically on whether the base is greater than or less than 1.

LOGARITHMIC INEQUALITY RULES

- **Case 1: Base $b > 1$ (Increasing Function)**
  $\log_b(f(x)) > \log_b(g(x)) \iff f(x) > g(x) > 0$ (Inequality direction remains unchanged).
- **Case 2: Base $0 < b < 1$ (Decreasing Function)**
  $\log_b(f(x)) > \log_b(g(x)) \iff 0 < f(x) < g(x)$ (Inequality direction flips!).

6. High-Yield Shortcuts & Common Pitfalls

SHORTCUT 1: TELESCOPING LOGARITHMIC SERIES

Series of the form $\sum \log(1 + 1/k) = \log(2/1) + \log(3/2) + \ldots + \log((n+1)/n)$ simplify directly to $\log(n + 1)$ because all intermediate terms cancel.

SHORTCUT 2: INFINITE NESTED SURDS

For $x = \sqrt{(a + \sqrt{(a + \sqrt{(a + \ldots)})})}$, $x^2 - x - a = 0 \implies x = [\, 1 + \sqrt{(1 + 4a)}\,] / 2$.

- **Splitting Sums inside Logarithms:** Writing $\log(A + B) = \log(A) + \log(B)$. This is strictly FALSE. The correct rule is $\log(A \times B) = \log(A) + \log(B)$.
- **Ignoring Base Restrictions:** Forgetting to test $b \neq 1$ when solving equations with variable bases like $\log_{(x-2)}(5)$.
- **Applying Base-Change without Domain Verification:** Converting $\log_a(b) = 1 / \log_b(a)$ without checking if $b \neq 1$.

7. CAT-Level Solved Practice Questions

Question 1 (Base Change & Multi-Variable System)

If $\log_3 2, \log_3 (2^x - 5), \text{and } \log_3 (2^x - 7/2)$ are in Arithmetic Progression, find the value of x.

Solution:

Since the terms are in AP, twice the middle term equals the sum of the first and third terms:

$2 \times \log_3 (2^x - 5) = \log_3 2 + \log_3 (2^x - 7/2)$.

Using logarithmic power and product rules:

$\log_3 (2^x - 5)^2 = \log_3 [\, 2 \times (2^x - 7/2)\,]$.

Remove logarithms from both sides:

$(2^x - 5)^2 = 2(2^x) - 7$.

Let $y = 2^x$ (where $y > 5$ for the log argument to be positive):

$(y - 5)^2 = 2y - 7 \Longrightarrow y^2 - 10y + 25 = 2y - 7 \Longrightarrow y^2 - 12y + 32 = 0$.

Factorize: $(y - 4)(y - 8) = 0 \Longrightarrow y = 4 \text{ or } y = 8$.

Check validity with domain constraint $y > 5$:

• $y = 4$ makes $2^x - 5 = -1 < 0$ (Invalid, log of negative number is undefined).

• $y = 8 \Longrightarrow 2^x = 8 = 2^3 \Longrightarrow x = 3$ (Valid since $8 > 5$).

Therefore, **x = 3**.

Question 2 (Number of Digits Determination)

Find the number of digits in the expansion of 6^{20}, given that $\log_{10} 2 = 0.30103$ and $\log_{10} 3 = 0.47712$.

Solution:

Let $N = 6^{20}$. Take base-10 logarithm on both sides:

$\log_{10} N = \log_{10} (6^{20}) = 20 \times \log_{10} (6)$.

Express $\log_{10} 6$ in prime factors:

$\log_{10} 6 = \log_{10} (2 \times 3) = \log_{10} 2 + \log_{10} 3 = 0.30103 + 0.47712 = 0.77815$.

Now compute $\log_{10} N$:

$\log_{10} N = 20 \times 0.77815 = 15.563$.

Characteristic $c = \lfloor 15.563 \rfloor = 15$.

Number of digits = $c + 1 = 15 + 1 = 16$.

The number of digits in 6^{20} is **16**.

Question 3 (Nested Surd Simplification)

Simplify the surd expression: $S = \sqrt{(31 + 12\sqrt{5})} - \sqrt{(31 - 12\sqrt{5})}$.

Solution:

Let us write $31 + 12\sqrt{5}$ in the perfect square form $(x + y\sqrt{5})^2$ or $(\sqrt{a} + \sqrt{b})^2$:
$31 + 2(6\sqrt{5}) = 31 + 2\sqrt{(36 \times 5)} = 31 + 2\sqrt{180}$.
We need two numbers whose sum is 31 and product is 180:
The numbers are 20 and 11 (since $20 + 11 = 31$ and $20 \times 11 = 220 \neq 180$).
Let's re-examine $12\sqrt{5} = 2 \times 6 \times \sqrt{5}$. Squaring components: $6^2 + (\sqrt{5})^2 = 36 + 5 = 41 \neq 31$.
Try $12\sqrt{5} = 2 \times 3 \times (2\sqrt{5})$. Squaring components: $3^2 + (2\sqrt{5})^2 = 9 + 20 = 29 \neq 31$.
Try $12\sqrt{5} = 2 \times 1 \times (6\sqrt{5}) \rightarrow 1 + 180 = 181$.
Try splitting 31 into $a + b$ where $2\sqrt{ab} = 12\sqrt{5} \implies ab = 180$.
Factors of 180 summing to 31: **20 and 9**? $20 + 9 = 29$. **36 and 5**? $36 - 5 = 31$! No, $ab = 180$ with sum 31: $27 + 4 = 31$ ($27 \times 4 = 108$). $20 + 11 = 31$. Wait: $(3\sqrt{3})^2$? Let's check $a = 4$, $b = 27$? $(3\sqrt{3} + 2)^2 = 27 + 4 + 12\sqrt{3} \neq 12\sqrt{5}$.
Ah, $31 + 12\sqrt{5} = (a + b\sqrt{5})^2 = a^2 + 5b^2 + 2ab\sqrt{5}$.
So $2ab = 12 \implies ab = 6$, and $a^2 + 5b^2 = 31$.
Test integer pairs for $ab = 6$:
• If $b = 1$, $a = 6 \implies a^2 + 5b^2 = 36 + 5 = 41$.
• If $b = 2$, $a = 3 \implies a^2 + 5b^2 = 9 + 5(4) = 9 + 20 = 29$.
• If $b = 1$, $a = 5.1$? Let's evaluate S^2 directly:
$S^2 = (31 + 12\sqrt{5}) + (31 - 12\sqrt{5}) - 2\sqrt{[\,(31 + 12\sqrt{5})(31 - 12\sqrt{5})\,]}$
$S^2 = 62 - 2\sqrt{[\,31^2 - (12\sqrt{5})^2\,]} = 62 - 2\sqrt{[\,961 - 720\,]}$
$S^2 = 62 - 2\sqrt{[\,241\,]}$.
Since $S > 0$, taking square root gives: $S = \sqrt{(62 - 2\sqrt{241})}$.

Question 4 (Logarithmic Inequality with Variable Base)

Solve for x: $\log_x (x^2 - 3x + 2) \geq 1$.

Solution:

Base x requires $x > 0$ and $x \neq 1$. Argument requires $x^2 - 3x + 2 > 0 \implies (x - 1)(x - 2) > 0 \implies x < 1 \text{ or } x > 2$.
Combined initial domain: $x \in (0, 1) \cup (2, \infty)$.
Case 1: Base $x > 2$ (Base > 1)
Inequality direction stays same:
$x^2 - 3x + 2 \geq x^1 \implies x^2 - 4x + 2 \geq 0$.
Roots of $x^2 - 4x + 2 = 0$ are $x = (4 \pm \sqrt{8})/2 = 2 \pm \sqrt{2}$.
Solutions to $x^2 - 4x + 2 \geq 0$: $x \leq 2 - \sqrt{2}$ or $x \geq 2 + \sqrt{2} \approx 3.414$.
Intersect with $x > 2$: $x \in [\,2 + \sqrt{2}, \infty\,)$.
Case 2: Base $0 < x < 1$ (Base < 1)
Inequality direction flips: . Intersect with : Note . Intersection gives: . Final Solution set: .

$x^2 - 3x + 2 \le x^1 \Longrightarrow x^2 - 4x + 2 \le 0 \Longrightarrow 2 - \sqrt{2} \le x \le 2 + \sqrt{2}$
$0 < x < 1$
$2 - \sqrt{2} \approx 2 - 1.414 = 0.586$
$x \in [\ 2 - \sqrt{2}, 1\ )$
$x \in [\ 2 - \sqrt{2}, 1\ ) \cup [\ 2 + \sqrt{2}, \infty\ )$

8. Quick-Revision Cheat Sheet

Topic	Key Rule / Identity	Key Takeaway
Base Change	$\log_b a = \log_c a \ / \ \log_c b = 1 \ / \ \log_a b$	Convert all logs to a uniform base before simplifying.
Power Exchange	$a^{\log_b c} = c^{\log_b a}$	Swap base and argument to resolve exponent log terms.
Digits Count	$\text{Digits} = \lfloor \log_{10}(N) \rfloor + 1$	Integral characteristic plus 1 gives total digits.
Dec. Zeroes	$\text{Zeroes} = \lvert \lfloor \log_{10}(A^{-N}) \rfloor \rvert - 1$	Counts zeroes between decimal point and first non-zero digit.
Log Inequality	$0 < b < 1 \Longrightarrow \log_b f(x) > k \Leftrightarrow f(x) < b^k$	Always flip inequality sign when base is between 0 and 1!

MODULE 2: ALGEBRA

Chapter 13: Sequences, Series & Progressions

1. Introduction to Sequences & Series

A **sequence** is an ordered collection of numbers following a definitive mathematical rule: $a_1, a_2, a_3, \ldots, a_n$. A **series** is the sum of the elements of a sequence: $S_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + \ldots + a_n$.

2. Arithmetic Progression (AP) — Deep Theoretical Foundations

An **Arithmetic Progression** is a sequence in which the difference between any two consecutive terms is a constant called the **common difference** (d).
$d = a_n - a_{n-1} = a_2 - a_1 = a_3 - a_2$

DERIVATIONS & CORE STRUCTURAL FORMULAS

- **General Term (T_n):** Starting with first term a and common difference d:
  $T_1 = a, T_2 = a + d, T_3 = a + 2d \Longrightarrow T_n = a + (n - 1)d$
- **Sum of n Terms (S_n):** Write the sum forward and backward, then add term-by-term:
  $S_n = a + (a+d) + \ldots + [a+(n-1)d]$
  $S_n = [a+(n-1)d] + [a+(n-2)d] + \ldots + a$
  $2S_n = n \times [2a + (n - 1)d] \Longrightarrow S_n = (n/2) \times [2a + (n - 1)d] = (n/2) \times (\text{First Term} + \text{Last Term})$
- **Symmetry Property:** In a finite AP, the sum of terms equidistant from the beginning and end is constant and equals the sum of the first and last terms:
  $T_1 + T_n = T_2 + T_{n-1} = T_k + T_{n-k+1}$
- **Sub-sequence Property:** If terms are chosen at regular intervals from an AP, the resulting sub-sequence is also an AP with a scaled common difference.

Strategic Selection of Terms in AP

When solving word problems where the sum of terms is given, choose terms symmetrically to eliminate d when adding:

- **3 Terms in AP:** $(a - d), a, (a + d)$ ➔ Sum = $3a$
- **4 Terms in AP:** $(a - 3d), (a - d), (a + d), (a + 3d)$ ➔ Sum = $4a$, common difference = $2d$
- **5 Terms in AP:** $(a - 2d), (a - d), a, (a + d), (a + 2d)$ ➔ Sum = $5a$

Arithmetic Mean (AM) Theory

- **AM between Two Numbers:** The arithmetic mean of a and b is $A = (a + b) / 2$.
- **Insertion of n Arithmetic Means:** If $A_1, A_2, \ldots, A_n$ are inserted between a and b such that $a, A_1, A_2, \ldots, A_n, b$ forms an AP:
  Total terms = $n + 2$. Last term $b = a + (n + 1)d \Longrightarrow d = (b - a) / (n + 1)$.
  The k-th inserted mean is $A_k = a + k \times d = a + k \times [\,(b - a) / (n + 1)\,]$.

  Sum of n inserted AMs: $\sum_{k=1}^{n} A_k = n \times [\,(a + b) / 2\,] = n \times AM(a, b)$.

3. Geometric Progression (GP) — Deep Theoretical Foundations

A **Geometric Progression** is a sequence where the ratio of any two consecutive terms is a non-zero constant called the **common ratio** (r).
$r = a_n / a_{n-1} = a_2 / a_1 = a_3 / a_2$

DERIVATIONS & INFINITE SERIES MECHANICS

- **General Term (T_n):** With first term a and common ratio r:
  $T_1 = a, T_2 = ar, T_3 = ar^2 \Longrightarrow T_n = a \times r^{n-1}$
- **Sum of n Terms (S_n):**
  $S_n = a + ar + ar^2 + ... + ar^{n-1}$
  $r \times S_n = ar + ar^2 + ... + ar^{n-1} + ar^n$
  Subtracting the two equations gives: $(1 - r)\, S_n = a(1 - r^n) \Longrightarrow S_n = a(r^n - 1) / (r - 1)$ for $r \neq 1$.
- **Sum of an Infinite GP (S_∞):** If $|r| < 1$, as $n \to \infty$, $r^n \to 0$.
  $S_\infty = a / (1 - r)$. (If $|r| \geq 1$, the infinite series diverges).
- **Product Symmetry Property:** In a finite GP, the product of terms equidistant from the beginning and end is constant:
  $T_1 \times T_n = T_2 \times T_{n-1} = T_k \times T_{n-k+1} = a^2\, r^{n-1}$

Strategic Selection of Terms in GP

- **3 Terms in GP:** $a/r, a, ar$ ➔ Product = a^3
- **4 Terms in GP:** $a/r^3, a/r, ar, ar^3$ ➔ Product = a^4, common ratio = r^2
- **5 Terms in GP:** $a/r^2, a/r, a, ar, ar^2$ ➔ Product = a^5

Geometric Mean (GM) Theory

- **GM between Two Positive Numbers:** $G = \sqrt{(a \times b)}$.
- **Insertion of n Geometric Means:** Inserting $G_1, G_2, ..., G_n$ between a and b yields sequence with $n + 2$ terms.
  Last term $b = a \times r^{n+1} \Longrightarrow r = (b / a)^{1 / (n + 1)}$.
  Product of n inserted GMs: $\prod_{k=1}^{n} G_k = (a \times b)^{n / 2} = [\, GM(a, b)\,]^n$.

4. Harmonic Progression (HP) & Harmonic Mean

A sequence $h_1, h_2, ..., h_n$ is in **Harmonic Progression** if their reciprocals $1/h_1, 1/h_2, ..., 1/h_n$ form an Arithmetic Progression.

HP CHARACTERISTICS

- **No Direct Sum Formula:** There is no direct general algebraic formula for the sum of n terms of an HP. Questions are solved by converting to AP.
- **General Term:** $T_n = 1 / [\, a + (n - 1)d\,]$, where $1/a$ is first term and d is common difference of reciprocal AP.
- **Harmonic Mean (HM):** HM of a and b is $H = 2ab / (a + b)$.

- **Harmonic Mean of n numbers:** $HM = n / [\ 1/x_1 + 1/x_2 + \ldots + 1/x_n\ ]$.

5. The Fundamental AM-GM-HM Inequality Chain

For any collection of n positive real numbers $x_1, x_2, \ldots, x_n$:
$AM \geq GM \geq HM$

RIGOROUS INEQUALITY FRAMEWORK

$$(x_1 + x_2 + \ldots + x_n) / n \geq (x_1 \times x_2 \times \ldots \times x_n)^{1/n} \geq n / (1/x_1 + 1/x_2 + \ldots + 1/x_n)$$

- **Equality Condition:** $AM = GM = HM$ if and only if $x_1 = x_2 = \ldots = x_n$.
- **Relation for Two Numbers:** For any two positive real numbers a and b:
  $GM^2 = AM \times HM \Longrightarrow (\sqrt{(ab)})^2 = [\ (a + b)/2\ ] \times [\ 2ab/(a + b)\ ] = ab$.
  Thus, AM, GM, HM of two positive numbers themselves form a Geometric Progression!

6. Arithmetico-Geometric Progression (AGP)

An **Arithmetico-Geometric Progression** is formed by multiplying corresponding terms of an AP and a GP:
$S_n = a + (a + d)r + (a + 2d)r^2 + (a + 3d)r^3 + \ldots + [a + (n - 1)d]r^{n-1}$

DERIVATION OF AGP SUM (SHIFT-AND-SUBTRACT METHOD)

Multiply S_n by common ratio r and shift terms one position right:

$$S_n = a + (a + d)r + (a + 2d)r^2 + \ldots + [a + (n - 1)d]r^{n-1}$$
$$r\,S_n = \quad ar \quad + (a + d)r^2 + \ldots + [a + (n - 2)d]r^{n-1} + [a + (n - 1)d]r^n$$

Subtracting the two equations:
$(1 - r)\,S_n = a + d(r + r^2 + r^3 + \ldots + r^{n-1}) - [a + (n - 1)d]r^n$
Sum of internal GP $= d \times r(1 - r^{n-1}) / (1 - r)$.

Infinite AGP Sum ($|r| < 1$): As $n \to \infty$, terms with $r^n \to 0$:
$S_\infty = a / (1 - r) + (d \times r) / (1 - r)^2$

7. Special Series & Telescoping Techniques

STANDARD NATURAL NUMBER SUMMATION FORMULAS

- $\sum_{k=1}^{n} k = 1 + 2 + 3 + \ldots + n = n(n+1)/2$
- $\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 = n(n+1)(2n+1)/6$
- $\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3 = [\, n(n+1)/2 \,]^2 = (\sum k)^2$
- $\sum_{k=1}^{n} (2k-1) = 1 + 3 + 5 + \ldots + (2n-1) = n^2$ (Sum of first n odd numbers)
- $\sum_{k=1}^{n} (2k) = 2 + 4 + 6 + \ldots + 2n = n(n+1)$ (Sum of first n even numbers)

The Method of Differences & Telescoping Sums

When the general term T_k of a series can be expressed as a difference of two consecutive function terms $T_k = V_k - V_{k+1}$ (or $V_{k+1} - V_k$), summing over k causes intermediate terms to cancel completely:
$S_n = (V_1 - V_2) + (V_2 - V_3) + \ldots + (V_n - V_{n+1}) = V_1 - V_{n+1}$

SHORTCUT: PARTIAL FRACTION FACTORIAL RULE FOR FRACTIONS

For terms like $T_k = 1 / [\, k(k+1)(k+2) \ldots (k+r-1) \,]$ (product of r terms in AP):
Split by subtracting first factor from last factor in numerator:
$T_k = [\, 1/(r-1) \,] \times [\, 1 / \{k(k+1)\ldots(k+r-2)\} - 1 / \{(k+1)(k+2)\ldots(k+r-1)\} \,]$

8. Common Pitfalls & High-Yield Exam Strategies

- **Dividing by zero in GP common ratio:** Assuming $S_n = a(r^n-1)/(r-1)$ holds for $r = 1$. When $r = 1$, all terms are equal to a, so $S_n = n \times a$.
- **Unchecked AM-GM Positivity Condition:** Applying AM ≥ GM to expressions containing variables that can take negative real values.
- **Misidentifying Number of Terms (n):** In series starting from T_0 or T_2, using n incorrectly in $a + (n-1)d$. Always verify term count via $n = (\text{Last} - \text{First})/d + 1$.

9. CAT-Level Solved Practice Questions

Question 1 (AP Sum Ratio & Term Ratio Conversion)

The ratio of the sum of m terms to the sum of n terms of an AP is $m^2 : n^2$. Find the ratio of its k-th term to its p-th term.

Solution:

Given: $S_m / S_n = m^2 / n^2$.

Using formula: $[(m/2)\{2a + (m - 1)d\}] / [(n/2)\{2a + (n - 1)d\}] = m^2 / n^2$.
Simplify by dividing numerators and denominators by m and n respectively:
$[2a + (m - 1)d] / [2a + (n - 1)d] = m / n$ --- (Eq 1).
We want to find the ratio of k-th term to p-th term: $T_k / T_p = [a + (k - 1)d] / [a + (p - 1)d]$.
Multiply numerator and denominator by 2 to match structure of Eq 1:
$T_k / T_p = [2a + (2k - 2)d] / [2a + (2p - 2)d]$.
Compare this with Eq 1: Substitute $m = 2k - 1$ and $n = 2p - 1$ into Eq 1:
$T_k / T_p = (2k - 1) / (2p - 1)$.
The required ratio is **$(2k - 1) : (2p - 1)$**.

Question 2 (Infinite AGP Derivation)

Find the sum of the infinite series: $S = 2 + 5/3 + 8/9 + 11/27 + 14/81 + \ldots$

Solution:

Identify component series:
Numerators: $2, 5, 8, 11, 14, \ldots$ (AP with $a = 2$, $d = 3$).
Denominators: $1, 3, 9, 27, 81, \ldots$ (GP with common ratio $r = 1/3$).
Apply infinite AGP formula: $S_\infty = a / (1 - r) + (d \times r) / (1 - r)^2$.
First term component: $2 / (1 - 1/3) = 2 / (2/3) = 3$.
Difference component: $(3 \times 1/3) / (1 - 1/3)^2 = 1 / (2/3)^2 = 1 / (4/9) = 9/4 = 2.25$.
Total Sum $S = 3 + 9/4 = 21 / 4 = 5.25$.

Question 3 (Complex Telescoping Partial Fraction)

Evaluate the sum to n terms: $S_n = 1/(1\times4) + 1/(4\times7) + 1/(7\times10) + \ldots + 1/[(3n - 2)(3n + 1)]$.

Solution:

General term is $T_k = 1 / [(3k - 2)(3k + 1)]$.
Notice difference between factors in denominator: $(3k + 1) - (3k - 2) = 3$.
Rewrite T_k by multiplying and dividing by 3:
$T_k = (1/3) \times [(3k + 1) - (3k - 2)] / [(3k - 2)(3k + 1)] = (1/3) \times [1/(3k - 2) - 1/(3k + 1)]$.
Summing from $k = 1$ to n:
. Intermediate terms cancel completely: .

$S_n = (1/3) \times [\ (1/1 - 1/4) + (1/4 - 1/7) + (1/7 - 1/10) + \ldots + (1/(3n-2) - 1/(3n+1))\ ]$

$S_n = (1/3) \times [\ 1 - 1/(3n+1)\ ] = (1/3) \times [\ (3n+1-1)/(3n+1)\ ] = n/(3n+1)$

10. Quick-Revision Cheat Sheet

Progression	n-th Term (T_n)	Sum Formula (S_n)	Mean Definition
AP	$a + (n-1)d$	$(n/2)[2a + (n-1)d]$	$AM = (a+b)/2$
GP	$a \times r^{n-1}$	$a(r^n - 1)/(r-1)$ | $S_\infty = a/(1-r)$	$GM = \sqrt{(ab)}$
HP	$1/[a + (n-1)d]$	Convert to AP reciprocals	$HM = 2ab/(a+b)$
AGP Infinite	$[a + (n-1)d]\ r^{n-1}$	$S_\infty = a/(1-r) + (dr)/(1-r)^2$	N/A
Sum of Squares	n^2	$n(n+1)(2n+1)/6$	N/A
Sum of Cubes	n^3	$[\ n(n+1)/2\ ]^2$	N/A

MODULE 3: NUMBER SYSTEM

Chapter 14: Divisibility Rules, Prime Numbers & Properties of Numbers

1. Classification & Structure of Real Numbers

The real number system ($\mathbb{R}$) is divided into two mutually exclusive sets: **Rational Numbers ($\mathbb{Q}$)**, which can be expressed as a fraction p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$, and **Irrational Numbers ($\mathbb{Q}'$)**, whose decimal expansions are non-terminating and non-repeating.

TAXONOMY OF NUMBERS

- **Natural Numbers ($\mathbb{N}$):** Counting numbers $\{1, 2, 3, 4, \ldots\}$.
- **Whole Numbers ($\mathbb{W}$):** $\{0, 1, 2, 3, 4, \ldots\}$.
- **Integers ($\mathbb{Z}$):** $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$.
- **Prime Numbers:** Natural numbers greater than 1 that have exactly two distinct positive divisors: 1 and themselves.
- **Composite Numbers:** Natural numbers greater than 1 having more than two positive divisors. (Note: 1 is neither prime nor composite).
- **Co-prime / Relatively Prime Numbers:** Pair of integers a, b such that $\gcd(a, b) = 1$. They need not be individual primes (e.g., 8 and 15).

2. Deep Derivation of Divisibility Rules

Divisibility rules are derived directly from place-value modular arithmetic:

$N = a_n 10^n + a_{n-1} 10^{n-1} + \ldots + a_1 10^1 + a_0 10^0$

DIVISIBILITY RULES MECHANICS

- **Divisibility by 2^k (2, 4, 8, 16, ...):** Since $10^k \equiv 0 \pmod{2^k}$, divisibility depends entirely on the last k digits of N.
  • By 2: Last digit divisible by 2.
  • By 4: Last 2 digits form a number divisible by 4.
  • By 8: Last 3 digits form a number divisible by 8.
- **Divisibility by 5^k (5, 25, 125, ...):** Same as powers of 2; depends on last k digits.
- **Divisibility by 3 and 9:** Since $10 \equiv 1 \pmod 3$ and $10 \equiv 1 \pmod 9$, $10^k \equiv 1 \pmod 9$.
  $N \equiv (a_n + a_{n-1} + \ldots + a_1 + a_0) \pmod 9$. A number is divisible by 3 or 9 iff the sum of its digits is divisible by 3 or 9 respectively.
- **Divisibility by 11:** Since $10 \equiv -1 \pmod{11}$, $10^k \equiv (-1)^k \pmod{11}$.
  $N \equiv (a_0 - a_1 + a_2 - a_3 + \ldots) \pmod{11}$.
  A number is divisible by 11 iff the difference between the sum of digits at odd places and even places is a multiple of 11.

Universal Osculator Method (Divisibility for 7, 13, 17, 19, 23, 29)

To test divisibility of N by prime P, find an osculator multiplier k such that $10k \pm 1$ is a multiple of P:

Prime (P)	Osculator Equation	Osculator Multiplier (k)	Operation on Last Digit (d) and Remaining Number (R)
7	$10 \times 2 + 1 = 21$	-2 (Negative)	$R - 2d$
13	$10 \times 4 - 1 = 39$	$+4$ (Positive)	$R + 4d$
17	$10 \times 5 + 1 = 51$	-5 (Negative)	$R - 5d$
19	$10 \times 2 - 1 = 19$	$+2$ (Positive)	$R + 2d$
23	$10 \times 7 - 1 = 69$	$+7$ (Positive)	$R + 7d$
29	$10 \times 3 - 1 = 29$	$+3$ (Positive)	$R + 3d$

3. Prime Numbers & Fundamental Theorem of Arithmetic

FUNDAMENTAL THEOREM OF ARITHMETIC

Every integer $N > 1$ can be uniquely represented as a product of prime powers, up to the order of factors:
$N = p_1^a \times p_2^b \times p_3^c \dots \times p_k^{k_n}$
where $p_1, p_2, \dots, p_k$ are distinct prime numbers and $a, b, c, \dots$ are positive integer exponents.

Properties & Distribution of Primes

- **Forms of Prime Numbers:** All prime numbers greater than 3 can be written in the form $6k \pm 1$ (where $k \in \mathbb{N}$).
  Note: Converse is not true! Not all numbers of form $6k \pm 1$ are prime (e.g., 25, 35).
- **Testing Primality of a Number N:**
  1. Find approximate integer value $k = \lfloor \sqrt{N} \rfloor$.
  2. Test divisibility of N by all prime numbers less than or equal to k.
  3. If N is not divisible by any of these primes, N is prime; otherwise, it is composite.
- **Primes Density under 100:** There are exactly 25 prime numbers between 1 and 100:
  $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97\}$.

4. Special Properties of Integers & Consecutive Products

KEY INTEGER PROPERTIES

- **Product of n Consecutive Integers:** Always divisible by $n!$.
  For example, $n(n+1)(n+2)$ is always divisible by $3! = 6$.
- **Square of a Prime:** If $p > 3$ is prime, $p^2 - 1$ is always divisible by 24.
  Proof sketch: $p^2 - 1 = (p-1)(p+1)$. Since p is odd, $p-1$ and $p+1$ are consecutive even numbers, so one is divisible by 4 and the other by 2 (divisible by 8). Since p is not divisible by 3, one of $(p-1)$ or $(p+1)$ must be divisible by 3. Total factor = $8 \times 3 = 24$.
- **Sum / Difference Powers Theorem:**
  - $a^n - b^n$ is always divisible by $(a-b)$ for all natural numbers n.
  - $a^n - b^n$ is divisible by $(a+b)$ for **even** natural numbers n.
  - $a^n + b^n$ is divisible by $(a+b)$ for **odd** natural numbers n.
  - $a^n + b^n$ is NEVER divisible by $(a-b)$ or $(a+b)$ for even n.

5. Common Pitfalls & Mistakes

- **Assuming $6k \pm 1$ implies Primality:** Treating every number of form $6k \pm 1$ as prime without testing factors (e.g., $6(6) + 1 = 37$ is prime, but $6(6) - 1 = 35$ is composite).
- **Overlooking 2 as a Prime:** Forgetting that 2 is the ONLY even prime number and the smallest prime number.
- **Misapplying Power Algebraic Divisibility:** Claiming $x^n + y^n$ is divisible by $x + y$ when n is even. It holds strictly when n is odd!

6. CAT-Level Solved Practice Questions

Question 1 (Composite Divisibility & Missing Digits)

If the 8-digit number $342x18y6$ is completely divisible by 72, find the maximum value of $2x + 3y$.

Solution:

For divisibility by $72 = 8 \times 9$ (where $\gcd(8, 9) = 1$), the number must be divisible by both 8 and 9.
Condition 1: Divisibility by 8
The last 3 digits $8y6$ must be divisible by 8.
Test single digit values for y (0 to 9):
• $y = 1 \Longrightarrow 816 / 8 = 102$ (Valid).
• $y = 5 \Longrightarrow 856 / 8 = 107$ (Valid).
• $y = 9 \Longrightarrow 896 / 8 = 112$ (Valid).
Possible values of y are $\{1, 5, 9\}$.
Condition 2: Divisibility by 9
Sum of digits $= 3 + 4 + 2 + x + 1 + 8 + y + 6 = 24 + x + y$.
For $24 + x + y$ to be divisible by 9, $x + y$ must be 3 or 12.
To maximize $2x + 3y$, evaluate valid pairs (y, x):
• If $y = 9$: $x + 9 = 12 \Longrightarrow x = 3$. Value $= 2(3) + 3(9) = 6 + 27 = 33$.
• If $y = 5$: $x + 5 = 12 \Longrightarrow x = 7$. Value $= 2(7) + 3(5) = 14 + 15 = 29$.
• If $y = 1$: $x + 1 = 3 \Longrightarrow x = 2$ (Value = 7) or $x + 1 = 12 \Longrightarrow x = 11$ (invalid digit).
Maximum value of $2x + 3y$ is **33**.

Question 2 (Primality & Algebraic Factors)

For how many positive integer values of n is $n^4 - 16n^2 + 100$ a prime number?

Solution:

Rewrite the quadratic expression in n^2 by completing the square:
$n^4 - 16n^2 + 100 = (n^4 + 20n^2 + 100) - 36n^2$
$= (n^2 + 10)^2 - (6n)^2$. Apply difference of squares : . For to be prime, one of the two factors MUST equal 1 (or -1), and the other factor must be a prime. Since is a positive integer, . So we must have : . Substitute into the second factor: Second factor = (which is prime). Thus, , which is prime. There is exactly positive integer value for (which is).

$A^2 - B^2 = (A - B)(A + B)$

$P(n) = (n^2 - 6n + 10)(n^2 + 6n + 10)$

$P(n)$

$nn^2 + 6n + 10 > 1$

$n^2 - 6n + 10 = 1$

$n^2 - 6n + 9 = 0 \Longrightarrow (n - 3)^2 = 0 \Longrightarrow n = 3$

$n = 3$

$3^2 + 6(3) + 10 = 9 + 18 + 10 = 37$

$P(3) = 1 \times 37 = 37$

$\mathbf{1}nn = 3$

Question 3 (Power Sum Divisibility)

Find the remainder when $S = 1^{99} + 2^{99} + 3^{99} + 4^{99} + \ldots + 100^{99}$ is divided by 101.

Solution:

Group terms from the ends in pairs:

$S = (1^{99} + 100^{99}) + (2^{99} + 99^{99}) + \ldots + (50^{99} + 51^{99})$.

Recall that for odd power n, $a^n + b^n$ is divisible by $a + b$.

Here power $n = 99$ (odd).

Every pair $(k^{99} + (101 - k)^{99})$ is divisible by $k + (101 - k) = 101$.

Since each of the 50 pairs is a multiple of 101, the entire sum S is divisible by 101.

The remainder is **0**.

7. Quick-Revision Cheat Sheet

Topic	Rule / Property	Key Takeaway
Divisibility 3 & 9	Sum of digits divisible by 3 or 9	$N \equiv \sum \text{digits} \pmod 9$
Divisibility 11	$(\sum \text{Odd Pos} - \sum \text{Even Pos})$ is multiple of 11	Alternating sign sum of digits.
Prime Form	$6k \pm 1$ for all primes $p > 3$	Necessary condition, not sufficient.
Primality Test	Test prime factors up to $\lfloor \sqrt{N} \rfloor$	If no prime $\le \sqrt{N}$ divides N, then N is prime.
Power Algebraic Sum	$a^n + b^n$ divisible by $a + b$ for odd n	Pair terms $k^n + (M-k)^n$ in symmetry problems.

MODULE 3: NUMBER SYSTEM

Chapter 15: Factors, Multiples, HCF & LCM

1. The Prime Factorization Architecture

Every natural number $N > 1$ can be written in prime canonical form:
$N = p_1^a \times p_2^b \times p_3^c \dots \times p_k^{k_n}$
where $p_1, p_2, \dots, p_k$ are prime factors and $a, b, c, \dots$ are non-negative integer exponents. Any divisor d of N takes the form $d = p_1^x \times p_2^y \times p_3^z \dots$ where $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$, etc.

FACTOR COUNTING & SUMMATION FORMULAS

- **Total Number of Factors $T(N)$:**
  $T(N) = (a + 1)(b + 1)(c + 1)\dots$
- **Sum of All Factors $S(N)$:**
  $S(N) = [(p_1^{a+1} - 1) / (p_1 - 1)] \times [(p_2^{b+1} - 1) / (p_2 - 1)] \times [(p_3^{c+1} - 1) / (p_3 - 1)] \dots$
- **Product of All Factors $P(N)$:**
  $P(N) = N^{T(N) / 2}$
- **Number of Pairs of Factors:**
  • If N is not a perfect square: $T(N) / 2$ pairs.
  • If N is a perfect square: $(T(N) + 1) / 2$ pairs (where $\sqrt{N} \times \sqrt{N}$ counts as a pair).
- **Number of Ways to Express N as Product of Two Co-prime Factors:**
  2^{k-1}, where k is the number of distinct prime factors of N.

2. Special Subsets of Factors

For $N = 2^a \times p_2^b \times p_3^c \dots$ (where p_2, p_3 are odd primes):

EVEN, ODD & POWER FACTORS

- **Number of Odd Factors:** Set power of 2 exponent to 0 (i.e. 1 choice):
  $T_{odd}(N) = (b + 1)(c + 1)\dots$
- **Number of Even Factors:** Exponent of 2 can range from 1 to a (a choices):
  $T_{even}(N) = a \times (b + 1)(c + 1)\dots = T(N) - T_{odd}(N)$
- **Perfect Square Factors:** Exponents must be even multiples of 2 (0, 2, 4, ...):
  $T_{sq}(N) = [\lfloor a/2 \rfloor + 1] \times [\lfloor b/2 \rfloor + 1] \times [\lfloor c/2 \rfloor + 1] \dots$
- **Perfect Cube Factors:** Exponents must be multiples of 3 (0, 3, 6, ...):
  $T_{cube}(N) = [\lfloor a/3 \rfloor + 1] \times [\lfloor b/3 \rfloor + 1] \times [\lfloor c/3 \rfloor + 1] \dots$

3. Euler's Totient Function $\phi(N)$

Euler's Totient Function $\phi(N)$ counts the number of positive integers less than or equal to N that are relatively prime (co-prime) to N.

TOTIENT PROPERTIES & SUM OF COPRIMES

- **Totient Formula:** $\phi(N) = N \times (1 - 1/p_1) \times (1 - 1/p_2) \times \ldots \times (1 - 1/p_k)$
- **Sum of Numbers Coprime to N and Less Than N:**
  $\text{Sum} = (N / 2) \times \phi(N)$

4. Highest Common Factor (HCF) & Lowest Common Multiple (LCM)

HCF (GCD): The largest positive integer that divides each of the numbers without a remainder. Take the *lowest exponent* of all common prime factors.
LCM: The smallest positive integer that is divisible by each of the numbers. Take the *highest exponent* of all prime factors present.

CORE IDENTITIES & FRACTION RULES

- **Product Identity (Two Numbers A, B):**
  $A \times B = HCF(A, B) \times LCM(A, B)$
  Note: This identity holds strictly for 2 numbers, NOT for 3 or more numbers!
- **Co-prime Representation:** Let $H = HCF(A, B)$. Then $A = H \times x$ and $B = H \times y$, where $\gcd(x, y) = 1$. Hence, $LCM(A, B) = H \times x \times y$.
- **HCF & LCM of Fractions:**
  • $HCF(a/b, c/d, e/f) = HCF(a, c, e) / LCM(b, d, f)$
  • $LCM(a/b, c/d, e/f) = LCM(a, c, e) / HCF(b, d, f)$
  Note: Fractions must be reduced to their simplest form before applying the formula!

5. Standard Word Problem Models for Remainders in HCF & LCM

Problem Type	Condition / Question Description	Mathematical Solution Formula
HCF Type 1	Largest number dividing A, B, C leaving remainders r_1, r_2, r_3 respectively.	$HCF(A - r_1, B - r_2, C - r_3)$
HCF Type 2	Largest number dividing A, B, C leaving the *same remainder r* (unspecified).	$HCF(\lvert A - B \rvert, \lvert B - C \rvert, \lvert C - A \rvert)$
LCM Type 1	Smallest number divided by A, B, C leaving the *same remainder r* in each case.	$LCM(A, B, C) \times k + r$ (Smallest for $k = 1$)
LCM Type 2	Smallest number divided by A, B, C leaving remainders r_1, r_2, r_3 such that $(A - r_1) = (B - r_2) = (C - r_3) = K$ (constant negative remainder).	$LCM(A, B, C) \times k - K$

6. Common Pitfalls & Mistakes

- **Applying $A \times B \times C = HCF \times LCM$:** Extending the 2-number product rule to 3 numbers. This is incorrect.
- **Failing to Simplify Fractions before HCF/LCM:** Computing $\text{HCF}(2/4, 3/6)$ without simplifying them to $1/2$ first, leading to invalid fractional answers.
- **Confusing Coprime Pairs with Prime Factors:** Counting ways to split N into co-prime factors using 2^{k-1}, where k is mistaken for total factors instead of *distinct prime factors*.

7. CAT-Level Solved Practice Questions

Question 1 (Factor Count & Perfect Squares)

Find the number of factors of $N = 2^5 \times 3^4 \times 5^3$ that are multiples of 12 but not multiples of 24.

Solution:

Prime factor representation: $N = 2^5 \times 3^4 \times 5^3$.
A factor $d = 2^x \times 3^y \times 5^z$ is a multiple of $12 = 2^2 \times 3^1$ if $x \geq 2$ and $y \geq 1$.
It is a multiple of $24 = 2^3 \times 3^1$ if $x \geq 3$ and $y \geq 1$.
Therefore, to be a multiple of 12 but NOT 24, we must have strictly $x = 2$ (exactly 1 choice for power of 2).
For power of 3 (y): $1 \leq y \leq 4 \Longrightarrow$ 4 choices ($y \in \{1, 2, 3, 4\}$).
For power of 5 (z): $0 \leq z \leq 3 \Longrightarrow$ 4 choices ($z \in \{0, 1, 2, 3\}$).
Total valid factors = $1 \times 4 \times 4 = 16$ factors.

Question 2 (Totient & Coprime Sum Application)

Find the sum of all natural numbers less than 360 that are coprime to 360.

Solution:

Step 1: Prime factorize 360:
$360 = 2^3 \times 3^2 \times 5^1$.
Step 2: Calculate Euler's Totient $\phi(360)$:
$\phi(360) = 360 \times (1 - 1/2) \times (1 - 1/3) \times (1 - 1/5)$
$= 360 \times (1/2) \times (2/3) \times (4/5) = 360 \times (8 / 30) = 96$.
There are 96 numbers less than 360 that are coprime to 360.
Step 3: Calculate sum of these coprimes using formula $\text{Sum} = (N / 2) \times \phi(N)$:
$\text{Sum} = (360 / 2) \times 96 = 180 \times 96 = 17{,}280$.
The sum of all coprimes is **17,280**.

Question 3 (HCF & LCM System Pair Counting)

How many ordered pairs of positive integers (A, B) exist such that HCF(A, B) = 12 and LCM(A, B) = 720?

Solution:

Let $A = 12x$ and $B = 12y$, where x and y are positive integers such that $\gcd(x, y) = 1$.
Using $LCM(A, B) = HCF(A, B) \times x \times y$:
$720 = 12 \times x \times y \implies x \times y = 60$.
We need to find the number of coprime pairs (x, y) whose product is 60.
Prime factorization of 60: $60 = 2^2 \times 3^1 \times 5^1$.
Distinct prime factors $k = 3$ (which are 2, 3, 5).
Number of unordered coprime factor pairs = $2^{k-1} = 2^{3-1} = 2^2 = 4$ pairs.
Listing the coprime pairs (x, y):
1. $(1, 60)$ 2. $(3, 20)$ 3. $(4, 15)$ 4. $(5, 12)$.
Since ordered pairs (A, B) are requested, each pair gives 2 ordered arrangements:
Total ordered pairs = $4 \times 2 = 8$ ordered pairs.

8. Quick-Revision Cheat Sheet

Concept	Formula / Rule	Key Insight
Total Factors	$T(N) = (a + 1)(b + 1)(c + 1)...$	Add 1 to each prime exponent and multiply.
Factor Product	$P(N) = N^{T(N)/2}$	Square root of N raised to total factor count.
Totient $\phi(N)$	$N \prod (1 - 1/p_i)$	Counts coprimes less than N. Sum = $(N/2)\phi(N)$.
Coprime Pairs Product	2^{k-1} (Unordered)	k is count of *distinct prime factors*.
HCF/LCM Fractions	HCF(Fractions) = HCF(Num) / LCM(Den)	Always simplify fractions before calculating!

MODULE 3: NUMBER SYSTEM

Chapter 16: Remainders

1. Fundamental Principles of Modular Arithmetic

According to the Division Algorithm, for any dividend integer N and positive divisor D, there exist unique integers q (quotient) and r (remainder) such that:
$N = D \times q + r$, where $0 \le r < D$.

CORE MODULAR AXIOMS & NEGATIVE REMAINDERS

- **Additive Rule:** $(A + B) \bmod D = [\,(A \bmod D) + (B \bmod D)\,] \bmod D$.
- **Multiplicative Rule:** $(A \times B) \bmod D = [\,(A \bmod D) \times (B \bmod D)\,] \bmod D$.
- **Exponential Rule:** $(A^k) \bmod D = [\,(A \bmod D)^k\,] \bmod D$.
- **Negative Remainder Concept:** If a number leaves a positive remainder r when divided by D, its equivalent negative remainder is $r' = r - D$.
  Example: $14 \bmod 5 = +4$, which is equivalent to $4 - 5 = -1$. Negative remainders dramatically simplify calculations involving high powers (e.g., $(-1)^n$).
- **Cancellation & Scaling Rule:** If $(k \times A) / (k \times D)$ leaves remainder R', then the actual remainder of A / D is $R = R' / k$.
  Crucial Rule: Always re-multiply the final remainder by the canceled common factor k!

2. Binomial Expansion Theorem for Remainders

Expanding $(a + b)^n$ using the binomial theorem reveals that all terms except the last are divisible by a:
$(a + b)^n = a^n + nC1\ a^{n-1}\ b + \ldots + nC(n-1)\ a\ b^{n-1} + b^n$

BINOMIAL REMAINDER RULES

- $(ax + 1)^n \bmod a = 1^n = 1$
- $(ax - 1)^n \bmod a = (-1)^n$:
  • If n is **even**, remainder = $+1$.
  • If n is **odd**, remainder = $-1 \equiv (a - 1)$.
- $(a^n + b^n) \bmod (a + b) = 0$ for all **odd** natural numbers n.
- $(a^n - b^n) \bmod (a + b) = 0$ for all **even** natural numbers n.
- $(a^n - b^n) \bmod (a - b) = 0$ for **all** natural numbers n.

3. Euler's Totient Theorem & Fermat's Little Theorem

EULER'S GENERAL THEOREM

If N and a are positive co-prime integers ($\gcd(a, N) = 1$), then:
$a^{\phi(N)} \equiv 1 \pmod{N}$
where $\phi(N) = N \times \prod (1 - 1/p_i)$ is Euler's Totient Function.

Power Reduction Technique: To calculate $a^K \bmod N$, reduce exponent K modulo $\phi(N)$:
$a^K \bmod N = a^{(K \bmod \phi(N))} \bmod N$.

FERMAT'S LITTLE THEOREM (SPECIAL CASE FOR PRIME DIVISORS)

If p is a prime number and a is an integer not divisible by p ($\gcd(a, p) = 1$), then $\phi(p) = p - 1$:
$a^{p-1} \equiv 1 \pmod{p}$
Alternatively: $a^p \equiv a \pmod{p}$ for any integer a.

4. Wilson's Theorem for Factorial Remainders

Wilson's Theorem gives an exact congruency for factorials modulo prime numbers:

WILSON'S IDENTITIES

- For any prime number p:
  $(p - 1)! \equiv -1 \equiv (p - 1) \pmod{p}$
- Dividing both sides by $(p - 1)$:
  $(p - 2)! \equiv 1 \pmod{p}$
- Dividing further:
  $(p - 3)! \equiv -(p - 1)/2 \equiv (p + 1)/2 \pmod{p}$

5. Chinese Remainder Theorem (CRT)

The Chinese Remainder Theorem solves systems of simultaneous linear congruences with pairwise coprime moduli.

CRT METHODOLOGY

Suppose a number N satisfies:

$N \equiv r_1 \pmod{m_1}$
$N \equiv r_2 \pmod{m_2}$

where $\gcd(m_1, m_2) = 1$. Let $M = m_1 \times m_2$.

1. Write general solution for first congruence: $N = m_1 k + r_1$.
2. Substitute this expression into second congruence: $(m_1 k + r_1) \equiv r_2 \pmod{m_2}$.
3. Solve for smallest non-negative integer k.
4. Substitute k back to find minimum value N_0. The general solution is $N \equiv N_0 \pmod{M}$.

6. Cyclicity Method of Remainders

When finding $a^n \bmod D$ where $\gcd(a, D) \neq 1$, compute initial powers of $a \bmod D$ to identify the repeating cycle pattern.

- Find cycle period c such that $a^{k+c} \equiv a^k \pmod{D}$.
- Divide power n by cycle length c to determine position in the repeating cycle.

7. Common Pitfalls & Mistakes

- **Forgetting to Restore Canceled Factors:** Canceling a common factor k between numerator and denominator to simplify calculations, but failing to multiply the resulting remainder by k at the end.
- **Misapplying Fermat's Theorem to Composite Moduli:** Using $a^{D-1} \equiv 1 \pmod{D}$ when D is composite. Fermat's Theorem applies strictly when D is prime. Use Euler's Totient Theorem for composite moduli.
- **Ignoring Co-primality for Euler's Theorem:** Applying $a^{\phi(N)} \equiv 1 \pmod{N}$ when $\gcd(a, N) \neq 1$. Always ensure base and divisor are coprime before taking power modulo $\phi(N)$.

8. CAT-Level Solved Practice Questions

Question 1 (Euler's Totient Power Reduction & Nested Exponents)

Find the remainder when $32^{32^{32}}$ is divided by 7.

Solution:

Divisor $D = 7$ (prime), base $32 \equiv 4 \pmod{7}$.
Since $\gcd(4, 7) = 1$, apply Euler/Fermat theorem:
$\phi(7) = 6 \Longrightarrow 4^6 \equiv 1 \pmod{7}$.
We need to reduce exponent $E = 32^{32}$ modulo 6:
$E = 32^{32} \bmod 6$.
Notice $32 \equiv 2 \pmod{6}$. So $32^{32} \equiv 2^{32} \pmod{6}$.
Compute powers of 2 mod 6:
$2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 2, 2^4 \equiv 4 \pmod{6}$.
For all even exponents k, $2^k \equiv 4 \pmod{6}$.
Since exponent 32 is even, $32^{32} \equiv 4 \pmod{6}$.
Substitute reduced exponent back into original expression:
$32^{32^{32}} \equiv 4^4 \pmod{7}$.
$4^4 = 256$.
$256 \bmod 7 = 256 - 7 \times 36 = 256 - 252 = 4$.
The remainder is **4**.

Question 2 (Wilson's Theorem Application)

Find the remainder when 96! is divided by 97.

Solution:

Here $p = 97$, which is a prime number.
Applying Wilson's Theorem:
$(p - 1)! \equiv -1 \equiv (p - 1) \pmod{p}$
Substitute $p = 97$:
$(97 - 1)! \equiv -1 \pmod{97} \implies 96! \equiv 96 \pmod{97}$.
The remainder is **96** (or -1).

Question 3 (Chinese Remainder Theorem System)

Find the smallest positive integer N which leaves a remainder of 3 when divided by 5, and a remainder of 5 when divided by 7.

Solution:

We have simultaneous congruences:
$N \equiv 3 \pmod{5}$ --- (Eq 1)
$N \equiv 5 \pmod{7}$ --- (Eq 2)
From Eq 1, express N as $N = 5k + 3$ for integer k.
Substitute into Eq 2:
$5k + 3 \equiv 5 \pmod{7} \implies 5k \equiv 2 \pmod{7}$.
Multiply both sides by 3 (since $5 \times 3 = 15 \equiv 1 \pmod{7}$):
$15k \equiv 6 \pmod{7} \implies k \equiv 6 \pmod{7}$.
Smallest non-negative value for k is $k = 6$.
Substitute $k = 6$ back into N:
$N = 5(6) + 3 = 30 + 3 = 33$.
Check: $33 \bmod 5 = 3$, and $33 \bmod 7 = 5$.
Smallest positive integer N is **33**.

9. Quick-Revision Cheat Sheet

Theorem / Tool	Mathematical Formula	Condition / Applicability
Euler's Theorem	$a^{\phi(N)} \equiv 1 \pmod{N}$	$\gcd(a, N) = 1$
Fermat's Theorem	$a^{p-1} \equiv 1 \pmod{p}$	p is prime, $\gcd(a, p) = 1$
Wilson's Theorem	$(p - 1)! \equiv -1 \pmod{p}$	p is prime
Binomial Form	$(ax + 1)^n \equiv 1 \pmod{a}$	Any natural number n
CRT Combination	$N \equiv N_0 \pmod{m_1 m_2}$	$\gcd(m_1, m_2) = 1$

MODULE 3: NUMBER SYSTEM

Chapter 17: Factorials, Highest Power of Primes & Base Systems

1. Factorial Fundamentals & Highest Power of Primes (Legendre's Formula)

The factorial of a non-negative integer n, denoted by $n!$, is the product of all positive integers less than or equal to n:
$n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$ (with $0! = 1$).

> **LEGENDRE'S THEOREM (HIGHEST POWER OF A PRIME P IN $N!$)**
>
> The exponent of the highest power of a prime number p that divides $n!$, denoted $E_p(n!)$, is given by:
> $E_p(n!) = \lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \lfloor n/p^3 \rfloor + \ldots + \lfloor n/p^k \rfloor$
> where $p^k \le n$ and $\lfloor x \rfloor$ represents the greatest integer function (floor function).

Highest Power of a Composite Number C in $n!$

To find the highest power of a composite number $C = p_1^a \times p_2^b \ldots$ dividing $n!$:

1. Find the highest power of each prime factor p_i in $n!$ using Legendre's Formula: $E_{p_i}(n!)$.
2. Determine the constraint imposed by each prime component: $\lfloor E_{p_i}(n!) / a_i \rfloor$.
3. The highest power of composite C is the **minimum** among these constrained values:
   $E_C(n!) = \min(\lfloor E_{p_1}(n!)/a \rfloor, \lfloor E_{p_2}(n!)/b \rfloor, \ldots)$

2. Trailing Zeroes Analysis

A trailing zero in an integer is created by a factor of $10 = 2 \times 5$.

> **TRAILING ZEROES FORMULAS**
>
> - **Trailing Zeroes in $n!$:** In any factorial, the prime factor 2 occurs far more frequently than 5 ($E_2(n!) > E_5(n!)$).
>   Hence, number of trailing zeroes in $n!$ equals the power of 5:
> $Z(n!) = E_5(n!) = \lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
> - **Trailing Zeroes in Base b:** In base b, the number of trailing zeroes of $n!$ corresponds to the highest power of b that divides $n!$.

3. Base Systems Architecture & Conversions

A number system with base (or radix) b uses b distinct digits $(0, 1, \ldots, b-1)$. A number represented as $(a_k\, a_{k-1} \ldots a_1\, a_0)_b$ has the positional value:
$N = a_k \times b^k + a_{k-1} \times b^{k-1} + \ldots + a_1 \times b^1 + a_0 \times b^0$

CONVERSION ALGORITHMS

- **Base b to Base 10:** Expand the digits as a polynomial sum in powers of b.
- **Base 10 to Base b:** Successively divide the base 10 number by b, recording remainders from bottom to top (least significant to most significant digit).
- **Direct Conversion between Base b and Base b^k:** Group digits in sets of k starting from the right. (e.g., Binary to Octal $2^3 = 8$: group binary digits in sets of 3).

4. Modular Base Arithmetic & Divisibility in Base b

BASE B DIVISIBILITY AXIOMS

- **Divisibility by $(b - 1)$:** A number in base b is divisible by $(b - 1)$ if and only if the sum of its digits is divisible by $(b - 1)$. (Analogous to test for 9 in Base 10).
- **Divisibility by $(b + 1)$:** A number in base b is divisible by $(b + 1)$ if the alternating sum of its digits is divisible by $(b + 1)$. (Analogous to test for 11 in Base 10).
- **Divisibility by factors of base b:** Depends on last digit(s).

5. Common Pitfalls & High-Yield Strategies

SHORTCUT 1: LAST NON-ZERO DIGIT OF $N!$

To find the last non-zero digit of $n!$ in Base 10:
$L(n!) = [\ 2^{(n \bmod 5)} \times L(\ \lfloor n/5 \rfloor\ !\ ) \times (n \bmod 5)!\ ] \bmod 10$ (for $n \geq 5$).

- **Applying Legendre's Formula Directly to Composite Numbers:** Using $\lfloor n / 6 \rfloor + \lfloor n / 36 \rfloor$ to find highest power of 6 in $n!$. Legendre's formula holds ONLY for prime numbers! For composite numbers, compute prime powers (e.g., powers of 2 and 3) first.
- **Ignoring Exponent Powers in Composite Base:** Finding highest power of $12 = 2^2 \times 3$ in $n!$ by taking $E_2(n!)$ instead of $\lfloor E_2(n!) / 2 \rfloor$.
- **Base Conversion Digit Value Errors:** Writing base b representations using digits greater than or equal to b (e.g., using digit 7 in Base 7). Base b digits are strictly bounded in $[0, b - 1]$.

6. CAT-Level Solved Practice Questions

Question 1 (Highest Power of Composite Number in Factorial)

Find the highest power of 24 that completely divides $100!$.

Solution:

Step 1: Prime factorize 24:

$24 = 2^3 \times 3^1$.

Step 2: Find highest power of prime factor 2 in 100! using Legendre's Formula:

$E_2(100!) = \lfloor 100/2 \rfloor + \lfloor 100/4 \rfloor + \lfloor 100/8 \rfloor + \lfloor 100/16 \rfloor + \lfloor 100/32 \rfloor + \lfloor 100/64 \rfloor$

$E_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$.

Maximum contribution of 2 to forming $24 = 2^3 \times 3$ is $\lfloor 97 / 3 \rfloor = 32$.

Step 3: Find highest power of prime factor 3 in 100! using Legendre's Formula:

$E_3(100!) = \lfloor 100/3 \rfloor + \lfloor 100/9 \rfloor + \lfloor 100/27 \rfloor + \lfloor 100/81 \rfloor$

$E_3(100!) = 33 + 11 + 3 + 1 = 48$.

Maximum contribution of 3 is $\lfloor 48 / 1 \rfloor = 48$.

Step 4: Take the minimum of the two constraints:

$E_{24}(100!) = \min(32, 48) = 32$.

The highest power of 24 dividing 100! is **32**.

Question 2 (Trailing Zeroes in Non-Decimal Base)

Find the number of trailing zeroes when 60! is written in Base 12.

Solution:

In base 12, a trailing zero is formed by each factor of $12 = 2^2 \times 3^1$.

We need to find $E_{12}(60!)$.

Find $E_2(60!)$:

$E_2(60!) = \lfloor 60/2 \rfloor + \lfloor 60/4 \rfloor + \lfloor 60/8 \rfloor + \lfloor 60/16 \rfloor + \lfloor 60/32 \rfloor = 30 + 15 + 7 + 3 + 1 = 56$.

Constraint from power of 2: $\lfloor 56 / 2 \rfloor = 28$.

Find $E_3(60!)$:

$E_3(60!) = \lfloor 60/3 \rfloor + \lfloor 60/9 \rfloor + \lfloor 60/27 \rfloor = 20 + 6 + 2 = 28$.

Constraint from power of 3: $\lfloor 28 / 1 \rfloor = 28$.

Taking minimum: $E_{12}(60!) = \min(28, 28) = 28$.

Number of trailing zeroes in Base 12 is **28**.

Question 3 (Base System Equation & Conversion)

If $(34)_b \times (25)_b = (1126)_b$ in base b, find the value of base b.

Solution:

Convert all terms from base b to decimal (base 10):

$(34)_b = 3b + 4$. Set up equation: . Rearrange to cubic equation in : . Since digit '6' appears in , base must be strictly greater than 6 (). Test : . Test units digit check directly: (leaves digit 6 carry 2). Tens digit: . Re-evaluate expansion: . Let's test : LHS = . Test : . Wait, let's re-verify product . RHS . If was 3-digit: (No positive solution). Testing : , . Product = . Convert 475 to base 7: rem 6; rem 4; rem 2 $\Longrightarrow$. Testing : , . Product = . Convert 588 to base 8: rem 4; rem 1; rem 1 $\Longrightarrow$. Testing with correct RHS equation roots: gives exact match for . Thus, for equation , base .

$(25)_b = 2b + 5$

$(1126)_b = 1 \cdot b^3 + 1 \cdot b^2 + 2 \cdot b + 6$

$(3b + 4)(2b + 5) = b^3 + b^2 + 2b + 6$

$6b^2 + 15b + 8b + 20 = b^3 + b^2 + 2b + 6$

$6b^2 + 23b + 20 = b^3 + b^2 + 2b + 6$

b

$b^3 - 5b^2 - 21b - 14 = 0$

$(1126)_b bb \geq 7$

$b = 7$

$7^3 - 5(7^2) - 21(7) - 14 = 343 - 5(49) - 147 - 14 = 343 - 245 - 147 - 14 = 343 - 406 \neq 0$

$b = 74 \times 5 = 20 = 2 \times 7 + 6$

$3 \times 5 + 4 \times 2 + 2 = 15 + 8 + 2 = 25 = 3 \times 7 + 4 \neq 2$

$b^3 - 5b^2 - 21b - 14 = 0$

$b = 7343 - 245 - 147 - 14 = -63$

$b = 8$

$8^3 - 5(8^2) - 21(8) - 14 = 512 - 320 - 168 - 14 = 512 - 502 = 10 \neq 0$

$(3b+4)(2b+5) = 6b^2 + 23b + 20$

$(1126)_b = b^3 + b^2 + 2b + 6$

$(126)_bb^2 + 2b + 6 = 6b^2 + 23b + 20 \implies 5b^2 + 21b + 14 = 0$

$b = 7(34)_7 = 25(25)_7 = 1925 \times 19 = 475$

$475 / 7 = 6767 / 7 = 99 / 7 = 1(1246)_7$

$b = 8(34)_8 = 28(25)_8 = 2128 \times 21 = 588$

$588 / 8 = 7373 / 8 = 99 / 8 = 1(1114)_8$

$b = 7b = 7(34)_7 \times (25)_7 = (1246)_7$

$(34)_b \times (25)_b = (1246)_bb = 7$

7. Quick-Revision Cheat Sheet

Concept	Formula / Axiom	Key Takeaway
Legendre's Formula	$E_p(n!) = \sum \lfloor n / p^k \rfloor$	Applies strictly to prime numbers p.
Trailing Zeroes (Base 10)	$Z(n!) = E_5(n!)$	Count powers of 5 in $n!$.
Composite Power in $n!$	$\min(\lfloor E_{p_i}(n!) / a_i \rfloor)$	Evaluate bottleneck prime power constraint.
Base b Divisibility by $b-1$	Sum of digits divisible by $b - 1$	Direct generalization of Rule for 9 in Base 10.
Base Range Rule	Digits in Base $b \in [0, b - 1]$	Base b can never contain digit $\geq b$.

MODULE 4: GEOMETRY & MENSURATION

Chapter 18: Lines, Angles & Triangles

1. Fundamentals of Lines, Angles & Parallel Transversals

Geometry begins with intersecting and parallel lines. When two parallel lines $L_1 \parallel L_2$ are intersected by a transversal line T:

ANGLE RELATIONSHIPS FOR PARALLEL LINES

- **Corresponding Angles:** Equal to each other ($\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 3 = \angle 7$, $\angle 4 = \angle 8$).
- **Alternate Interior Angles:** Equal to each other ($\angle 3 = \angle 6$, $\angle 4 = \angle 5$).
- **Alternate Exterior Angles:** Equal to each other ($\angle 1 = \angle 8$, $\angle 2 = \angle 7$).
- **Consecutive Interior Angles (Co-interior):** Supplementary ($\angle 3 + \angle 5 = 180^\circ$, $\angle 4 + \angle 6 = 180^\circ$).

2. Fundamental Triangle Inequalities & Angle Theorems

For a triangle with side lengths a, b, c opposite to vertices A, B, C respectively:

TRIANGLE EXISTENCE & CLASSIFICATION LAWS

- **Triangle Inequality Theorem:** Sum of any two sides must be strictly greater than the third side:
  $a + b > c, \quad b + c > a, \quad c + a > b$
  Equivalent compact form: $|a - b| < c < a + b$.
- **Classification by Angles (assuming largest side is c):**
  - **Acute-angled Triangle:** $a^2 + b^2 > c^2$ (All angles < 90°).
  - **Right-angled Triangle:** $a^2 + b^2 = c^2$ (One angle = 90°).
  - **Obtuse-angled Triangle:** $a^2 + b^2 < c^2$ (One angle > 90°).
- **Exterior Angle Theorem:** An exterior angle of a triangle equals the sum of the two opposite interior angles ($\angle \text{Ext} = \angle A + \angle B$).
- **Angle-Side Scale Inequality:** Larger sides face larger angles ($a > b \iff \angle A > \angle B$).

3. Congruence & Similarity Principles

TESTS OF CONGRUENCE ($\Delta ABC \cong \Delta DEF$)

Two triangles are congruent if they are identical in shape and size: SAS, SSS, ASA/AAS, RHS.

SIMILARITY THEOREM ($\Delta ABC \sim \Delta DEF$)

Two triangles are similar if their corresponding angles are equal and corresponding sides are in proportion:

- **Ratio of Linear Dimensions:** If $\Delta ABC \sim \Delta DEF$ with scale factor k:
  $AB / DE = BC / EF = AC / DF = h_1 / h_2 = m_1 / m_2 = r_1 / r_2 = R_1 / R_2 = k$
- **Ratio of Areas:** $\text{Area}(\Delta ABC) / \text{Area}(\Delta DEF) = k^2 = (AB / DE)^2$.
- **Thales' Theorem (Basic Proportionality Theorem):** If a line is drawn parallel to one side of a triangle intersecting the other two sides, it divides those sides in the same ratio:
  If $DE \parallel BC$, then $AD / DB = AE / EC$ and $\Delta ADE \sim \Delta ABC$.

4. The Five Centers of a Triangle

Center	Point of Concurrence Of	Key Formulas & Geometric Properties
Incenter (I)	Internal Angle Bisectors	• Center of inscribed circle (Inradius $r = \text{Area} / s$, where $s = (a+b+c)/2$). • $\angle BIC = 90^\circ + \angle A / 2$. • Angle Bisector Theorem: $BD / DC = c / b$.
Circumcenter (O)	Perpendicular Bisectors of Sides	• Center of circumscribed circle (Circumradius $R = abc / (4 \times \text{Area})$). • Angle subtended at center: $\angle BOC = 2\ \angle A$. • Lies inside (acute), on hypotenuse midpoint (right), outside (obtuse).
Centroid (G)	Medians (lines to side midpoints)	• Divides each median in ratio $2 : 1$ ($AG : GD = 2 : 1$). • Divides triangle into 6 smaller triangles of equal area. • Median length formula via Apollonius Theorem.
Orthocenter (H)	Altitudes (perpendiculars to sides)	• $\angle BHC = 180^\circ - \angle A$. • Lies inside (acute), at right-angle vertex (right), outside (obtuse).
Excenter (I_a)	One internal & two external angle bisectors	• Exradius $r_a = \text{Area} / (s - a)$. • $\angle BI_aC = 90^\circ - \angle A / 2$.

EULER'S LINE & DISTANCE THEOREM

- In any triangle, the Orthocenter (H), Centroid (G), and Circumcenter (O) are collinear on **Euler's Line**.
- Centroid divides HO in ratio $HG : GO = 2 : 1$.
- Distance between Circumcenter O and Incenter I: $OI^2 = R^2 - 2Rr$ (Euler's Distance Formula).

5. Advanced Metric Theorems (Apollonius, Stewart & Ceva)

ESSENTIAL ADVANCED EQUATIONS

- **Apollonius' Theorem (Median Length):** If AD is the median to side BC:
  $AB^2 + AC^2 = 2 \times (AD^2 + BD^2) = 2 \times (AD^2 + (BC/2)^2)$
- **Stewart's Theorem (General Cevian Length):** If d is a cevian from A dividing BC into segments m (adjacent to c) and n (adjacent to b), where $m + n = a$:
  $b^2 m + c^2 n = a (d^2 + m n)$
- **Ceva's Theorem:** Three cevians AD, BE, CF are concurrent if and only if:
  $(AF / FB) \times (BD / DC) \times (CE / EA) = 1$

- **Mass Point Geometry (MPG):** Assign weights to vertices such that mid-point system balances: $w_A \times AF = w_B \times FB$. Total weight at $F = w_A + w_B$.

6. Common Pitfalls & Mistakes

- **Confusing Incenter and Circumcenter Angle Formulas:** Using $2\angle A$ for Incenter ($\angle BIC$) instead of $90° + \angle A/2$. Remember $2\angle A$ applies to Circumcenter ($\angle BOC$)!
- **Misapplying Similarity Ratios to Areas:** Stating that if side ratio is $2:3$, area ratio is $2:3$. Area ratio is the *square* of side ratio ($4:9$).
- **Assuming Right-Angle Vertex is Incenter:** For a right triangle, the orthocenter is at the 90° vertex, and circumcenter is at the hypotenuse midpoint. The incenter lies strictly inside the triangle.

7. CAT-Level Solved Practice Questions

Question 1 (Mass Point Geometry / Cevian Ratios)

In ΔABC, D is a point on BC such that $BD : DC = 3 : 2$, and E is a point on AC such that $AE : EC = 1 : 4$. Let AD and BE intersect at P. Find the ratios $AP : PD$ and $BP : PE$.

Solution:

Assign mass points to vertices A, B, C using Mass Point Geometry:
On segment BC: $BD : DC = 3 : 2$. To balance D, set $3 \times \text{Mass}(B) = 2 \times \text{Mass}(C)$.
Let $\text{Mass}(B) = 2$ and $\text{Mass}(C) = 3 \Longrightarrow \text{Mass}(D) = 2 + 3 = 5$.
On segment AC: $AE : EC = 1 : 4$. To balance E, set $1 \times \text{Mass}(A) = 4 \times \text{Mass}(C)$.
Since $\text{Mass}(C) = 3$, we get $\text{Mass}(A) = 4 \times 3 = 12 \Longrightarrow \text{Mass}(E) = 12 + 3 = 15$.
Now find P on line segment AD:
$AP : PD = \text{Mass}(D) : \text{Mass}(A) = 5 : 12$.
Find P on line segment BE:
$BP : PE = \text{Mass}(E) : \text{Mass}(B) = 15 : 2$.
Therefore, **AP : PD = 5 : 12** and **BP : PE = 15 : 2**.

Question 2 (Apollonius' Theorem Application)

In ΔABC, $AB = 7$ cm, $AC = 9$ cm, and $BC = 8$ cm. Find the length of the median AD drawn to side BC.

Solution:

Since AD is the median, D is the midpoint of BC, so $BD = DC = BC / 2 = 8 / 2 = 4$ cm.
Apply Apollonius' Theorem:
$AB^2 + AC^2 = 2 \times (AD^2 + BD^2)$
Substitute known lengths:
$7^2 + 9^2 = 2 \times (AD^2 + 4^2)$
$49 + 81 = 2 \times (AD^2 + 16)$
$130 = 2 \times (AD^2 + 16) \Longrightarrow 65 = AD^2 + 16$ cm. The length of median is .

$AD^2 = 65 - 16 = 49 \Longrightarrow AD = \sqrt{49} = 7$
AD**7 cm**

Question 3 (Inradius & Right Triangle Geometry)

A right-angled triangle has sides of length 15 cm, 20 cm, and 25 cm. Find the distance between its Orthocenter and Incenter.

Solution:

Step 1: Identify triangle centers:
In a right triangle with legs $a = 15$, $b = 20$ and hypotenuse $c = 25$:
The **Orthocenter H** is situated exactly at the right-angle vertex $(0, 0)$.
Step 2: Calculate Inradius r:
For a right triangle, $r = (a + b - c) / 2 = (15 + 20 - 25) / 2 = 10 / 2 = 5$ cm.
Step 3: Distance from right-angle vertex to Incenter:
The Incenter I is at coordinates $(r, r) = (5, 5)$ relative to the vertex at origin $(0, 0)$.
Distance $HI = \sqrt{(r^2 + r^2)} = r\sqrt{2} = 5\sqrt{2}$ cm.
The distance between the Orthocenter and Incenter is **$5\sqrt{2}$ cm**.

8. Quick-Revision Cheat Sheet

Concept	Key Formula / Rule	Key Takeaway
Triangle Inequality	$\|a - b\| < c < a + b$	Condition for triangle to exist.
Area Scale Factor	$\text{Area}_1 / \text{Area}_2 = k^2$	Square of side ratio for similar triangles.
Inradius & Circumradius	$r = \text{Area} / s$, $R = abc / (4 \times \text{Area})$	Right triangle: $r = (a + b - c)/2$, $R = c/2$.
Apollonius Theorem	$b^2 + c^2 = 2(m^2 + (a/2)^2)$	Relates sides to median length m.
Euler's Line	$HG : GO = 2 : 1$	Orthocenter, Centroid, Circumcenter are collinear.

MODULE 4: GEOMETRY & MENSURATION

Chapter 19: Quadrilaterals, Polygons & Circles

1. Hierarchy & Properties of Quadrilaterals

A quadrilateral is a 4-sided polygon whose interior angles sum to 360°. Its special forms follow strict hierarchy:

CORE QUADRILATERAL METRIC FORMULAS

- **General Quadrilateral:** Area = $(1/2) \times d \times (h_1 + h_2)$, where d is a diagonal and h_1, h_2 are perpendicular offsets from opposite vertices. If diagonals intersect at angle θ, Area = $(1/2) \times d_1\, d_2 \sin\theta$.
- **Parallelogram:** Opposite sides equal & parallel. Diagonals bisect each other.
  Area = $\text{base} \times \text{height} = a\, b \sin\theta$. Parallelogram Law: $2(a^2 + b^2) = d_1^2 + d_2^2$.
- **Rhombus:** All sides equal, diagonals bisect each other at right angles (90°).
  Area = $(1/2) \times d_1\, d_2$. Side length $a = (1/2)\sqrt{(d_1^2 + d_2^2)}$.
- **Trapezium:** One pair of opposite sides parallel ($a \parallel b$).
  Area = $(1/2) \times (a + b) \times h$. Line connecting midpoints of non-parallel sides = $(a + b) / 2$.
- **Cyclic Quadrilateral:** Vertices lie on a circle. Opposite angles are supplementary ($\angle A + \angle C = 180°$).
  • **Brahmagupta's Formula:** Area = $\sqrt{[\,(s-a)(s-b)(s-c)(s-d)\,]}$, where $s = (a + b + c + d) / 2$.
  • **Ptolemy's Theorem:** $d_1 \times d_2 = a\, c + b\, d$ (Product of diagonals equals sum of products of opposite sides).

2. Polygons — Angles, Diagonals & Regular Metrics

GENERAL & REGULAR POLYGON FORMULAS (N SIDES)

- **Sum of Interior Angles:** $S_{int} = (n - 2) \times 180°$
- **Sum of Exterior Angles:** $S_{ext} = 360°$ (for any convex polygon)
- **Number of Diagonals:** $N_{diag} = n(n - 3) / 2$
- **Each Interior Angle of Regular Polygon:** $I = [\,(n - 2) \times 180°\,] / n$
- **Each Exterior Angle of Regular Polygon:** $E = 360° / n$
- **Area of Regular Polygon with side a:**
  Area = $(n / 4) \times a^2 \times \cot(180° / n)$
- **Regular Hexagon ($n = 6$):** Composed of 6 congruent equilateral triangles of side a.
  Area = $6 \times [\,(\sqrt{3} / 4)\, a^2\,] = (3\sqrt{3} / 2)\, a^2$. Long diagonal = $2a$, short diagonal = $a\sqrt{3}$.

3. Circle Theorems, Chords & Secant Properties

CORE CIRCLE THEOREMS

- **Perpendicular from Center:** The perpendicular drawn from the center of a circle to a chord bisects the chord.
- **Angles Subtended by Arc:**
  • Angle subtended by an arc at center is twice the angle subtended at any point on the circumference: $\angle\text{Center} = 2\ \angle\text{Circumference}$.
  • Angles in the same segment of a circle are equal. Angle in a semicircle is a right angle (90°).
- **Intersecting Chords Theorem:** If two chords AB and CD intersect at point P (inside or outside circle): $PA \times PB = PC \times PD$
- **Tangent-Secant Theorem:** If secant PAB and tangent PT are drawn from external point P: $PT^2 = PA \times PB$
- **Alternate Segment Theorem:** The angle between a tangent and a chord through the point of contact equals the angle subtended by the chord in the alternate segment ($\angle BAT = \angle BCA$).

4. Common Tangents & Touch Conditions for Circles

For two circles with radii R and r whose centers are separated by distance d:

Relative Circle Position	Distance Condition (d)	Number of Common Tangents	Tangent Length Formulas
Disjoint (Non-intersecting)	$d > R + r$	4 (2 direct, 2 transverse)	Direct: $L_{DCT} = \sqrt{(d^2 - (R - r)^2)}$ Transverse: $L_{TCT} = \sqrt{(d^2 - (R + r)^2)}$
Touching Externally	$d = R + r$	3 (2 direct, 1 common vertical)	Direct: $L_{DCT} = 2\sqrt{(R \times r)}$
Intersecting at 2 points	$\lvert R - r\rvert < d < R + r$	2 (2 direct)	Common chord length = $2 \times \sqrt{(R^2 - x^2)}$
Touching Internally	$d = \lvert R - r\rvert$	1 (single common tangent)	N/A

5. Common Pitfalls & Mistakes

- **Applying Brahmagupta's Formula to Non-Cyclic Quadrilaterals:** Using $\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}$ for general quadrilaterals. It is valid ONLY when opposite angles sum to 180° (Cyclic).
- **Confusing Direct and Transverse Tangent Formulas:** Direct tangent uses $(R - r)^2$, while Transverse tangent uses $(R + r)^2$. Transverse tangents are always shorter than direct tangents.
- **Using Incorrect Hexagon Diagonals:** Confusing long diagonal ($2a$, joining opposite vertices through center) with short diagonal ($a\sqrt{3}$, joining alternate vertices).

6. CAT-Level Solved Practice Questions

Question 1 (Ptolemy's Theorem & Cyclic Quadrilateral)

A cyclic quadrilateral ABCD has side lengths AB = 3, BC = 4, CD = 4, and DA = 5. Find the product of the lengths of its diagonals AC × BD.

Solution:

Since ABCD is a cyclic quadrilateral, apply Ptolemy's Theorem:
AC × BD = (AB × CD) + (BC × DA).
Substitute the given side lengths:
AC × BD = (3 × 4) + (4 × 5)
AC × BD = 12 + 20 = 32.
The product of the diagonal lengths AC × BD is **32**.

Question 2 (Intersecting Secants & Tangents)

From an external point P, a tangent PT and a secant PAB are drawn to a circle. If PT = 12 cm and PA = 8 cm, find the length of chord AB.

Solution:

Apply the Tangent-Secant Theorem:
$PT^2 = PA \times PB$.
Substitute PT = 12 and PA = 8:
$12^2 = 8 \times PB \Longrightarrow 144 = 8 \times PB \Longrightarrow PB = 144 / 8 = 18$ cm.
Since PB = PA + AB:
$18 = 8 + AB \Longrightarrow AB = 18 - 8 = 10$ cm.
The length of chord AB is **10 cm**.

Question 3 (Regular Hexagon Inscribed Circle Area Ratio)

Find the ratio of the area of a regular hexagon to the area of its incircle (inscribed circle).

Solution:

Let side length of regular hexagon be a.
Area of regular hexagon = $(3\sqrt{3} / 2)\, a^2$.
The incircle touches the midpoints of the side of the hexagon.
Inradius r = height of an equilateral triangle of side a = $(\sqrt{3} / 2)\, a$.
Area of incircle = $\pi r^2 = \pi \times [\, (\sqrt{3} / 2)\, a \,]^2 = \pi \times (3/4)\, a^2 = (3\pi / 4)\, a^2$.
Ratio of Area(Hexagon) to Area(Incircle):
Ratio = $[\, (3\sqrt{3} / 2)\, a^2 \,] / [\, (3\pi / 4)\, a^2 \,] = (3\sqrt{3} / 2) \times (4 / 3\pi) = 2\sqrt{3} / \pi$.
The ratio of areas is **2√3 : π**.

7. Quick-Revision Cheat Sheet

Concept	Key Formula / Rule	Key Takeaway
Cyclic Quad Area	$\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}$	Valid strictly when $\angle A + \angle C = 180^\circ$.
Ptolemy Theorem	$d_1 d_2 = ac + bd$	Diagonal product equals sum of opposite side products.
Polygon Diagonals	$N_{diag} = n(n-3)/2$	Total diagonals in n-sided polygon.
Tangent Secant	$PT^2 = PA \times PB$	Product of secant segments equals tangent squared.
Direct Tangent	$L_{DCT} = \sqrt{(d^2 - (R-r)^2)}$	Transverse uses $(R+r)^2$ instead.

MODULE 4: GEOMETRY & MENSURATION

Chapter 20: Coordinate Geometry & Locus

1. Fundamental Distance & Section Mechanics

Coordinate geometry maps geometric figures into an algebraic Cartesian plane defined by orthogonal axes X and Y.

CORE DISTANCE & DIVISION FORMULAS

- **Distance Formula:** Distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$:
  $d = \sqrt{[\,(x_2 - x_1)^2 + (y_2 - y_1)^2\,]}$
- **Section Formula (Internal Division $m : n$):** Coordinates of point R dividing PQ internally:
  $R = (\,(m\,x_2 + n\,x_1)\,/\,(m + n),\ (m\,y_2 + n\,y_1)\,/\,(m + n)\,)$
- **Section Formula (External Division $m : n$):**
  $R = (\,(m\,x_2 - n\,x_1)\,/\,(m - n),\ (m\,y_2 - n\,y_1)\,/\,(m - n)\,)$
- **Midpoint Formula:** $M = (\,(x_1 + x_2)/2,\ (y_1 + y_2)/2\,)$
- **Centroid of Triangle:** $G = (\,(x_1 + x_2 + x_3)/3,\ (y_1 + y_2 + y_3)/3\,)$
- **Incenter of Triangle:** $I = (\,(a\,x_1 + b\,x_2 + c\,x_3)/(a + b + c),\ (a\,y_1 + b\,y_2 + c\,y_3)/(a + b + c)\,)$

2. Straight Lines — Equations, Slopes & Angular Metrics

The slope (inclination) of line passing through (x_1, y_1) and (x_2, y_2) is $m = \tan\theta = (y_2 - y_1)\,/\,(x_2 - x_1)$.

STANDARD LINE FORMS & ANGULAR RELATIONS

- **Slope-Intercept Form:** $y = mx + c$ (c is y-intercept).
- **Point-Slope Form:** $y - y_1 = m(x - x_1)$.
- **Two-Point Form:** $y - y_1 = [\,(y_2 - y_1)/(x_2 - x_1)\,]\,(x - x_1)$.
- **Intercept Form:** $x/a + y/b = 1$ (a, b are x and y-intercepts). Area under line = $(1/2)|ab|$.
- **General Linear Form:** $Ax + By + C = 0$. Slope $m = -A\,/\,B$.
- **Angle Between Two Lines (m_1, m_2):** $\tan\theta = |\,(m_1 - m_2)\,/\,(1 + m_1\,m_2)\,|$.
  - **Parallel Lines ($L_1 \parallel L_2$):** $m_1 = m_2$.
  - **Perpendicular Lines ($L_1 \perp L_2$):** $m_1 \times m_2 = -1$.

3. Perpendicular Distances & Polygon Area Mechanics

- **Perpendicular Distance from Point (x_0, y_0) to Line $Ax + By + C = 0$:**
  $p = |\,A\,x_0 + B\,y_0 + C\,|\,/\,\sqrt{(A^2 + B^2)}$
- **Distance Between Parallel Lines ($Ax + By + C_1 = 0$ & $Ax + By + C_2 = 0$):**
  $d = |\,C_1 - C_2\,|\,/\,\sqrt{(A^2 + B^2)}$
- **Shoelace Formula (Area of Polygon):** For vertices (x_1, y_1), (x_2, y_2), ..., (x_n, y_n):
  $\text{Area} = (1/2)\,|\,(x_1y_2 + x_2y_3 + \ldots + x_ny_1) - (y_1x_2 + y_2x_3 + \ldots + y_nx_1)\,|$

4. Circles in Coordinate Geometry

A circle is the locus of a point that moves such that its distance from a fixed point (center) remains constant (radius).

STANDARD & GENERAL EQUATIONS OF A CIRCLE

- **Standard Form:** Center (h, k) and radius R:
  $(x - h)^2 + (y - k)^2 = R^2$
- **General Equation:** $x^2 + y^2 + 2gx + 2fy + c = 0$.
  • Center = $(-g, -f)$
  • Radius = $R = \sqrt{(g^2 + f^2 - c)}$
- **Diametric Form:** Circle with endpoints of diameter (x_1, y_1) and (x_2, y_2):
  $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
- **Position of Point (x_1, y_1) relative to Circle $S = 0$:**
  Evaluate $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$:
  • $S_1 > 0 \Longrightarrow$ Outside • $S_1 = 0 \Longrightarrow$ On circle • $S_1 < 0 \Longrightarrow$ Inside

5. Locus Concept & Transformations

REFLECTION & TRANSFORMATION RULES

- **Reflection of Point (x, y):**
  • Across X-axis: $(x, -y)$ • Across Y-axis: $(-x, y)$ • Across Origin: $(-x, -y)$
  • Across line $y = x$: (y, x) • Across line $y = -x$: $(-y, -x)$
- **Reflection of Point (x_0, y_0) across Line $Ax + By + C = 0$:**
  Image (x, y) satisfies: $(x - x_0)/A = (y - y_0)/B = -2(A x_0 + B y_0 + C) / (A^2 + B^2)$.
- **Locus Derivation Procedure:**
  1. Assume coordinates of moving point as $P(h, k)$.
  2. Write given geometric conditions in mathematical form using h, k.
  3. Eliminate parameter variables to get an equation purely in h and k.
  4. Replace $h \rightarrow x$ and $k \rightarrow y$ to obtain the locus equation.

6. Common Pitfalls & Mistakes

- **Sign Errors in Perpendicular Line Slopes:** Taking slope of perpendicular line as $1/m$ instead of negative reciprocal $-1/m$.
- **Forgetting Absolute Value in Distance and Area Formulas:** Omitting magnitude bars in Shoelace formula or point-to-line distance, yielding negative areas or distances.
- **Inconsistent Coefficient Matching in Parallel Line Distance:** Using $|C_1 - C_2|/\sqrt{(A^2+B^2)}$ without ensuring coefficients of x and y are identical in both line equations.

7. CAT-Level Solved Practice Questions

Question 1 (Area of Triangle via Shoelace Formula)

Find the area of the triangle formed by the intersection of lines $y = x$, $y = 2x$, and $y = 3x - 4$.

Solution:

Step 1: Find vertices by solving pairwise line intersections:
• Intersection 1 ($y = x$ and $y = 2x$): $x = 2x \Rightarrow x = 0$, $y = 0$ ➔ Vertex $A(0, 0)$.
• Intersection 2 ($y = x$ and $y = 3x - 4$): $x = 3x - 4 \Rightarrow 2x = 4 \Rightarrow x = 2$, $y = 2$ ➔ Vertex $B(2, 2)$.
• Intersection 3 ($y = 2x$ and $y = 3x - 4$): $2x = 3x - 4 \Rightarrow x = 4$, $y = 8$ ➔ Vertex $C(4, 8)$.
Step 2: Apply Shoelace Formula for $(0, 0)$, $(2, 2)$, $(4, 8)$:
Area $= (1/2) \mid (0 \times 2 + 2 \times 8 + 4 \times 0) - (0 \times 2 + 2 \times 4 + 8 \times 0) \mid$
Area $= (1/2) \mid (0 + 16 + 0) - (0 + 8 + 0) \mid = (1/2) \mid 16 - 8 \mid = (1/2) \times 8 = 4$ square units.
The area of the triangle is **4 square units**.

Question 2 (Reflection of Point across a Line)

Find the reflection of the point $P(2, 3)$ across the line $x + 2y - 3 = 0$.

Solution:

Here $x_0 = 2$, $y_0 = 3$, $A = 1$, $B = 2$, $C = -3$.
Apply the reflection image formula:
$(x - 2)/1 = (y - 3)/2 = -2\,[\,(1(2) + 2(3) - 3) / (1^2 + 2^2)\,]$.
Calculate RHS factor:
Numerator $= 2 + 6 - 3 = 5$.
Denominator $= 1 + 4 = 5$.
Factor $= -2 \times (5 / 5) = -2$.
Solve for x and y:
$(x - 2)/1 = -2 \Rightarrow x - 2 = -2 \Rightarrow x = 0$.
$(y - 3)/2 = -2 \Rightarrow y - 3 = -4 \Rightarrow y = -1$.
The reflection of point $P(2, 3)$ across the line is **(0, −1)**.

Question 3 (Locus Derivation)

Find the locus of a point P which moves such that the sum of the squares of its distances from $A(1, 0)$ and $B(-1, 0)$ is always equal to 10.

Solution:

Let coordinates of moving point be $P(h, k)$.
Given condition: $PA^2 + PB^2 = 10$.
Express PA^2 and PB^2 using distance squared formula:
$PA^2 = (h - 1)^2 + (k - 0)^2 = h^2 - 2h + 1 + k^2$. . Add the two squared distances: . Set equal to 10: . Replace and : Locus equation: (which represents a circle centered at origin with radius 2).

$PB^2 = (h - (-1))^2 + (k - 0)^2 = (h + 1)^2 + k^2 = h^2 + 2h + 1 + k^2$

$PA^2 + PB^2 = (h^2 - 2h + 1 + k^2) + (h^2 + 2h + 1 + k^2) = 2h^2 + 2k^2 + 2$

$2h^2 + 2k^2 + 2 = 10 \Longrightarrow 2h^2 + 2k^2 = 8 \Longrightarrow h^2 + k^2 = 4$
$h \rightarrow xk \rightarrow y$
$x^2 + y^2 = 4$

8. Quick-Revision Cheat Sheet

Concept	Key Formula / Rule	Key Takeaway
Perpendicular Distance	$p = \lvert Ax_0 + By_0 + C\rvert \,/\, \sqrt{(A^2 + B^2)}$	Distance from point to line.
Parallel Lines Distance	$d = \lvert C_1 - C_2\rvert \,/\, \sqrt{(A^2 + B^2)}$	Ensure A and B coefficients match first.
Shoelace Area	$(1/2)\lvert(x_1y_2+\ldots) - (y_1x_2+\ldots)\rvert$	Polygon area from ordered coordinate vertices.
Circle Center/Radius	Center $(-g, -f)$, $R = \sqrt{(g^2 + f^2 - c)}$	For general form $x^2 + y^2 + 2gx + 2fy + c = 0$.
Perpendicular Slope	$m_1 \times m_2 = -1$	Negative reciprocal of original slope.

MODULE 4: GEOMETRY & MENSURATION

Chapter 21: 2D Mensuration & 3D Mensuration

1. 2D Mensuration Essentials & Arc Sector Dynamics

2D mensuration deals with calculating perimeters and bounded surface areas of two-dimensional closed plane figures.

SECTOR, SEGMENT & ANNULUS METRICS

- **Arc Length of Circle (l):** For central angle θ (in degrees):
  $l = (\theta / 360^\circ) \times 2\pi r$
- **Area of Sector (A_{sec}):** $A_{sec} = (\theta / 360^\circ) \times \pi r^2 = (1/2) \times l \times r$
- **Area of Segment:** $A_{seg} = \text{Area of Sector} - \text{Area of Triangle} = (\theta / 360^\circ)\, \pi r^2 - (1/2)\, r^2 \sin\theta$
- **Annulus (Ring) Area between Concentric Circles ($R > r$):**
  $\text{Area} = \pi(R^2 - r^2)$

2. 3D Mensuration — Standard Prismatic & Pyramidal Solids

Three-dimensional solids are broadly categorized into two structural families: **Prism-type (uniform cross-section)** and **Pyramid-type (tapering to an apex)**.

Solid Figure	Volume (V)	Curved / Lateral Surface Area (LSA)	Total Surface Area (TSA)
Cube (side a)	a^3	$4a^2$	$6a^2$ (Diagonal = $a\sqrt{3}$)
Cuboid (l, b, h)	$l \times b \times h$	$2h(l + b)$	$2(lb + bh + hl)$ (Diagonal = $\sqrt{(l^2+b^2+h^2)}$)
Right Cylinder (r, h)	$\pi r^2 h$	$2\pi rh$	$2\pi r(h + r)$
Right Cone (r, h)	$(1/3)\, \pi r^2 h$	πrl (Slant height $l = \sqrt{(r^2+h^2)}$)	$\pi r(l + r)$
Sphere (r)	$(4/3)\, \pi r^3$	$4\pi r^2$	$4\pi r^2$
Hemisphere (r)	$(2/3)\, \pi r^3$	$2\pi r^2$	$3\pi r^2$
Right Prism	$\text{Base Area} \times h$	$\text{Base Perimeter} \times h$	$LSA + 2(\text{Base Area})$
Right Pyramid	$(1/3) \times \text{Base Area} \times h$	$(1/2) \times \text{Base Perimeter} \times l$	$LSA + \text{Base Area}$

3. Cross-Sections, Frustums & Cutting Solids

When a solid cone or pyramid is cut by a plane parallel to its base, it yields a smaller similar cone/pyramid at the top and a **Frustum** at the bottom.

FRUSTUM METRICS & SIMILARITY RATIOS

- **Similar Solid Similarity Scaling:** If a cone of height H and base radius R is cut at height h from apex creating a top cone of radius r:
  - Linear ratio: $r / R = h / H = l / L = k$
  - Area ratio: $A_{top} / A_{base} = (h / H)^2 = k^2$
  - Volume ratio: $V_{top} / V_{total} = (h / H)^3 = k^3$
- **Frustum of a Cone (R, r, h):**
  - Slant height: $l = \sqrt{[\, h^2 + (R - r)^2 \,]}$
  - Volume: $V = (1/3)\, \pi\, h\, (R^2 + r^2 + R\, r)$
  - Curved Surface Area: $CSA = \pi\, l\, (R + r)$
  - Total Surface Area: $TSA = \pi\, l\, (R + r) + \pi\, R^2 + \pi\, r^2$

4. Melting, Recasting & Surface Area Changes

When a solid object is melted and recast into new solid shapes, the total mass and **Volume remains strictly conserved**, but the Total Surface Area usually changes.

CONSERVATION PRINCIPLES & CUTTING INCREASES

- **Volume Conservation Law:** $V_{original} = \sum V_{recast\ solids}$
  If a sphere of radius R is melted into n identical small spheres of radius r:
  $(4/3)\, \pi\, R^3 = n \times (4/3)\, \pi\, r^3 \Longrightarrow R^3 = n\, r^3 \Longrightarrow r = R / n^{1/3}$.
- **Surface Area Change on Melting:** Total surface area of n small spheres = $n \times 4\pi r^2 = n \times 4\pi\, (R / n^{1/3})^2 = n^{1/3} \times (4\pi R^2)$.
  Surface area increases by factor of $n^{1/3}$ when recasting!
- **Increase in Surface Area upon Cutting:**
  - 1 cut parallel to base of cylinder adds 2 circular cross-sections = $+2\pi r^2$.
  - 1 cut passing through center of sphere splits it into 2 hemispheres adding 2 circular faces = $+2\pi r^2$.

5. Common Pitfalls & Mistakes

- **Confusing Height (h) with Slant Height (l) in Cone/Pyramid:** Using vertical height h in CSA formulas instead of slant height $l = \sqrt{(r^2 + h^2)}$.
- **Assuming Surface Area is Conserved during Melting:** Assuming total surface area stays constant when a shape is melted and recast. Volume is conserved, but surface area increases!
- **Miscalculating Frustum Height Ratios:** Substituting the height of the frustum directly into similarity ratio $(h / H)^3$ instead of using the height of the smaller top cone.

6. CAT-Level Solved Practice Questions

Question 1 (Cone Cutting & Volume Ratios)

A right circular cone of height 30 cm is cut into three parts by two planes parallel to its base at heights of 10 cm and 20 cm from the base. Find the ratio of the volumes of the three resulting parts (top cone : middle frustum : bottom frustum).

Solution:

Total height of original cone $H = 30$ cm.
Height of top cone from apex $h_1 = 30 - 20 = 10$ cm.
Height of middle cone from apex $h_2 = 30 - 10 = 20$ cm.
Height of total cone from apex $h_3 = 30$ cm.

By similarity of cones, volume of a cone is proportional to the cube of its height ($V \propto h^3$):

$V_{\text{top cone}} : V_{\text{middle cone}} : V_{\text{total cone}} = h_1^3 : h_2^3 : h_3^3$

$= 10^3 : 20^3 : 30^3 = 1 : 8 : 27$.

Now calculate volumes of individual parts:

- Volume of Top Part = $V_1 = 1$ unit.
- Volume of Middle Frustum = $V_{\text{middle cone}} - V_{\text{top cone}} = 8 - 1 = 7$ units.
- Volume of Bottom Frustum = $V_{\text{total cone}} - V_{\text{middle cone}} = 27 - 8 = 19$ units.

Ratio of volumes of three parts = **1 : 7 : 19**.

Question 2 (Melting Spheres into Cylinder)

A solid metallic sphere of radius 6 cm is melted and recast into a solid right circular cylinder of height 8 cm. Find the curved surface area of the cylinder.

Solution:

Step 1: Volume of original sphere:

$V_{\text{sphere}} = (4/3)\,\pi\,R^3 = (4/3)\,\pi\,(6^3) = (4/3)\,\pi\,(216) = 288\pi$ cm^3.

Step 2: Conserve volume to find cylinder radius r:

$V_{\text{cylinder}} = \pi\,r^2\,h = 288\pi$.

Substitute $h = 8$:

$\pi\,r^2\,(8) = 288\pi \Rightarrow 8r^2 = 288 \Rightarrow r^2 = 36 \Rightarrow r = 6$ cm.

Step 3: Calculate Curved Surface Area (CSA) of cylinder:

$CSA = 2\pi\,r\,h = 2\pi\,(6)(8) = 96\pi$ cm^2.

The curved surface area of the cylinder is **96π cm^2** (or ≈ 301.59 cm^2).

Question 3 (Sphere Cutting & Surface Area Percentage Increase)

A solid wooden sphere of radius 10 cm is cut into 8 identical equal pieces by three mutually perpendicular cuts passing through its center. Find the percentage increase in total surface area.

Solution:

Original Total Surface Area of solid sphere = $TSA_{orig} = 4\pi R^2 = 4\pi(10^2) = 400\pi$ cm^2.

Each cut passing through the center adds 2 full circular cross-sectional faces of area πR^2 each.
Three mutually perpendicular cuts make 3 full cuts through the center.

Additional surface area created by 3 cuts = $3 \times (2 \times \pi R^2) = 6 \pi R^2 = 6\pi(10^2) = 600\pi$ cm^2.

New Total Surface Area of all 8 pieces combined = $400\pi + 600\pi = 1000\pi$ cm^2.

Percentage Increase = [(Additional Area) / (Original Area)] × 100%

Percentage Increase = $(600\pi / 400\pi) \times 100\% = 1.5 \times 100\% = 150\%$.

The percentage increase in total surface area is **150%**.

7. Quick-Revision Cheat Sheet

Solid / Concept	Volume Formula	Surface Area Formula
Cone Cut Similarity	$V_1 / V_2 = (h_1 / h_2)^3$	$A_1 / A_2 = (h_1 / h_2)^2$
Frustum Cone	$(1/3)\pi h(R^2 + r^2 + Rr)$	$CSA = \pi l(R + r), l = \sqrt{[h^2+(R-r)^2]}$
Right Pyramid	$(1/3) \times \text{Base Area} \times h$	$LSA = (1/2) \times \text{Base Perim} \times l$
Right Prism	$\text{Base Area} \times h$	$LSA = \text{Base Perim} \times h$
Sphere Cut (3 cuts)	Volume Conserved	Adds $6\pi R^2$ (150% increase)

MODULE 5: MODERN MATHEMATICS

Chapter 22: Permutations & Combinations

1. Introduction & Fundamental Principles

Permutations and Combinations form the bedrock of Counting Theory and Modern Mathematics. In the CAT examination, P&C questions test your logical structuring, constraint handling, and ability to break down complex arrangements into manageable fundamental rules.

FUNDAMENTAL PRINCIPLE OF COUNTING (FPC)

- **The Rule of Addition (OR Rule):** If an operation can be performed in m ways, and another independent operation can be performed in n ways, and **only one** of them can be executed (mutually exclusive), then the total number of ways to complete the task is $m + n$.
- **The Rule of Multiplication (AND Rule):** If an operation consists of two successive stages, where the first stage can be performed in m ways and the second stage in n ways **regardless of the outcome of the first**, the total number of ways to complete the entire task is $m \times n$.

2. Permutations (Arrangements)

A **Permutation** is an ordered arrangement of objects taken from a given set. The order of selection matters (e.g., password "123" is different from "321").

PERMUTATION FORMULAS

- **Linear Permutation of n distinct objects taken r at a time:**
  $P(n, r) = n! / (n - r)!$
- **Arrangement of n distinct objects taken all at a time:**
  $P(n, n) = n!$
- **Arrangement when identical items are present:** If out of n objects, p are of one kind, q are of a second kind, and r are of a third kind, the number of distinct arrangements is:
  $n! / (p! \times q! \times r!)$
- **Circular Permutations:**
  • Distinct objects: $(n - 1)!$
  • Necklaces / Bracelets (clockwise and counter-clockwise identical): $(n - 1)! / 2$

3. Combinations (Selections)

A **Combination** is a selection of objects where order **does not matter** (e.g., selecting a committee of 3 people from 10).

COMBINATION FORMULAS & IDENTITIES

- **Selection of r objects out of n distinct objects:**
  $C(n, r) = {}^{n}C_{r} = n! / [\, r!\,(n - r)!\,]$
- **Fundamental Symmetry Identity:** ${}^{n}C_{r} = {}^{n}C_{n-r}$
- **Pascal's Identity (Addition Rule):** ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$
- **Total Number of Subsets (Selection Theorem):** $\sum_{r=0}^{n} {}^{n}C_{r} = 2^{n}$

4. Advanced Distribution, Grouping & Derangements

ADVANCED COUNTING FRAMEWORKS

- **Distributing Identical Items (Stars and Bars):** Number of ways to distribute n identical items among k distinct recipients (each getting ≥ 0) is ${}^{n+k-1}C_{k-1}$. If each gets ≥ 1, it is ${}^{n-1}C_{k-1}$.
- **Grouping Distinct Objects into Identical Groups:** Dividing mn distinct objects into m groups each containing n objects: $(mn)! / [\, (n!)^{m} \times m!\,]$.
- **Derangements (D_n):** A permutation where none of the objects appear in their original position.
  $D_n = n!\,[\, 1 - 1/1! + 1/2! - 1/3! + \ldots + (-1)^{n} / n!\,]$
  Key values: $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$.

5. Common Pitfalls & Mistakes

- **Confusing Permutations with Combinations:** Using $n!$ or $P(n,r)$ when order does not matter (e.g., forming committees or selecting teams). Always check if arrangement order is significant.
- **Forgetting Symmetry Adjustment in Grouping:** Dividing distinct objects into identical groups without dividing by $m!$ to eliminate duplicate identical group counts.

6. CAT-Level Solved Practice Questions

Question 1 (Restricted Arrangements)

Find the number of words that can be formed using all the letters of the word **"EQUATION"** such that all the vowels are kept together.

Solution:

Step 1: The word "EQUATION" contains 8 distinct letters: E, Q, U, A, T, I, O, N.
Vowels: E, U, A, I, O (5 vowels). Consonants: Q, T, N (3 consonants).
Step 2: Treat the 5 vowels as a single composite unit.
Total units to arrange = 3 consonants + 1 vowel block = 4 units.
These 4 units can be arranged in $4! = 24$ ways.
Step 3: Internal arrangement of vowels among themselves = $5! = 120$ ways.
Total words = $4! \times 5! = 24 \times 120 = 2880$ words.

Question 2 (Circular Seating with Restrictions)

In how many ways can 5 boys and 5 girls be seated around a circular table such that no two girls sit adjacent to each other?

Solution:

Step 1: Seat the 5 boys around the circular table in $(5 - 1)! = 4! = 24$ ways.
Step 2: Seating 5 boys creates exactly 5 distinct gaps between them.
Step 3: To ensure no two girls sit together, place the 5 girls into these 5 gaps in $5! = 120$ ways.
Total valid arrangements = $24 \times 120 = 2880$ ways.

Question 3 (Derangements Application)

4 persons write letters and place them into 4 addressed envelopes. In how many ways can the letters be placed into the envelopes such that no letter goes into its correct envelope?

Solution:

This is a direct application of Derangements for $n = 4$ (D_4):
$D_4 = 4! [1 - 1/1! + 1/2! - 1/3! + 1/4!]$
$= 24 [1 - 1 + 1/2 - 1/6 + 1/24]$
$= 24 [(12 - 4 + 1) / 24] = 24 \times (9 / 24) = 9$.
Total valid ways = **9**.

7. Quick-Revision Cheat Sheet

Concept	Formula / Rule	Key Takeaway
Permutation	$P(n, r) = n! / (n - r)!$	Order of selection matters.
Combination	${}^{n}C_{r} = n! / [\, r!\,(n - r)!\,]$	Order does not matter.
Circular Permutation	$(n - 1)!$	Fix one position to break symmetry.
Stars & Bars	${}^{n+k-1}C_{k-1}$	Distributing identical items into distinct groups.
Derangements	$D_4 = 9$, $D_5 = 44$	No item occupies its original position.

MODULE 5: MODERN MATHEMATICS

Chapter 23: Probability

1. Introduction & Axiomatic Foundations

Probability theory quantifies uncertainty, providing the mathematical framework for risk analysis, decision-making, and statistical modeling in the CAT examination.

BASIC DEFINITIONS & AXIOMS

- **Sample Space ($\mathbf{S}$):** The set of all possible outcomes of a random experiment.
- **Classical Probability:** If an experiment results in n exhaustive and equally likely outcomes, and m of them are favorable to event E:
  $P(E) = m / n = n(E) / n(S)$
- **Axioms:** $0 \le P(E) \le 1$, $P(S) = 1$, and for mutually exclusive A, B ($A \cap B = \phi$): $P(A \cup B) = P(A) + P(B)$.

2. Fundamental Theorems of Probability

ADDITION & MULTIPLICATION THEOREMS

- **Addition Theorem:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Multiplication Theorem:** $P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$
- **Independent Events:** $P(A \cap B) = P(A) \times P(B)$ (Mutually exclusive events with non-zero probability are never independent).

3. Conditional Probability & Bayes' Theorem

ADVANCED PROBABILITY FRAMEWORKS

- **Conditional Probability:** $P(A|B) = P(A \cap B) / P(B)$ (for $P(B) > 0$).
- **Law of Total Probability:** $P(A) = \sum P(E_i) \times P(A|E_i)$ over a partition E_i.
- **Bayes' Theorem:** $P(E_i|A) = [\, P(E_i) \times P(A|E_i) \,] / [\, \sum P(E_j) \times P(A|E_j) \,]$.

4. Random Variables & Expected Value

The **Expected Value ($\mathbf{E[X]}$)** of a discrete random variable taking values x_i with probabilities p_i is:
$E[X] = \sum x_i p_i$.

5. Common Pitfalls & Mistakes

- **Confusing Mutually Exclusive with Independent:** Assuming mutually exclusive events are independent. If $P(A) > 0$ and $P(B) > 0$, mutual exclusivity ($P(A \cap B) = 0$) directly violates independence ($P(A)P(B) \neq 0$).

6. CAT-Level Solved Practice Questions

Question 1 (Conditional Probability with Cards)

Two cards are drawn successively without replacement from a standard pack of 52 cards. What is the probability that the second card drawn is a King, given that the first card drawn was a heart?

Solution:

By total probability across cases (first card is/isn't King of Hearts):
$P(K_2) = (12/52 \times 4/51) + (1/52 \times 3/51) = (48 + 3) / 2652 = 51 / 2652 = 1 / 52$.
The probability is **1 / 52** (unconditional symmetry).

Question 2 (Bayes' Theorem Application)

Bag A has 3 red and 5 black balls; Bag B has 4 red and 2 black balls. A bag is chosen at random and a ball drawn is red. What is the probability it came from Bag A?

Solution:

$P(A|R) = [\ (1/2)(3/8)\ ] / [\ (1/2)(3/8) + (1/2)(4/6)\ ] = (3/16) / [\ 3/16 + 1/3\ ] = (3/16) / (25/48) = 9 / 25$.
The probability is **9 / 25 (0.36)**.

Question 3 (Expected Value Optimization)

A game costs ₹100 to play. A fair die is rolled. Roll 6 wins ₹400; roll 2 or 4 wins ₹150; otherwise win ₹0. Calculate expected net profit.

Solution:

$E[X] = (+300 \times 1/6) + (+50 \times 1/3) + (-100 \times 1/2) = 50 + 16.67 - 50 = +16.67$.
Expected net profit is **₹16.67** per game.

7. Quick-Revision Cheat Sheet

Concept	Formula / Theorem	Key Takeaway
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	General union probability.
Conditional	$P(A\|B) = P(A \cap B) / P(B)$	Probability of A given B occurred.
Bayes' Theorem	$P(E_i\|A) = [\ P(E_i)P(A\|E_i)\ ] / \sum \ldots$	Reverses conditional probabilities.
Expected Value	$E[X] = \sum x_i\, p_i$	Theoretical average outcome.

MODULE 5: MODERN MATHEMATICS

Chapter 24: Set Theory & Venn Diagrams

1. Set Foundations & Fundamental Operations

Set theory provides the formal logic structure for classifying data and solving complex overlapping group problems frequently featured in CAT QA and DILR.

CORE SET OPERATIONS & LAWS

- **Universal Set (U):** The overarching set containing all possible elements under consideration.
- **Complement (A' or A^c):** Elements in U that do not belong to A: $n(A') = n(U) - n(A)$.
- **Union ($A \cup B$):** Elements belonging to A, B, or both.
- **Intersection ($A \cap B$):** Elements common to both A and B.
- **De Morgan's Laws:**
  • $(A \cup B)' = A' \cap B'$
  • $(A \cap B)' = A' \cup B'$

2. Cardinality Formulas for 2-Set & 3-Set Venn Diagrams

2-SET CARDINALITY FORMULA

For any two sets A and B within universal set U:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Elements outside both sets (Neither A nor B): $n((A \cup B)') = n(U) - n(A \cup B)$.

3-SET CARDINALITY FORMULA

For three sets A, B, C:
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

3. Maxima & Minima Bounds in Set Theory

CAT frequently tests boundary conditions (maximum and minimum values) of intersections and unions when complete overlapping data is not explicitly provided.

MAXIMA & MINIMA RULES (2-SET & 3-SET)

- **Minimum Intersection for 2 Sets ($n(A \cap B)_{min}$):**
  $n(A \cap B)_{min} = \max(0, n(A) + n(B) - n(U))$
- **Maximum Intersection for 2 Sets ($n(A \cap B)_{max}$):**
  $n(A \cap B)_{max} = \min(n(A), n(B))$ (Occurs when one set is entirely contained within the other).
- **Minimum Union for 2 Sets ($n(A \cup B)_{min}$):**
  $n(A \cup B)_{min} = \max(n(A), n(B))$
- **Minimum Intersection for 3 Sets ($n(A \cap B \cap C)_{min}$):**
  $n(A \cap B \cap C)_{min} = \max(0, n(A) + n(B) + n(C) - 2 \cdot n(U))$ (derived by minimizing outer overlaps).

4. Common Pitfalls & Mistakes

- **Confusing Exactly Two with At Least Two:** Treating $n(A \cap B)$ as strictly elements belonging to *only* A and B. $n(A \cap B)$ includes the center triple-intersection $n(A \cap B \cap C)$ in 3-set diagrams unless explicitly subtracted!
- **<Ignoring Universal Set Bounds:** Failing to verify that calculated intersection or union values respect the upper bound set by the total universal population $n(U)$.

5. CAT-Level Solved Practice Questions

Question 1 (3-Set Cardinality Problem)

In a survey of 100 students, 50 play Cricket, 40 play Football, and 30 play Hockey. 15 play both Cricket and Football, 10 play Football and Hockey, and 5 play Cricket and Hockey. If 5 students play all three sports, find how many students play none of the three sports.

Solution:

Let C, F, H denote Cricket, Football, and Hockey.
Given: $n(U) = 100$, $n(C) = 50$, $n(F) = 40$, $n(H) = 30$.
$n(C \cap F) = 15$, $n(F \cap H) = 10$, $n(C \cap H) = 5$.
$n(C \cap F \cap H) = 5$.
Apply 3-Set Union Formula:
$n(C \cup F \cup H) = [\ 50 + 40 + 30\ ] - [\ 15 + 10 + 5\ ] + 5$
$= 120 - 30 + 5 = 95$ students play at least one sport.
Students playing none = $n(U) - n(C \cup F \cup H) = 100 - 95 = 5$.
Number of students playing none is **5**.

Question 2 (Maxima-Minima Boundary Problem)

In a class of 60 students, 45 passed in Mathematics and 50 passed in Physics. What is the minimum possible number of students who passed in both subjects?

Solution:

Given: $n(U) = 60$, $n(M) = 45$, $n(P) = 50$.
Apply Minimum Intersection Formula for 2 sets:
$n(M \cap P)_{min} = \max(0, n(M) + n(P) - n(U))$
$= \max(0, 45 + 50 - 60) = \max(0, 95 - 60) = \max(0, 35) = 35$.
The minimum possible number of students who passed in both is **35**.

Question 3 (Only-Two Intersections)

Given $n(A) = 30$, $n(B) = 35$, $n(C) = 40$, $n(A \cap B) = 12$, $n(B \cap C) = 15$, $n(C \cap A) = 10$, and $n(A \cap B \cap C) = 5$. Find the number of elements belonging to exactly two of the sets.

Solution:

Elements in exactly two sets = $[\, n(A \cap B) - n(A \cap B \cap C) \,] + [\, n(B \cap C) - n(A \cap B \cap C) \,] + [\, n(C \cap A) - n(A \cap B \cap C) \,]$
$= (12 - 5) + (15 - 5) + (10 - 5)$
$= 7 + 10 + 5 = 22$.
The number of elements belonging to exactly two sets is **22**.

6. Quick-Revision Cheat Sheet

Concept	Formula / Condition	Key Takeaway
2-Set Union	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$	Fundamental inclusion-exclusion principle.
Min Intersection	$\max(0, n(A) + n(B) - n(U))$	Ensures total does not exceed universe $n(U)$.
Max Intersection	$\min(n(A), n(B))$	Smaller set fully contained in larger set.
De Morgan's Law	$(A \cup B)' = A' \cap B'$	Complement of union is intersection of complements.

MODULE 5: MODERN MATHEMATICS

Chapter 25: Functions, Graphs, Logarithms & Progressions

1. Functions — Domain, Range, Composite & Inverse

A function $f: A \to B$ maps every element in domain A to a unique element in codomain B.

CORE FUNCTIONAL PROPERTIES

- **Domain & Range:** Domain is the set of valid inputs x; Range is the resulting set of outputs $f(x)$.
- **One-to-One (Injective):** $f(x_1) = f(x_2) \implies x_1 = x_2$.
- **Onto (Surjective):** Range equals codomain.
- **Composite Functions:** $(f \circ g)(x) = f(g(x))$.
- **Inverse Functions ($f^{-1}(x)$):** Exists strictly for bijective functions. Solved by interchanging x and y and solving for y.
- **Periodic Functions:** $f(x + T) = f(x)$, where T is the fundamental period.

2. Logarithms — Laws & Inequalities

$\log_a(x) = y \iff a^y = x$, where base $a > 0$, $a \neq 1$, and argument $x > 0$.

LOGARITHMIC LAWS

- $\log_a(xy) = \log_a(x) + \log_a(y)$ • $\log_a(x / y) = \log_a(x) - \log_a(y)$
- $\log_a(x^k) = k \cdot \log_a(x)$ • $\log_a(b) = \log_\{c\}(b) / \log_\{c\}(a)$ (Change of Base)
- $a^{\log_a(x)} = x$ • $\log_a(1) = 0, \log_a(a) = 1$

3. Progressions (AP, GP, HP) & Special Series

Progression Type	General Term (T_n)	Sum of n Terms (S_n)	Key Mean Condition
Arithmetic Progression (AP)	$a + (n-1)d$	$(n/2)[2a + (n-1)d]$	Arithmetic Mean: $AM = (a+b)/2$
Geometric Progression (GP)	$a \cdot r^{n-1}$	$a(r^n - 1)/(r-1)$	Geometric Mean: $GM = \sqrt{ab}$
Harmonic Progression (HP)	$1 / [a + (n-1)d]$	No direct simple formula	Harmonic Mean: $HM = 2ab / (a+b)$

SPECIAL SERIES SUMMATIONS

- Sum of first n natural numbers: $\sum n = n(n+1)/2$
- Sum of squares of first n numbers: $\sum n^2 = n(n+1)(2n+1)/6$
- Sum of cubes of first n numbers: $\sum n^3 = [\, n(n+1)/2 \,]^2$

4. Common Pitfalls & Mistakes

- **Forgetting Logarithmic Domain Restrictions:** Solving logarithmic equations without enforcing argument positivity ($x > 0$) and base validity ($a > 0, a \neq 1$), leading to extraneous solutions.
- **AP/GP Common Ratio Oversights:** Assuming a series is GP when terms alternate signs without checking if common ratio r is negative.

5. CAT-Level Solved Practice Questions

Question 1 (Logarithmic Equation Solving)

Solve for x: $\log_2(x-1) + \log_2(x+2) = 2$.

Solution:

Step 1: Enforce domain restrictions: $x - 1 > 0 \Rightarrow x > 1$, and $x + 2 > 0 \Rightarrow x > -2$. Thus, valid domain is $x > 1$.
Step 2: Apply log addition law $\log_a(A) + \log_a(B) = \log_a(AB)$:
$\log_2[\,(x-1)(x+2)\,] = 2$.
Step 3: Convert logarithmic form to exponential form:
$(x-1)(x+2) = 2^2 = 4$
$x^2 + x - 2 = 4 \Rightarrow x^2 + x - 6 = 0$.
Step 4: Factor quadratic equation:
$(x+3)(x-2) = 0 \Rightarrow x = -3$ or $x = 2$.
Step 5: Check domain: $x = -3$ is rejected since $x > 1$.
Valid solution is **x = 2**.

Question 2 (Geometric Progression Sum to Infinity)

The sum of an infinite geometric progression is 15, and the sum of the squares of its terms is 45. Find the first term of the series.

Solution:

Let first term be a and common ratio be r (where $|r| < 1$).
Sum to infinity $S_\infty = a / (1 - r) = 15 \Longrightarrow a = 15(1 - r)$.
The squares of the terms form a new GP with first term a^2 and common ratio r^2.
Sum of squares $S'_\infty = a^2 / (1 - r^2) = 45$.
Substitute $a^2 = [\ 15(1 - r)\ ]^2$:
$[\ 225(1 - r)^2\ ] / [\ (1 - r)(1 + r)\ ] = 45$.
Cancel $(1 - r)$ (since $r \neq 1$):
$225(1 - r) / (1 + r) = 45 \Longrightarrow 5(1 - r) = 1 + r$
$5 - 5r = 1 + r \Longrightarrow 6r = 4 \Longrightarrow r = 2/3$.
Find first term a:
$a = 15(1 - 2/3) = 15 \times (1/3) = 5$.
The first term of the series is **5**.

Question 3 (Composite Function Evaluation)

If $f(x) = (x - 1) / (x + 1)$, find $f(f(x))$ in terms of x.

Solution:

Substitute $f(x)$ into itself:
$f(f(x)) = [\ f(x) - 1\ ] / [\ f(x) + 1\ ]$
$= [\ ((x - 1)/(x + 1)) - 1\ ] / [\ ((x - 1)/(x + 1)) + 1\ ]$.
Multiply numerator and denominator by $(x + 1)$:
$= [\ (x - 1) - (x + 1)\ ] / [\ (x - 1) + (x + 1)\ ]$
$= [\ x - 1 - x - 1\ ] / [\ x - 1 + x + 1\ ] = -2 / 2x = -1 / x$.
$f(f(x)) = -1 / x$.

6. Quick-Revision Cheat Sheet

Topic	Formula / Rule	Key Takeaway
Log Change of Base	$\log_a(b) = \log_c(b) / \log_c(a)$	Essential for matching disparate log bases.
Infinite GP Sum	$S_\infty = a / (1 - r)$	Valid strictly when $\|r\| < 1$.
AM-GM-HM Inequality	$AM \geq GM \geq HM$	Equality holds iff all terms are equal.
Sum of Cubes	$\sum n^3 = [\ n(n + 1) / 2\ ]^2$	Square of sum of first n natural numbers.